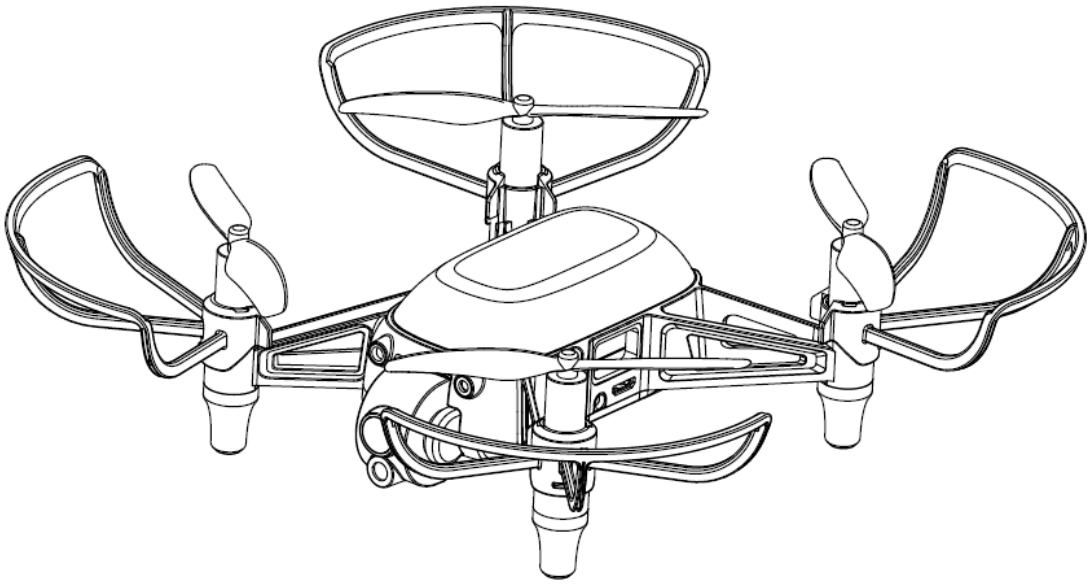
高巨创新
HIGH GREAT



Hula User Manual

(Revision date: January 30, 2024)

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Tips

Hula is a professional indoor drone for artificial intelligence and educational programming, with high requirements for the use methods and use environment. In order to quickly understand the basic use methods and various considerations of this Product and to improve the efficiency and reduce various risks during use, please read the *Hula User Manual* and the *Hula Quick Start Guide* carefully.

The Hula drone is suitable for ages 14 and above. Children should use it under the supervision of a guardian.

Description of symbols:



Note



Cautions

1. Hula Quick Start Guide

1.1 Disclaimer

1. Children under the age of 14 should use this Product under the supervision of an adult.
2. This Product is a multi-rotor aircraft that offers an easy flight experience when powered normally and with all components undamaged. Please visit the official website of Shenzhen HighGreat Innovation Technology Development Co., Ltd.(hereinafter referred to as "HighGreat Innovation") to obtain the complete *User Manual* as well as the latest instructions and warnings. HighGreat Innovation reserves the right to update all documents.
3. The information in this document affects your safety and your legal rights and responsibilities. Read this entire document and the *User Manual* carefully to ensure proper configuration before use. Failure to read and follow the instructions and warnings in this document may result in property loss, safety incidents, and personal injury risk. By using this Product, you hereby signify that you have read this disclaimer carefully and that you understand and agree to abide by the terms and conditions herein. Users undertake to be responsible for his / her own actions and all consequences arising therefrom. Users agree to use this Product only for legitimate purposes and to abide by these terms and conditions in this clause and any related policies or guidelines that may be established by HighGreat Innovation. You understand and agree that without flight records from the APP, HighGreat Innovation may be unable to analyze the causes of product damage or accidents and may not be able to provide you with HighGreat Innovation's after-sales service.
4. HighGreat Innovation assumes no responsibility for any loss or damage resulting from users' failure to follow the instructions in this document and the *User Manual* when using this Product. Subject to compliance with laws and regulations, HighGreat Innovation reserves the right to final interpretation of this document. HighGreat Innovation reserves the right, at any time, to update, revise, or terminate these terms without prior notice.
5. Please visit the official website for the complete User Manual and the latest instructions and warnings. Shenzhen HighGreat Innovation Technology Development Co., LTD. (hereinafter referred to as " HighGreat Innovation") reserves the right to update all documents.

1.2 Warnings

1. Please stay away from the high-speed rotating propellers to prevent cuts.
2. The propeller motors are heat-producing components. Please do not touch to avoid burns.
3. In order to ensure the electromagnetic environment of the aviation radio station, it is forbidden to use all kinds of model remote controls in the area with the airport runway as the center point and a radius of 5000 m. During the period when the radio control commands are issued by the relevant state departments, the model remote control shall be stopped as required.
4. Please use the dedicated batteries equipped. DO NOT short-circuit the positive and negative terminals of the battery.
5. The product package and the manual contain important information and should be retained.
6. Please use this Product within the operating environment range specified in the specifications.
7. DO NOT direct the laser at human eyes to avoid causing eye injuries. 

1.3 Flight limit

The user acknowledges and agrees to be responsible for all consequences arising from the use of the aircraft.

1. Fly NO higher than 10 meters above ground level.
2. Fly in conditions with wind speed below class 3. DO NOT fly outdoors during thunderstorms or typhoon weather.
3. Indoor environments can be complex. When flying indoors, please assess thoroughly whether the flight environment is safe.
4. Maintain a distance of at least 2 meters between the aircraft and any pedestrians when you operate the aircraft.
5. Use the aircraft in an environment with clear ground textures and without strong light exposure. Flying over water surfaces or mirrored surfaces is strictly prohibited.

1.4 Laws and regulations

Please comply with the local laws and regulations when using the aircraft to avoid possible injury and damage.

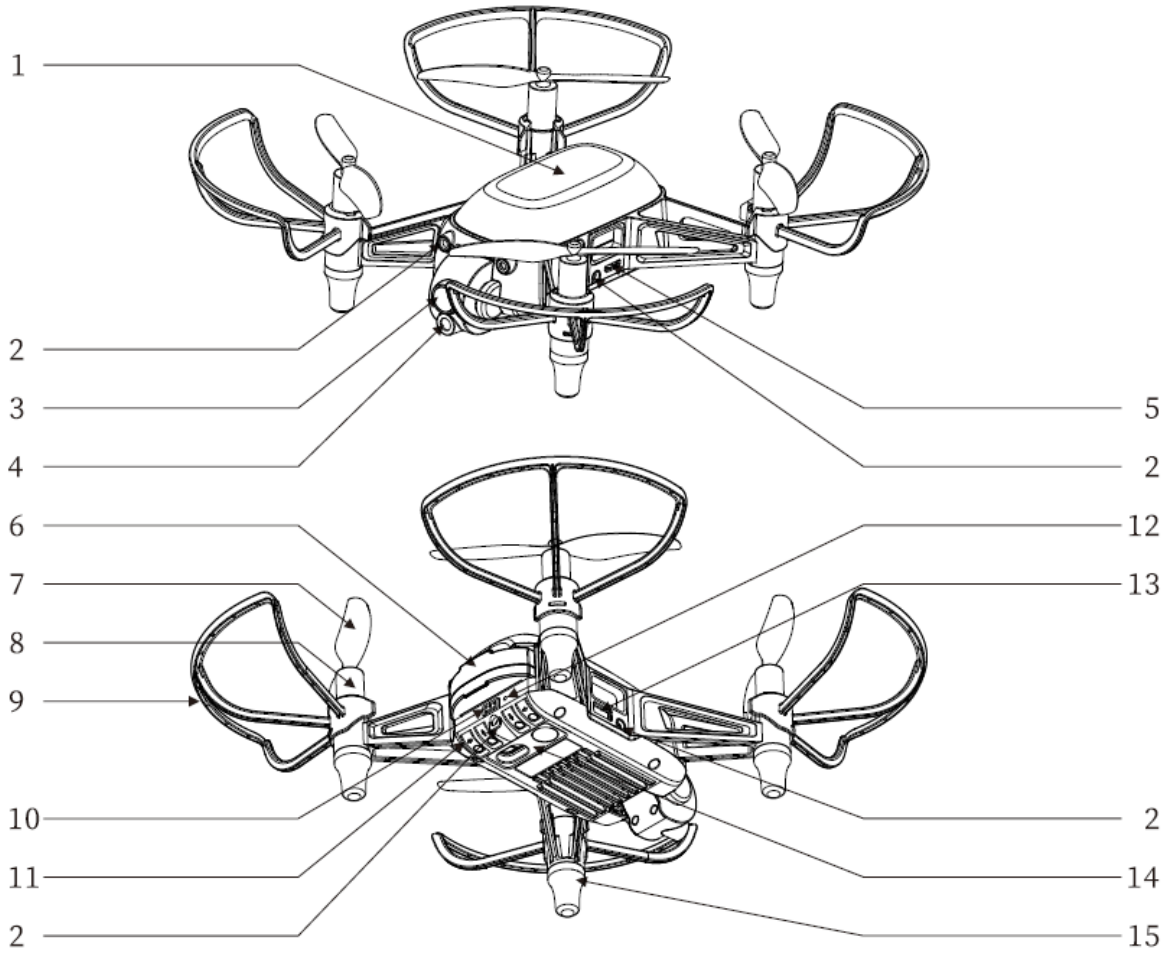
Please observe the following rules:

1. DO NOT fly the aircraft in the vicinity of manned aircraft. Land your aircraft immediately when necessary.
2. DO NOT fly the aircraft in or at densely populated areas, including but not limited to cities, sports stadiums, exhibitions, and concerts.
3. Make sure that your aircraft does not interfere with manned aircraft operations. Be aware of and avoid any other aircraft at all times.
4. DO NOT fly in the no-fly zones specified by local laws and regulations. No-fly zones include airports, borderlines, power plants, hydropower plants, prisons, heavily traveled roadways, government buildings, military facilities.
5. DO NOT fly the aircraft above the authorized altitude.
6. Keep your aircraft in sight at all times and arrange for an observer to assist you in monitoring the aircraft's position if needed.
7. DO NOT use the aircraft to transport dangerous or contraband items.
8. Make sure you understand the type of your flight operation (such as for recreation, for public use, or for commercial use). You have to obtain the corresponding license from the related agencies before the flight. If necessary, you can consult with local legal professionals for detailed definitions and explanations of flight activity categories. Please note that in certain regions and countries, the use of aircraft for any form of commercial activity is prohibited.
9. Respect the privacy rights of others when using the aircraft for photo taking. DO NOT conduct surveillance operations without authorization. These operations include but are not limited to conducting surveillance on any person, entity, event, performance, exhibition, or building.
10. Please be advised that in certain regions or countries, the recording of images and videos of any person, entity, event, performance, exhibition, and so on by means of a camera may contravene copyright or other legal rights, even if the image or video is not shot for commercial use. In certain regions and countries, the use of small aerial models for commercial purposes is also prohibited. Therefore, please carefully understand and comply with the local laws and regulations before use.
11. Shall not arbitrarily change the use scenario or use conditions of the aircraft, expand the transmission frequency range, increase the transmission power (including additional RF power amplifier), shall not arbitrarily change the transmission antenna.
12. Shall not cause harmful interference to other lawful radio stations (stations), nor shall protection from harmful interference be proposed;

13. Shall withstand interference from industrial, scientific and medical (ISM) applications radiating RF energy or other legitimate radio station (station) interference.
14. If harmful interference occurs to other legal radio stations (stations), the use shall be stopped immediately, and measures shall be taken to eliminate the interference before the use can continue.
15. The use of Hula in aircraft, radio observatories, meteorological radar stations, satellite earth stations (including measurement and control, ranging, receiving and navigation stations) and other military and civilian radio stations (stations), airports and other electromagnetic environmental protection areas in accordance with laws and regulations, relevant state regulations and standards shall comply with the provisions of the competent authorities of electromagnetic environmental protection and relevant industries.

1.5 Hula Introduction

Hula is a programmable educational aircraft designed specifically for teenagers. It supports various terminal devices for programming flight using Scratch and can also be wirelessly controlled for flight through the Hula APP. The aircraft is equipped with a visual positioning system, gimbal system, image transmission system, obstacle avoidance system, and flight control system. It can perform functions such as 1080P HD image transmission, photo-taking, video-recording, AI recognition, obstacle avoidance, line patrol, and various flight stunts.



1. Lamp cover	6. Battery	11. Obstacle Avoidance Adjustment Knob
2. Infrared Obstacle Avoidance Transmitter	7. Propeller	12. Reset Button
3. Gimbal/Front Camera	8. Motor	13. Expansion Serial Port
4. Laser Transmitter	9. Propeller Guard	14. Visual Positioning System
5. Micro USB Port	10. Power Button	15. Foot mats

1.6 Technical specifications

Specifications		Parameters
Aircraft	Weight(Battery included)	100 g (± 3 g)
	Axle distance	128 mm
	Aircraft dimensions	189.3 × 184.6 × 50 mm
	Motor specifications	L8.5 20
	Propeller blade	75 mm/3’’
	Wind resistance class*①	Below Class 3
	Positioning method	Optical flow and QR code
	Positioning accuracy	QR code: horizontal ± 5 cm, vertical ± 10cm
		Optical flow: horizontal ± 20 cm, vertical ± 10 cm
	Max tilt angle	20°
	Max horizontal speed*②	3 m/s
	Max climb speed	1.2 m/s
	Max descent speed	1 m/s
	Max flight height	10 m
	Max communication distance	50 m
	Antenna type	PCB Antenna
	Antenna Gain	2.4GHz:3.41dBi (MAX)　5.8GHz:2.31dBi (MAX)*③
	Communication frequency	2.412-2.462GHz，5.745-5.825GHz
	Transmitting power limit	≤14dBm(e.i.r.p)
	Max lighting power	1.5 W
	Flight time*④	9 min~10 min
	Operating temperature	0°C~40°C
Battery	Capacity	1200 mAh
	Voltage	3.8 V
	Type	Lithium-ion battery

	Weight	31 g
	Storage temperature*⑤	≤ 1 Month: -20°C~45°C
		≤ 3 Month: -20°C~30°C
		≤ 1 Year: 25°C ± 3°C
Laser	Wavelength	640nm
Camera	Photo	1920 × 1080P
	Video	720P/30 fps*⑥
	Field of view	71°
	Format	JPG, MP4
Charging box (optional)	Input voltage	5V 3A
	Charging current	1.4 A
Control software	Hula APP for mobile	IOS 15.0 or above and Android 11 or above
	Hula APP for PAD	IOS 15.0 or above and Android 11 or above
	Hula software version for PC	64-bit operating system (Windows 10 or above)

① The wind resistance data is obtained in the standard wind tunnel laboratory under hovering conditions.

② The maximum flight speed is determined by the actual flight environment. This speed listed here is the maximum flight speed in the optical flow positioning mode. The maximum flight speed in the APP control mode is 1.5 m/s. In programming mode, the recommended flight speed is 0.5 m/s~1 m/s.

③ The flight time is measured at a height of 2 m, with the lighting powered off, the low-power landing function disabled, and the drone hovering in windless conditions.

④ In the line patrol flight mode, the video resolution automatically switches to 360p/30 fps.

⑤ When the aircraft is used under low temperature conditions (< 5°C), the battery life is reduced to a normal phenomenon due to insufficient battery activity. It is recommended that the low-power landing function can be turned off for use under familiar conditions of lithium battery performance.

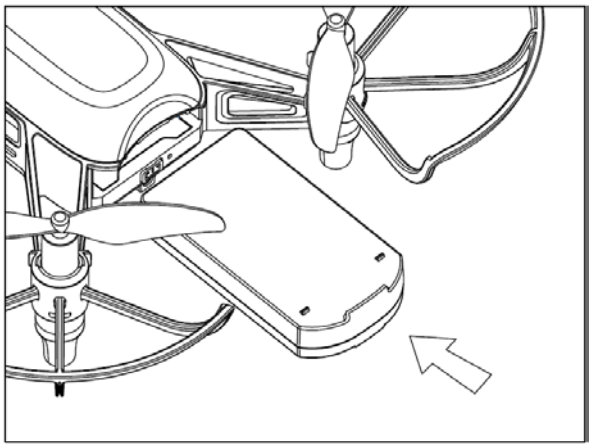
⑥ In patrol flight mode, the video resolution is automatically switched to 360p/30fps.

Hula noise measurement (Normalized to 1 m from drone) schedule:

Observation	Hover	Moving 1 m/s
Ground (vertically below)	71.1 dB (A)	75.2 dB (A)
Side (contour plane)	70.2 dB (A)	76.2 dB (A)
Note: The test environment was a full anechoic chamber.		

The above data is measured in a standard experimental environment, and may vary with the change of environment or firmware! The right of final interpretation belongs to Shenzhen HighGreat Innovation Technology Development Co., Ltd.

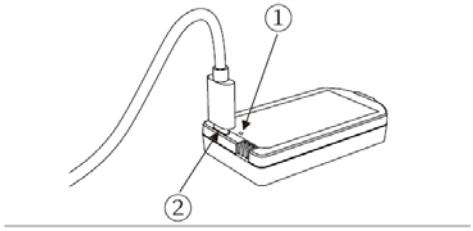
1.7 Install the battery

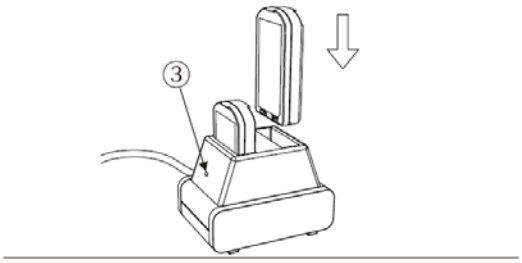


Insert the flight battery into the aircraft as shown.

Hold the aircraft body steady, place the side with a sticker of the battery facing downward, and insert it into the battery compartment on the back of the aircraft. Push it in until you hear a click, which indicates that the battery is properly installed.

1.8 Charge the battery





① Indicator for charging status on the battery, ②USB charging port, ③Indicator for charging status on the charging box.

Using USB direct charger:
Solid red, charging;
Red light being off, charged;

Using charging box: Solid red, charging;
Solid green, No pulgged in or charged;

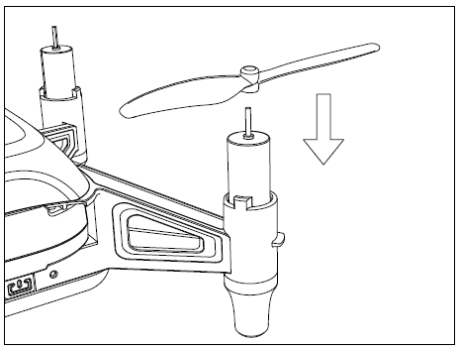
Hula supports two charging methods. You can use a standard Micro USB cable to connect the battery's Micro USB port to your own USB 5 V charger for charging, which takes about 1 hour and 40 minutes. Alternatively, you can insert the battery into the dedicated charging box for charging, which takes about 1 hour.

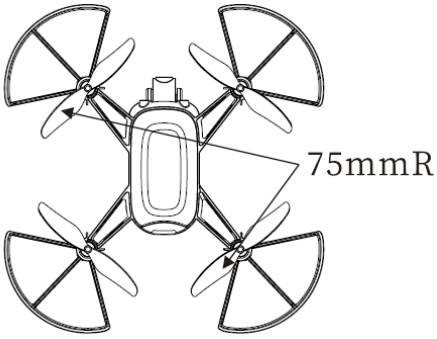
Please use the dedicated charging box equipped by HighGreat Innovation to charge the battery.

In case the battery gets wet, swells, leaks, emits unpleasant odors, deforms, or exhibits any other abnormalities, please stop using it immediately.

Please use a 5V 3A power adapter for the charging box.

1.9 Install the propellers

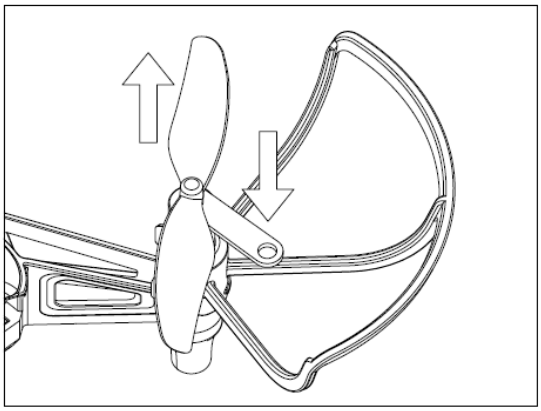




When mounting, ensure that the gap between the bottom of the propeller cap and the motor is no bigger than needed to insert the propeller removal tool.

Please note that the propeller blades with "75 mmR" markings should be installed on the top left and bottom right sides of the aircraft nose, while the propeller blades with "75 mm" markings should be installed on the top right and bottom left sides of the nose. Failing to follow this configuration may result in the aircraft not being able to take off properly.

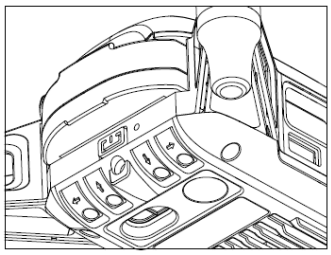
Detach the propellers



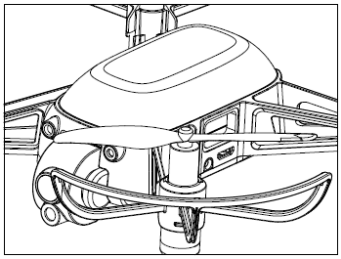
Insert the propeller removal tool between the propeller cap and the motor, press down on the other end of the removal tool, and then remove the propeller blade.

Always detach the propellers using the propeller removal tool. DO NOT remove the propellers by hand, or you may be seriously hurt and may damage the motors.

1.10 Direct connection mode for aircraft connection



Press and hold the power button on the back of the aircraft for 2 seconds to power on/off the aircraft.



After the aircraft is powered on, it is in the direct connection mode if the lamp cover flashes with purple light.

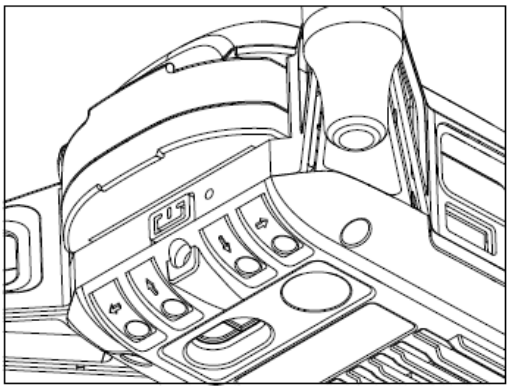


Connect your PC/mobile device to the aircraft WiFi, and the initial password is 12345678.

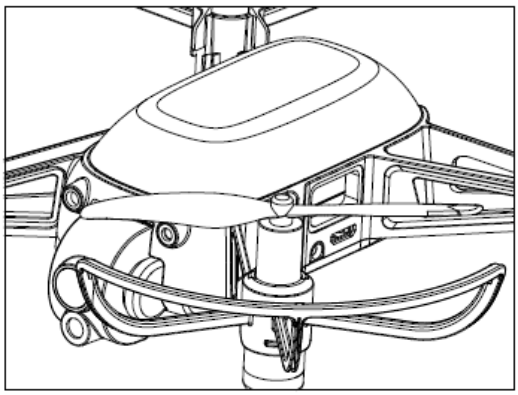
Run the Hula APP, and when the aircraft status indicator stays solid green and the APP indicates a successful connection to the aircraft, it means the aircraft is successfully connected.

By default, the aircraft is set to the direct connection mode. If you have switched modes, press the power button three times to switch back to the direct connection mode.

1.11 Mode switch



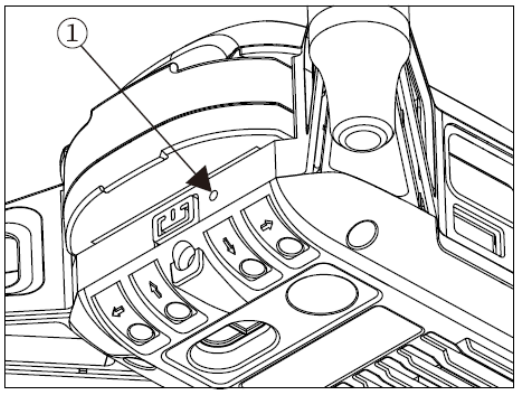
While the aircraft is powered on, you can press the power button three times quickly to switch between the direct connection mode and the networking mode.



When the aircraft is switched to the direct connection mode, the lamp cover will flash with purple light.
When the aircraft is switched to the networking mode, the lamp cover will flash with white light.

Direct connection mode: Connect the PC/mobile device directly to the aircraft WiFi and control the aircraft using the Hula APP.
Networking mode: Connect both the PC/mobile device and the aircraft to your own router WiFi. In networking mode, multiple aircrafts can network together to compete on the same stage.

1.12 Reset

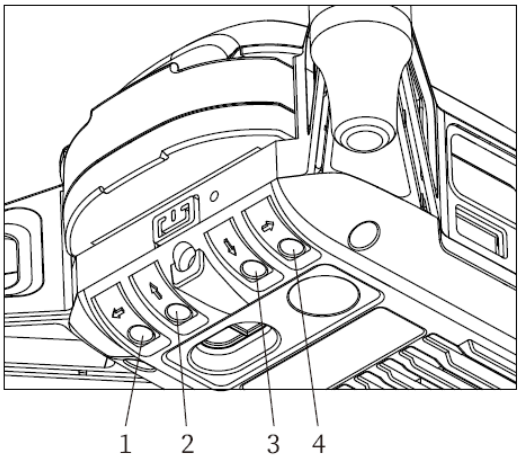


① Reset hole

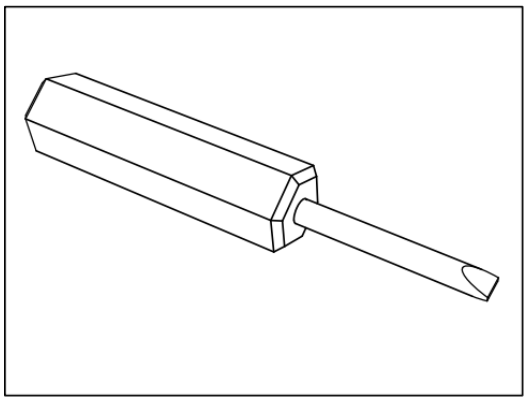
1. Press and hold the reset button inside the reset hole for 5 seconds, and the aircraft WiFi will be restored to factory settings. If the green light of the aircraft blinks fast after releasing, the aircraft is reset successfully.
2. Press and hold the reset button inside the reset hole for 10 seconds to revert the firmware version of the aircraft to the previous one. If the aircraft restarts automatically after releasing, the aircraft is reset successfully.

DO NOT perform the reset operation violently in case you may damage the internal components of the aircraft.

1.13 Four-way obstacle avoidance adjustment



①Left obstacle avoidance adjustment knob
②Rear obstacle avoidance adjustment knob
③Front side obstacle avoidance adjustment knob
④Right obstacle avoidance adjustment knob



1、Use a special screwdriver for obstacle avoidance adjustment knob, insert the adjustment screw hole, and turn the knob clockwise/counterclockwise to adjust the obstacle avoidance distance in each of the four directions.

2、The obstacle avoidance distance can be reduced by turning the obstacle avoidance adjustment knob clockwise. Turn the obstacle avoidance adjustment knob counterclockwise to increase the obstacle avoidance distance.

DO NOT twist the knob violently in case you may damage the internal components of the aircraft.

The infrared obstacle avoidance receiver is placed inside the lampshade, do not block the lampshade.

The obstacle avoidance distance has been adjusted before delivery, and the effective distance of obstacle avoidance is about 30-50cm. If it is necessary to adjust it again, please use the whiteboard to adjust it under ordinary lighting conditions in the room. The obstacle avoidance distance of the reflected object is different in material or color.

1.14 Instructions for handling glitches

1. If there is a glitch on the aircraft, please contact official channels for repair service.

In need of maintenance service, please contact your dealer, or contact HIGHGREAT customer service in the following ways:

Manufacturer: Shenzhen HighGreat Innovation Technology Development Co., Ltd.
Shenzhen HighGreat Innovation Technology Development Co., Ltd
Address: 2/F, Building 6, Yuanlingzi Industrial Zone, Hengping Road, Yuanshan Street, Longgang District, Shenzhen

E-mail: service@hg-fly.com
TEL: (+86)19924918168

The official website of HighGreat Innovation

WeChat Official Account

2. Install the Hula APP

The Hula APP is the companion control software for the Hula drone, designed to create an educational and entertaining drone programming platform. It explores innovative gameplay for drone battles, connects with virtual reality, sparks children's curiosity, and encourages them to explore from the heart. It offers a joyful learning experience in drone programming using artificial intelligence.

2.1 Android version

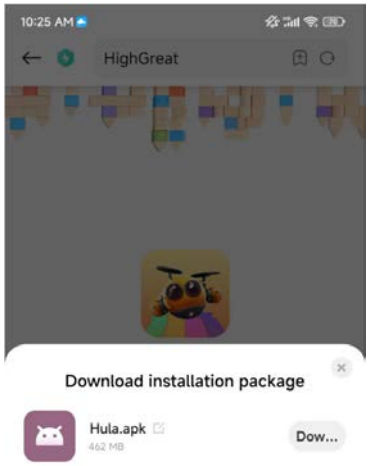
QR code for Android devices

Download the Android version of the Hula APP

Supports mobile phones and pads

Or you can use your Browser to open the download link at:

https://download.hg-fly.net/app/hula_app.html



After the download is finished, please follow the prompts to install it.

2.2 iOS version

QR code for iOS devices

Redirect to the App Store to download the iOS version of the Hula APP

Supports iPhone and iPad

Or you can use your browser to open the link at:

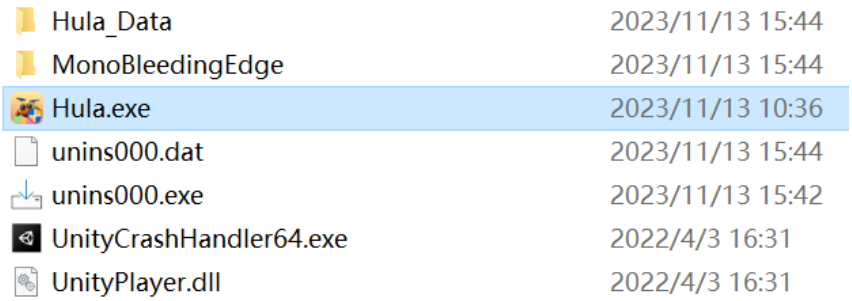
https://download.hg-fly.net/app/hula_app.html

Redirect to the Apple App Store to download and install it.

2.3 PC version



To download the PC version of the Hula exe, you can use your computer browser to open the download link at:
https://download.hg-fly.net/app/hula_pc.html



After the download is finished, find and double-click the installation package Hula.exe to install it. Turn off the firewall during installation. Install it in a directory named in English. Otherwise, it may fail to start. Supports Win10, not MacOs.

3. Register and log in



Domestic server: For domestic users, please select [English] to register/log in to the domestic server;

Foreign server: Foreign users please select [English] or [other language] to register/log in to a foreign server.

The account data of domestic and foreign servers does not communicate with each other. If you need cross-border access, you can change the language and re-register, and the login takes effect.

3.1 Register

Please provide a Chinese mobile phone number or email to register.

Step 1: Fill in a valid Chinese mobile phone number or email, receive the verification code, and enter it accurately.

Step 2: Move the slider to the rightmost, and check the box next to "I have read and accept the *User Agreement...*".

Step 3: Tap the Register and Login button to successfully register and login to the APP.

3.2 Log in

Verification Code Login: Enter your account number and get your verification code to log in.

WeChat Login: Authorize WeChat to use your personal information to quickly login.

After a successful login, your local login status will be maintained for 30 consecutive days.

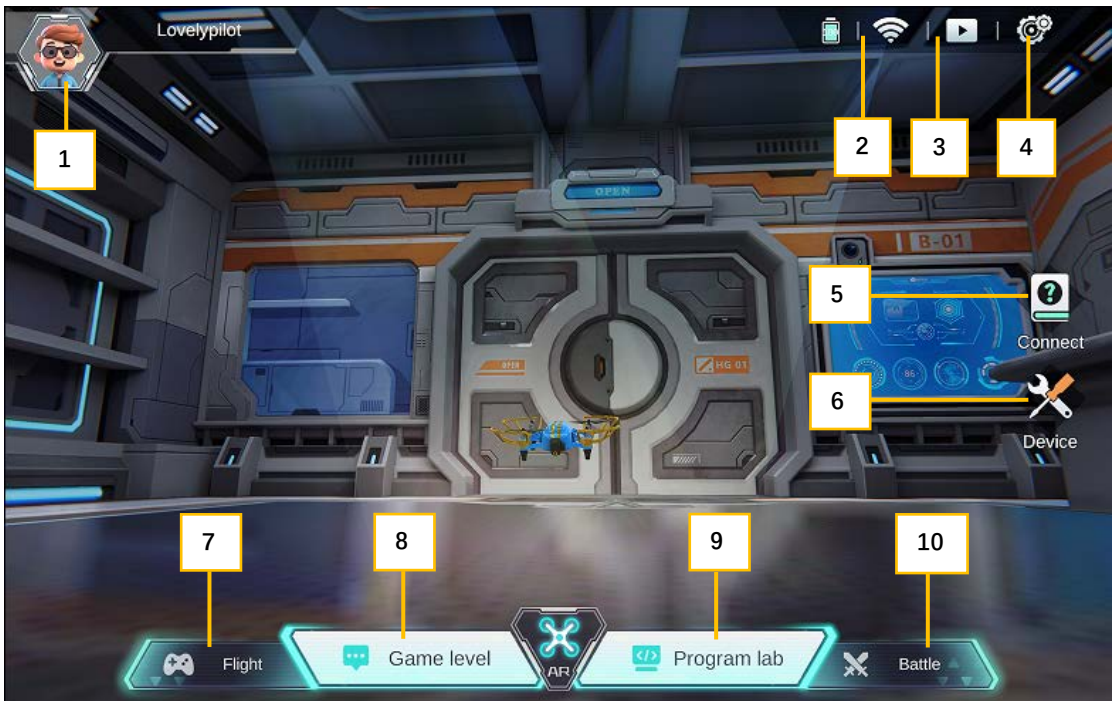
Password Login: Use your account number and password to log in. Password can be set through Forgot the Password or Profile-Modify the Password.

Visitor Login: You can quickly experience this APP with no need to register. Some features are restricted, and the visitor data cannot be saved.

3.3 Forget the password

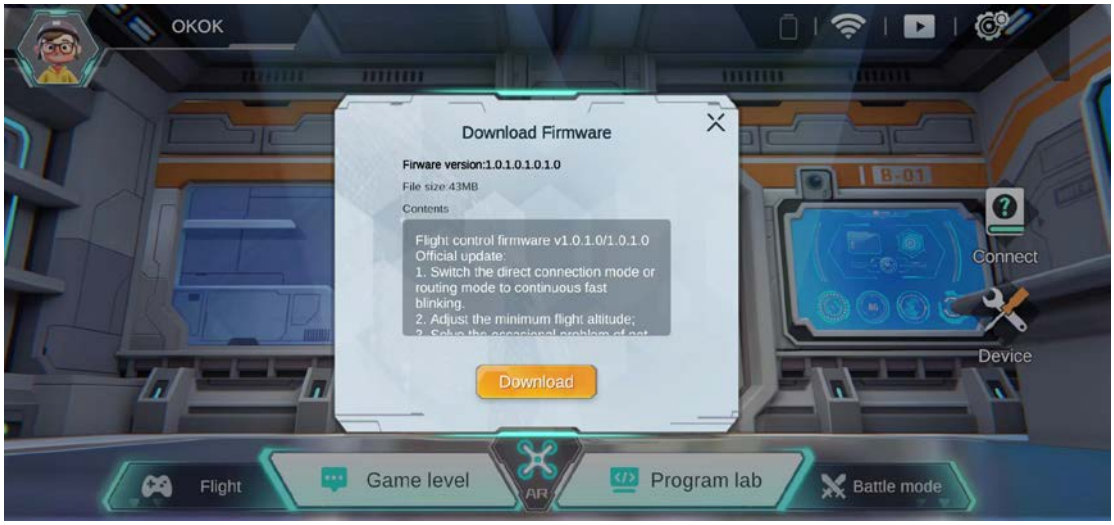
Enter your account number, get the verification code, then click the Next Step button, and you can set your password.

4. Home screen



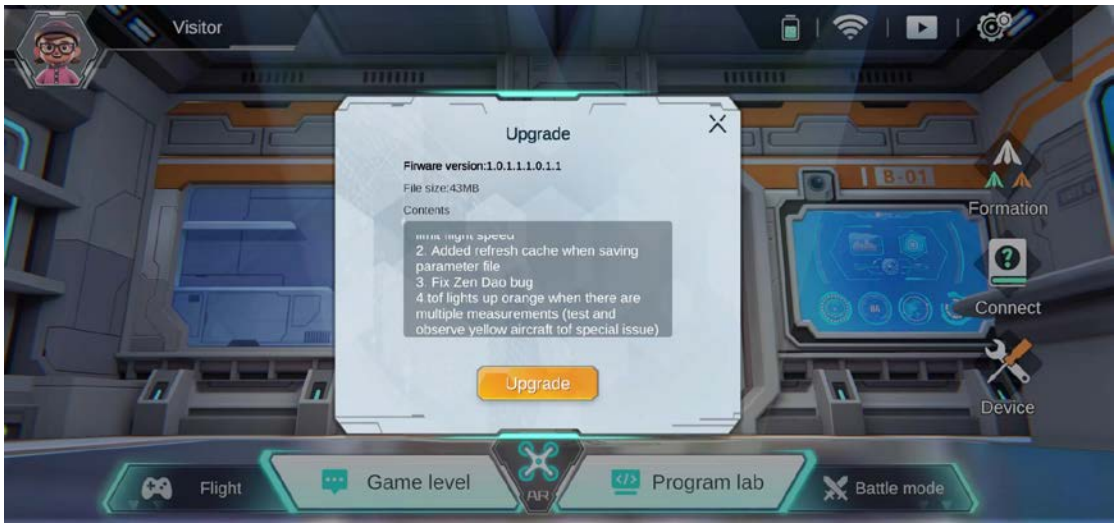
1. **Profile:** You can change your head avatar and nickname, modify your password, view messages, and seek technical support here.
2. **WiFi**: It displays the WiFi connection status. You can tap it to enter the WLAN settings interface on your mobile device.
3. **Album:** It manages all the photos and videos in your aircraft.
4. **Settings:** You can enter the Settings section here.
5. **Connection guide:** You can enter this section and connect to your aircraft WiFi and set the router WiFi according to this guide.
6. **Device:** You can enter the base and select your aircraft for battle. Next time you connect with the aircraft, the selected one will battle automatically.
7. **Flight:** You can enter the pilot interface to control the drone;
8. **Game level:** You can find entry-level breakthrough games of drone programming that are thoughtfully designed.
9. **Program lab:** You can control your aircraft through Scratch programming and explore your novel way to interact with artificial intelligence.
10. **Battle:** You can explore novel gameplay like the melee mode or the team battle mode.

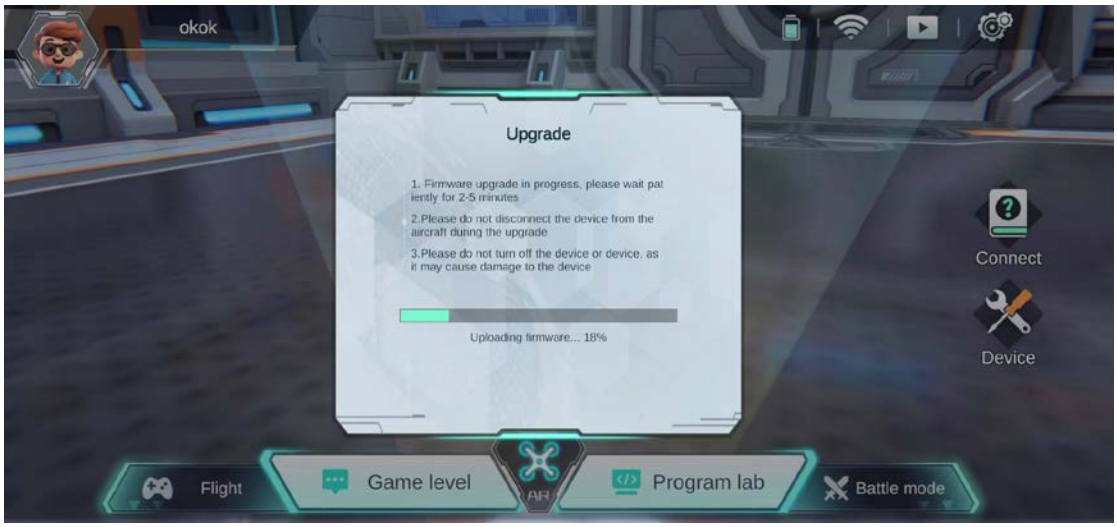
4.1 Flight control firmware upgrade

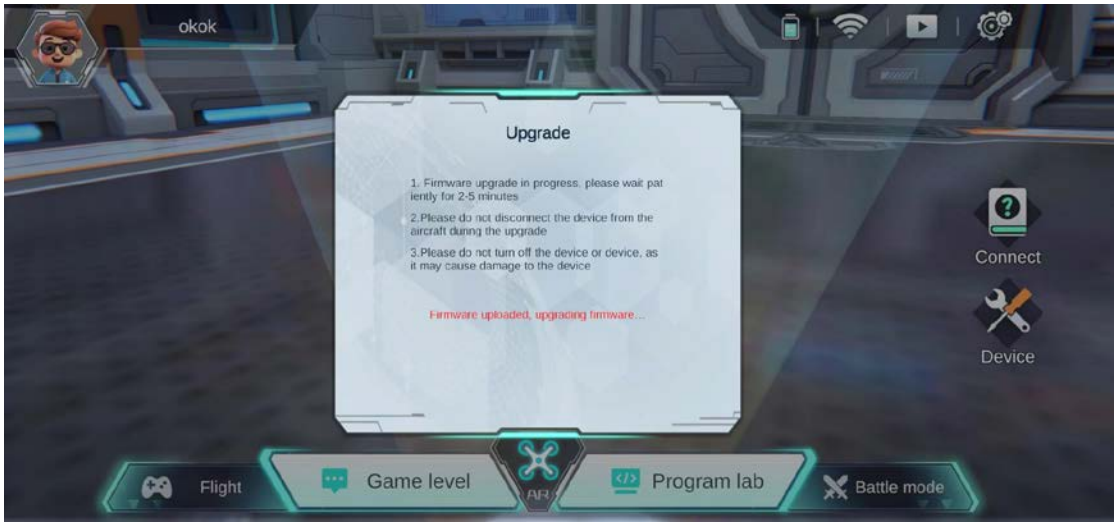


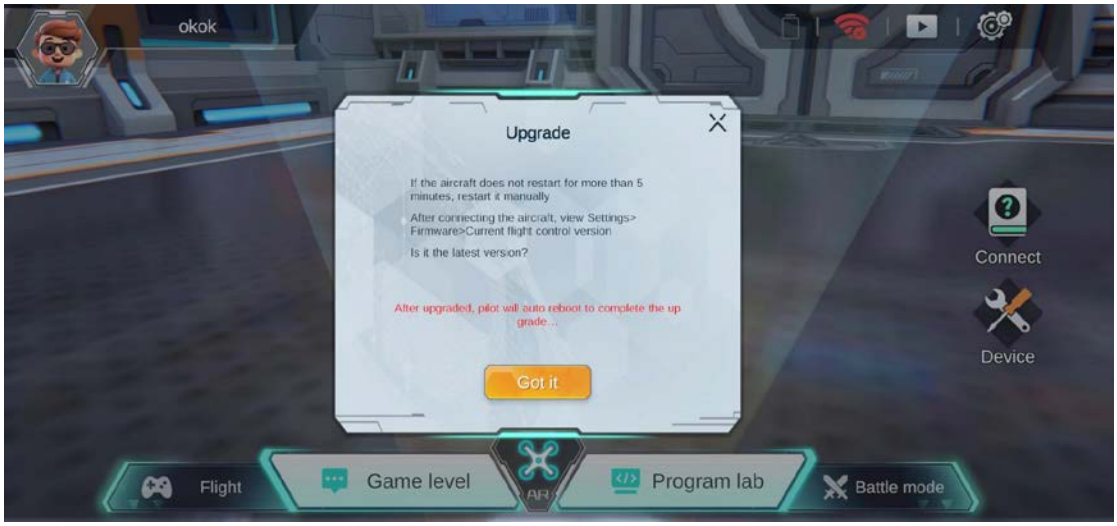
The server has been connected, and the APP will automatically detect the latest firmware version. Please click [Download].

After downloading, connect the aircraft and follow the prompts to upgrade the flight control firmware.




okok
Upgrade
1. Firmware upgrade in progress, please wait pat iently for 2-5 minutes
2.Please do not disconnect the device from the aircraft during the upgrade
3.Please do not turn off the device or device, as it may cause damage to the device
Uploading firmware... 18%
Connect
Device
Flight
Game level
AR
Program lab
Battle mode


okok
Upgrade
1. Firmware upgrade in progress, please wait pat iently for 2-5 minutes
2.Please do not disconnect the device from the aircraft during the upgrade
3.Please do not turn off the device or device, as it may cause damage to the device
Firmware uploaded, upgrading firmware...
Connect
Device
Flight
Game level
AR
Program lab
Battle mode


okok
Upgrade
If the aircraft does not restart for more than 5 minutes, restart it manually
After connecting the aircraft, view Settings> Firmware>Current flight control version
Is it the latest version?
After upgraded, pilot will auto reboot to complete the up grade...
Got it
Connect
Device
Flight
Game level
AR
Program lab
Battle mode

4.2 APP upgrade



When the server is connected and the APP is started, the latest APP version will be automatically detected.

For Android and Win10 systems, click [Upgrade Now], after downloading, please upgrade according to system guidelines;

If you click "Upgrade Now" on iOS, you will be redirected to the Apple Store. Please manually download the installation package to upgrade.

5. How to connect with the aircraft

You can connect with your aircraft through the **direct connection mode** or the **router mode**.

How can you determine the current mode of your aircraft: the direct connection mode or the router mode?

Method one: Observe the light signals



Light signal of the direct connection mode:

Purple light flashes when you power it on for self-check.



Light signal of the router mode:

White light flashes when you power it on for self-check.

Method two: Observe the WLAN list



Direct connection mode: After powering on the aircraft, you **can find** the aircraft WiFi name "Hula-XXXXXXXX" in the WLAN list on your mobile device.

Router mode: After powering on the aircraft, you **cannot find** the aircraft WiFi name "Hula-XXXXXXXX" in the WLAN list on your mobile device.

*The aircraft is set to the direct connection mode by default when it leaves the factory.

*Quickly press the power button of the aircraft 3 times in a row to switch between the direct connection mode and the router mode.

*If you have connected with the aircraft, the connection will be cut off when you switch the connection mode.

Hula Status signal:

Status	Color	light type	Explain
self-check	Red/green/blue	Flash	Power-on Self-test
Direct connection mode standby (No connected)	green	Breathing	The normal state of positioning using the visual positioning system or wireless positioning is not connected to the upper computer
Router mode standby (No connected)	white	Breathing	The normal state of positioning using the visual positioning system or wireless positioning is not connected to the upper computer

direct connection mode is connected	green	Steady on	WIFI、Remote control, direct connection mode is connected, green on
Router mode is connected	white	Steady on	Router mode， A router is connected with Hula, white on
Positioning system exception	yellow	Breathing	There is no visual positioning or wireless positioning
Data transfer, upgrade, pairing	blue	Flash	Dance Step, firmware package data transmission, upgrade, pairing completed, and then return to the original light
Infrared detected an obstacle	blue	double flash	When the aircraft is normally lit up, when the obstacle avoidance infrared sensor is turned on, it will detect the left and right directions before and after the detection. If there is an obstacle, the blue light will flash twice at intervals until it leaves the obstacle, and the normal light will be restored
Enter special flight mode	orange	Breathing	One key surround, bounce, line patrol, tracking, return to normal standby light after exit function
system reset	green	Flash	That is, restore factory Settings
low battery	red	Slow flash	Low Battery Clarm
The hardware is faulty or the self-check fails. Procedure	red	Steady on	Hardware failure, self-check until normal, and then restore to the original light
Specific object recognition	orange	Flash	1) The identification fails. Continue the identification. The orange flash is displayed. 2) Exit the recognition mode and return to the original light
The specific object was identified successfully	orange	Steady on	1) Recognition success, orange light on for 5 seconds, and then return to the original light, 2) The identification fails. Continue the identification. The orange flash is displayed. 3) Exit the recognition mode and return to the original light
The router mode is	Purple light	Flash	Flash the purple light when changing, when finished, then return to the green breath light

changed to the direct connection mode			
The direct connection mode is changed to the router mode	white light	Flash	Flash the white light when changing, when finished, then return to the white breath light; if the router fails to connect, The red lights flash for 3 seconds and then return to white breath light
IMU calibrates signals	Blue/orange/ green/red	Breathing/ Flash/ Steady on	Entering calibration IMU mode, the aircraft lights up with blue breath lights, After each successful calibration, the orange light flashes for 3 seconds, the blue breathing light is restored and the calibration continues All calibrated. Green flash five seconds later, The aircraft is placed in an abnormal attitude, showing a red light (attitude detection) Restore standby lights when the aircraft is in normal level position Calibration fails. The red light is on until recalibration
Formation mode controlled	Green	Breathing	In formation, HulaAPP is connected to Hula via routing(controlled), and the green breath light is on
Selected by the host computer	orange	Flash	In formation, light up when the aircraft is selected
Take-off check (cluster check)	Blue	Breathing	In formation, take off to check whether the dance steps, flight control and numbering are consistent
Check fail	red	Breathing	When formation, take off to check the steps, flight control, numbering inconsistency of the aircraft bright red breathing light
Permissible formation	green	Steady on	In formation, take off to check the dance steps, flight control, consistent numbering, sufficient power, healthy hardware, green on
In formation flight	Steps light	Steps light	Light the lights according to the dance steps
Set off light	off	off	Set lights off, but upgrade/calibration/ error states/obstacle avoidance/change Direct connection &Route mode lights except

5.1 Direct connection mode

Connect the APP to the aircraft WiFi directly, and this APP supports Single driver, Single device block programming, and Battle-Melee mode. In this mode, the aircraft will battle automatically.

5.1.1 Directly connect to the aircraft

Step 1: Press and hold the power button for 2 seconds to power on, and the aircraft will flash the self-check light of red, green, and blue.
Step 2: When it finishes the self-check, you continue to observe the light. Purple flash is the direct connection mode.
Step 3: White flash is the router mode. You need to press the power button 3 times in a row to switch to the direct connection mode.

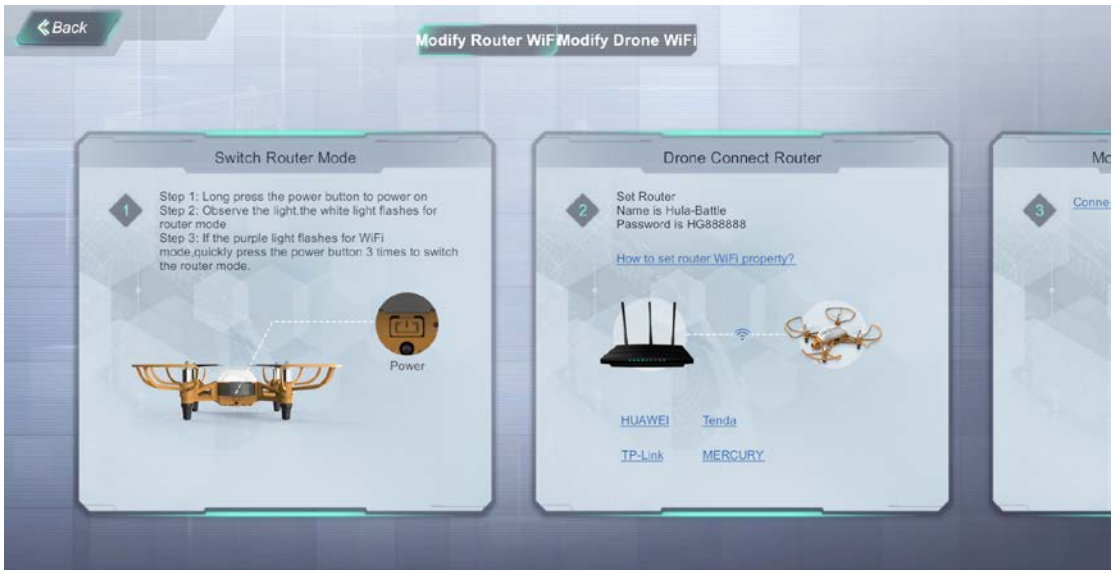
After the purple flashing, please connect to WiFi "Hula -XXXXXXXX", and the default password is 12345678. After the aircraft is successfully connected, the light will remain solid green.

If you cannot find the aircraft WiFi, please try switching the direct connection mode again, restarting the aircraft, or refreshing the WLAN list.

5.2 Router mode

This mode connects to aircraft through routing and supports multiple aircraft team battle and formation performance.
For networking, the router WiFi and the aircraft WiFi in routing mode must be the same. Please follow the connection guide to complete networking. You need to go to the equipment base to select your battle aircraft when you connect to the aircraft via router mode for the first time.

5.2.1 Switch the router mode



Step 1: Press and hold the power button for 2 seconds to power on, and the aircraft will flash the self-check light of red, green, and blue.

Step 2: When it finishes the self-check, you continue to observe the light. White flash is the router mode.

Step 3: Purple flash is the direct connection mode. You need to press the power button 3 times in a row to switch to the router mode.

5.2.2 Modify router WiFi

*The aircraft WiFi in routing mode is factory-set as follows:

Name: Hula-Battle

Password: HG888888

Please modify the router WiFi name and password to the corresponding one mentioned above.

After modification, the aircraft will connect to the router WiFi automatically. Then you can connect your mobile devices to the router WiFi mentioned above to complete networking.



Diagram of modifying the router WiFi

How to modify the router WiFi name and password?

To modify the router WiFi via the web page, please refer to:

https://jingyan.baidu.com/article/48a42057416c54a9242504e1.html

To modify the router WiFi via your mobile phone, please refer to:

https://resource.tp-link.com.cn/pc/docCenter/showDoc?id=1655112499847414

The above links are for reference only. Please refer to the User Manual of your router for instructions on how to modify the router's WiFi.

Please be aware that modifying the router WiFi may result in your smart home devices being disconnected from the WiFi, so you will need to reconfigure them.

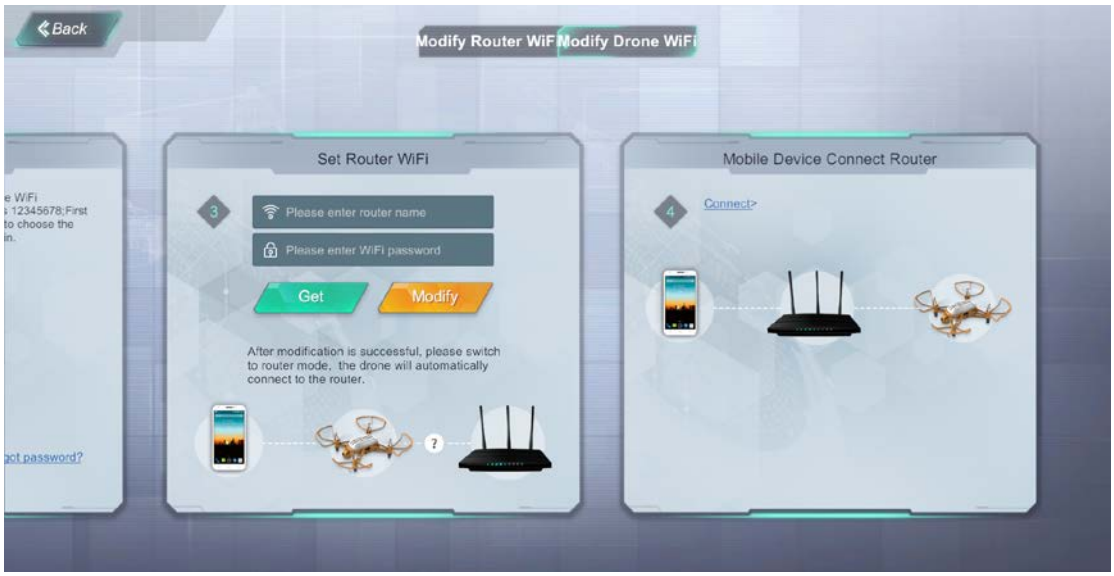
5.2.3 Modify aircraft WiFi

Step 1: Switch the aircraft to the direct connection mode.

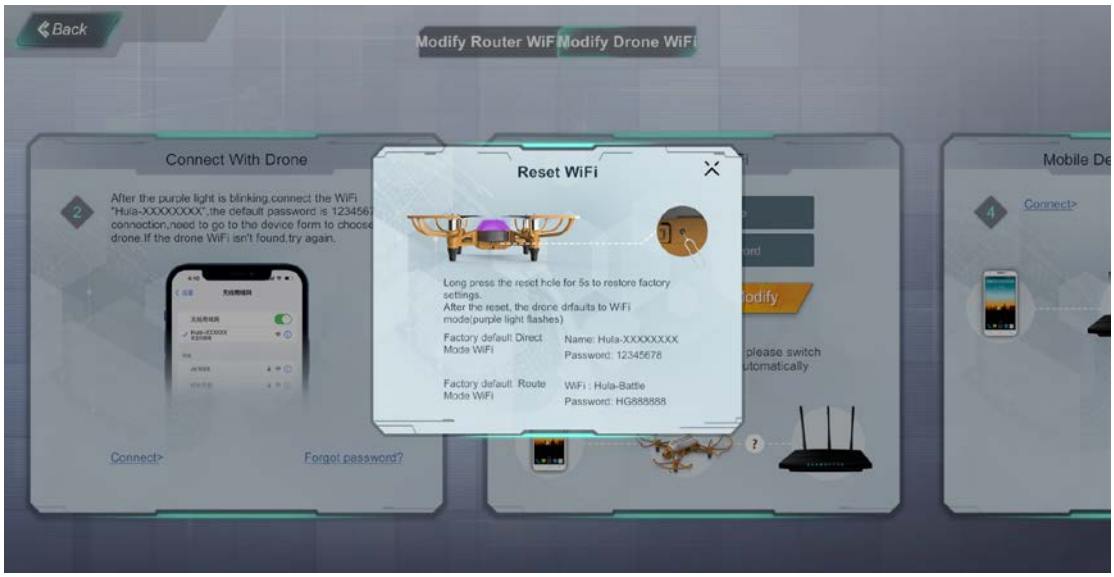
Step 2: Directly connect to the aircraft. Send the WiFi name and password of the router you want to connect to the aircraft.

Step 3: After the router WiFi name and password are successfully sent to the aircraft, please switch the aircraft to the router mode. The aircraft will automatically identify the router and connect to it.

Step 4: If there are multiple aircraft that need to connect to the same router, each aircraft needs to repeat the step 1 to step 3 mentioned above.



5.2.4 How to reset aircraft WiFi



Use the card pin and press the reset button next to the power button for 5 seconds to restore the factory WiFi（2.4G）. press 10 seconds to restore the flight control to the previous version.

5.3 Equipment base

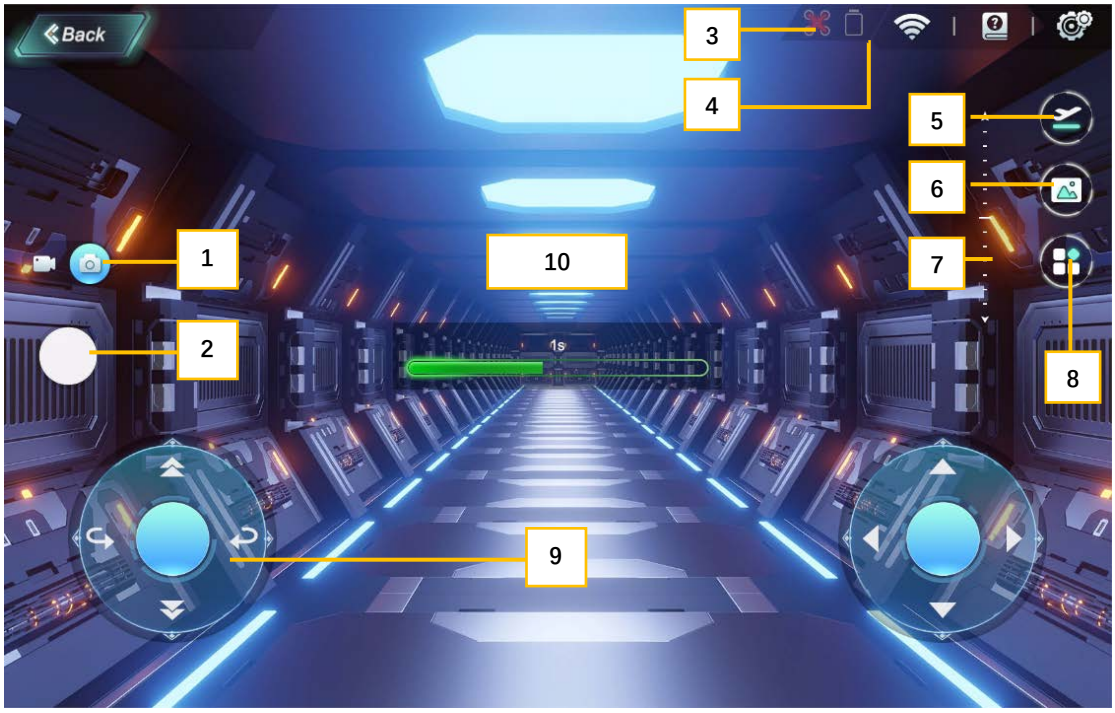


Aircraft list: The aircraft connected to the APP directly or to the router will be displayed in this list. Each aircraft corresponds to a unique SN code.

Battle: Select the aircraft you want to control, and the selected one will flash with a yellow light. After verifying the SN code is correct, tap [Battle]. This aircraft will be set as your default battle aircraft. In the direct connection mode, the selected aircraft will battle automatically, while in the router mode, the selected will only battle automatically after the router reconnects to the aircraft.

Cancel: In the router mode, you can tap the Cancel button and then change your battle aircraft, while in the direct connection mode, you can select another aircraft directly without canceling.

6. Single driver



1. Photo-taking/video-recording switch;
2. Photo-taking/video-recording button;
3. Aircraft health status: Red indicates the abnormal status or not connected, while green indicates the normal status. Tap to know more details about the aircraft status.
4. Remaining battery level;
5. Takeoff/Landing button;
6. Album;
7. Gimbal control slider;
8. Advanced functions;
9. Virtual joystick;
10. Image transmission;

6.1 Takeoff/Landing

Press and hold for 1 seconds to take off. After taking off, it will switch to the Landing button automatically.

Press ， than second confirmation to land.

The default takeoff height of the aircraft is 1 meter, and the flight height for optimal control experience is between 1 to 4 meters. The maximum flight height is 10 meters, and the minimum flight height is 0.5 meters.

Please stay at least 2 meters away from the aircraft during takeoff and landing. To avoid getting cut by the propellers, you are forbidden to fly the aircraft when it is held in hand.
The temperature of the heatsink and the motor at the bottom of the aircraft is high, so please avoid touching them to prevent burns.
Flight altitude below 0.5m there is a risk of crash, flying speed should not exceed 0.2 m/s.

6.2 Virtual joystick



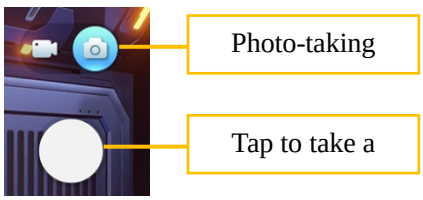

You can use your finger to tap or drag the virtual joystick to control the aircraft.

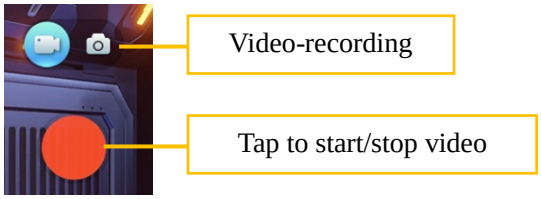


The default control way of the joystick is the American hand. If you want to switch to the Japanese hand, please go to Settings - Control - Control Way to make the adjustment.

For safety during flight, please avoid exiting the control interface, disconnecting from WiFi, or closing the APP while the aircraft is in flight.
If the aircraft loses connection, it will automatically make a controlled landing in its current position after a 10-second delay.

6.3 Photo-taking/video-recording





Taking into account the flight endurance and the video downloading speed, video recording is limited to a maximum of 8 minutes per session.

6.4 Advanced functions

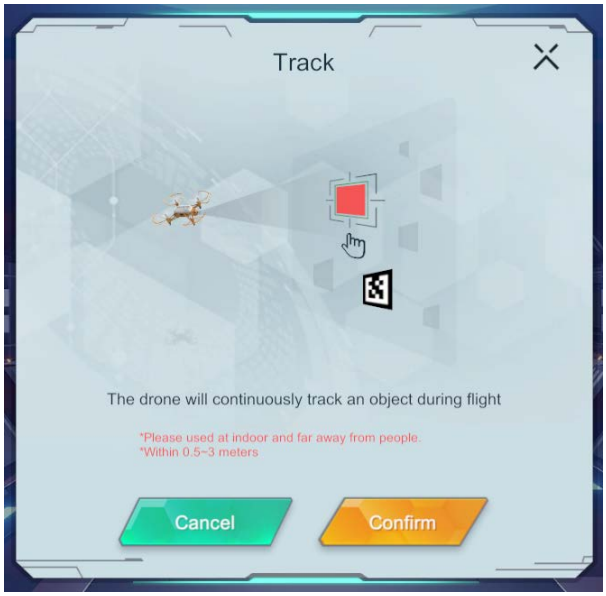
Advanced functions usage requirements:

For flight safety, please use advanced functions indoors in an open area and away from pedestrians.

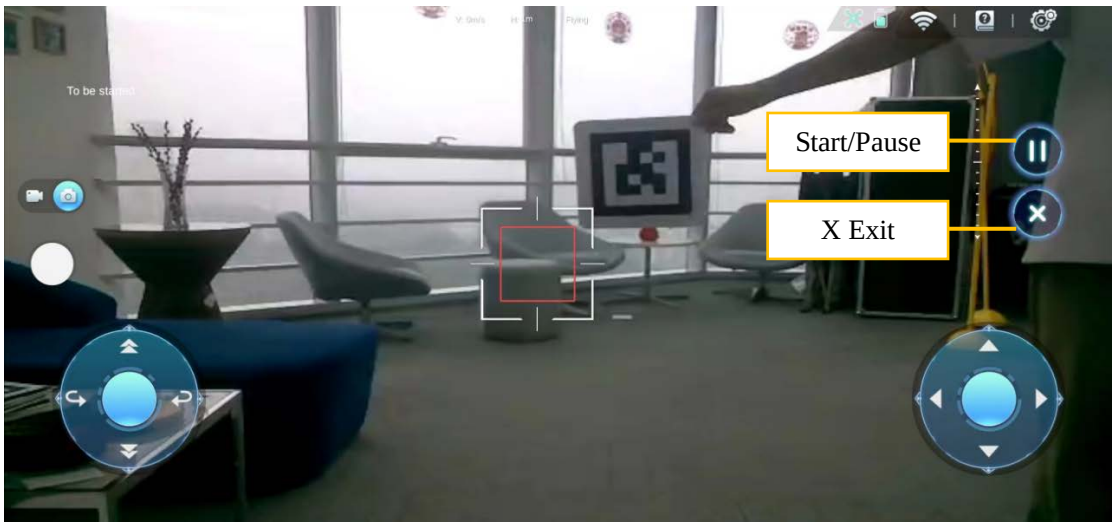
The battery level of the aircraft must be at least 30% or higher, and the flight height should be at least 0.5 meter.

When the aircraft is flying, you can press and hold the joystick for 1 second to pause the flight. Please do not exit the control interface, disconnect from WiFi, or close the APP.

6.4.1 Follow



The aircraft can choose to follow the QR code or target the color.



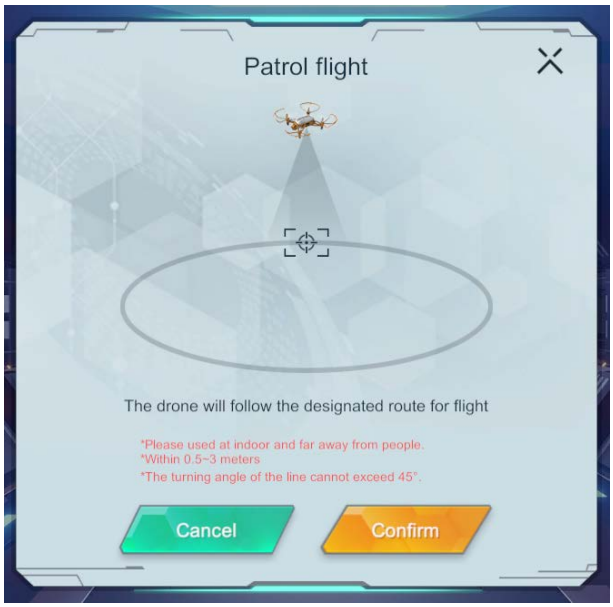
QR code following requirements: With the image transmission function enabled, it requires good indoor lighting, no backlight, and the QR code intact, unobstructed, and clearly visible. Also, the flight height of the aircraft should be between [1~2.2] meters, and the main camera should be adjusted to a horizontal position. The QR code should maintain a horizontal distance of [1~2.2] meters from the front camera for optimal recognition results.

Starting following: The aircraft uses the **front camera** to recognize the QR code and record the distance between the QR code and the aircraft's main camera. The aircraft will dynamically maintain its distance from the QR code to track it.

Following: You can hold the QR code and slowly move it back and forth and left to right to ensure stable QR code recognition. Please note that the QR code should remain within the aircraft's screen area. The optimal following effect is achieved with a QR code movement speed of less than or equal to 0.3 m/s horizontally and less than or equal to 0.5 m/s vertically. When following, if the QR code is not recognized, the aircraft will hover to locate the QR code. You can also use the joystick to adjust the position. The aircraft will resume following once the QR code is successfully recognized again.

Stop following: Tap the [Pause] button to stop following, and tap the [X] button to exit the following mode.

6.4.2 Line patrol flight



The aircraft will follow the designated route for flight.

Line patrol requirements: The optimal flight height of the aircraft is [1~2.2] meters. The best visual effect will be achieved by light lines on a dark background, and for other colors, the greater the contrast between the line and the background, the better the effect. The line width is recommended to be more than 5cm,The angle between lines at turns should be greater than 135°, meaning the turning angle of the aircraft nose at each turn cannot exceed 45°. Flying over indoor surfaces with good lighting and non-reflective textures produces the best flight results.



Start line patrol: Hover the aircraft above the line, tap the  button to start line patrol, and tap again to stop.

Line patrol timeout: If the line on the ground is not recognized within 30 seconds, line patrol will stop.

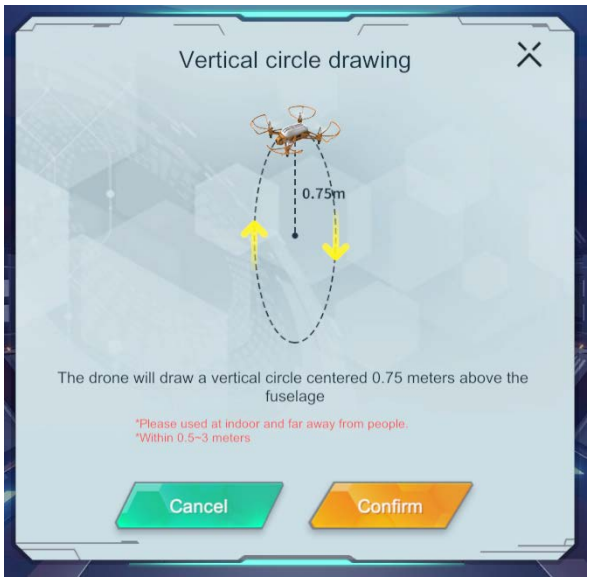
Exit line patrol: Tap the [X] button to exit the line patrol mode.

During the line patrol, you can use the joystick to control the aircraft.

During the line patrol, the image transmission defaults to 360 px. If you record videos or take photos during following or line patrol, the resolution of videos and photos will also be 360 px.

Avoid using this function in environments with poor lighting, where the line color is close to the background, or where the ground reflects light strongly, as it may affect the optical flow positioning function of the aircraft.

6.4.3 Draw a vertical circle

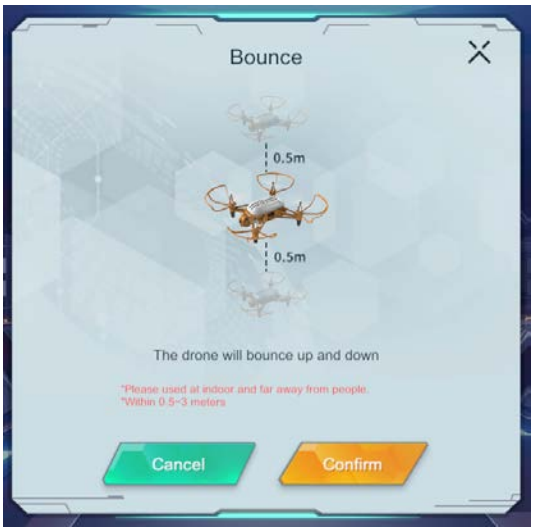


The aircraft will draw a vertical circle centered 0.75 meters above the fuselage.

Upon completing a vertical circle, the aircraft will hover in place.

When the aircraft is drawing a vertical circle, you can pause its movement by pressing and holding any direction on the joystick for 1 second.

6.4.4 One click bounce

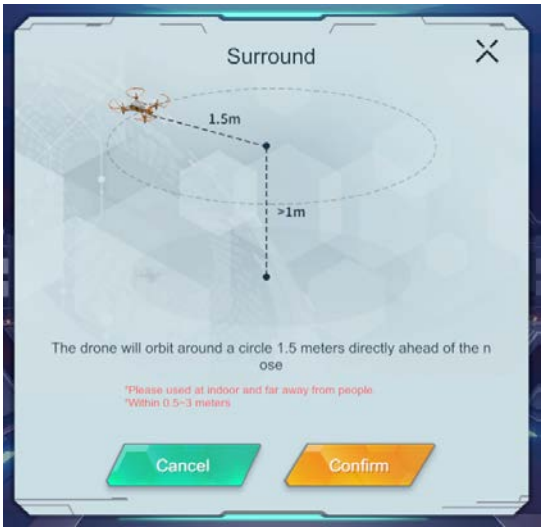


The aircraft will bounce up and down.

Upon completing one bounce cycle, the aircraft will hover in place.

When the aircraft is bouncing up and down, you can pause its movement by pressing and holding any direction on the joystick for 1 second.

6.4.5 One click fly around



The aircraft will orbit around a circle 1.5 meters directly ahead of the nose.

Upon completing one orbit, the aircraft will hover in place.

When the aircraft is orbiting around a circle, you can pause its movement by pressing and holding any direction on the joystick for 1 second.

6.4.6 Circle

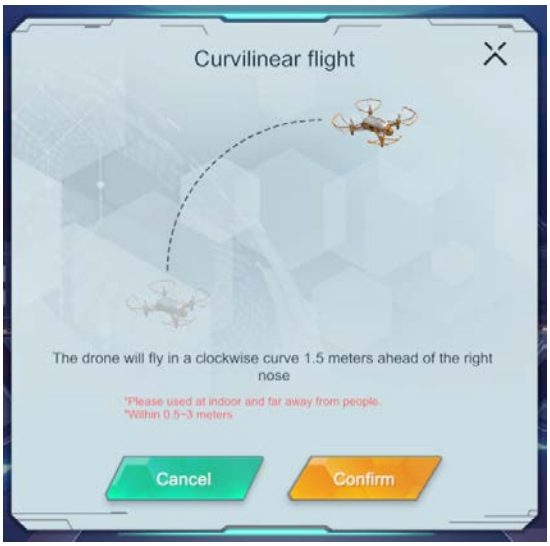


The aircraft will rotate clockwise for 1 turn in its current position.

Upon completing one turn, the aircraft will hover in place.

When the aircraft is rotating, you can pause its movement by pressing and holding any direction on the joystick for 1 second.

6.4.7 Curve



The aircraft will fly in a clockwise curve 1.5 meters ahead the right side of the nose.

Upon reaching a point 1.5 meters ahead the right side of its nose, the aircraft will hover in place.

When the aircraft is arc flying, you can pause its movement by pressing and holding any direction on the joystick for 1 second.

6.4.8 One click flip



The aircraft will roll in the direction of the finger slide.

6.5 Aircraft status

Details of aircraft status: Green indicates the normal status, and red indicates the abnormal status.



Flight control version: It displays the flight control version number of the current connected aircraft.

Battery level: It displays the current aircraft battery level. When the battery level is low, it will display red.

Signal: It displays the aircraft WiFi connection status.

IMU: It detects and measures the sensor status when the sensor is in acceleration and rotational movement.

Laser reception: It displays the status of the laser receiver in the aircraft.

Optical flow: It displays the status of the light-sensitive camera at the bottom of the aircraft. It will display red in poor lighting conditions or when obstructed.

Cumulative flight duration: It records the cumulative flight duration since the aircraft leaves the factory. It is recommended to replace the battery after the flight duration exceeds 30 hours.

Temperature: It displays the motherboard temperature of the aircraft.

Storage space: It displays the current remaining storage space on the aircraft.

ToF: It displays the status of the ToF sensor. When the flight height of the aircraft is in abnormal status, it will display red.

Barometer: It displays the state of the barometer.

It is normal for the ToF and the optical flow to display red when the aircraft is not in flight.

It is because when the aircraft is not in flight, the ground will obstruct the light, causing the ToF height to fall below its minimum flight height. You can take off with confidence.

6.6 Cautions

1. DO NOT direct the laser at human eyes to avoid causing eye injuries;
2. Please avoid touching the heatsink and the motor of the aircraft to prevent burns.
3. When the aircraft is flying, please maintain a safety distance of at least 2 meters from it.
4. Keep the optical flow camera and ToF module clean and free from obstructions.

5. It is intended for indoor flight under good lighting conditions only.
6. DO NOT fly in environments with wind speeds greater than or equal to 3 m/s.
7. Do NOT fly on water, mirror, dark environment, solid color ground without obvious texture and other environmental conditions;

7. Programming level

Programming games, let children get started quickly, skilled use of basic building blocks programming.

7.1 Levels interface



1. **Scratch block menu**
2. **Block programming area:** You can drag blocks to this area to program.
3. **Levels number:** 1-1 represents Chapter 1, Level 1, and 1-2 represents Chapter 1, Level 2.
4. **Reminder:** Levels instructions. Play multiple: adjustable from 1x to 3x. Play again: It resets the block, role, and stage.
5. **Task and target:** You can pass the level perfectly if you have collected enough props.
6. **Levels stage:** You can stretch to zoom the stage and rotate the perspective.
7. **Run/Stop button**
8. **Zoom in blocks, Zoom out blocks, and Center blocks**
9. **Step back and Step forward**
10. **Used blocks (optional blocks) and Clear blocks**

When passing the programming level, please ensure an Internet connection in case you lose your level progress data.

8. Programming lab

8.1 Single device programming

By Scratch programming to control the drone, you can first observe the rationality and safety of your flight paths through SIM run and then operate the flight in reality.

In the single-device programming mode, the aircraft will default to turning off its lights and the obstacle avoidance function.

For safety, programmed control aircraft, please use in indoor environment, DO NOT use in outdoor environment.

8.1.1 Programming Interface



1. **Block menu**
2. **Zoom in blocks, Zoom out blocks, and Center blocks**
3. **Step back and Step forward**
4. **Program:** Programming building blocks make up the program.
5. **Perspective switch:** You can switch to the roaming perspective, tracking perspective (default), FPV perspective, or overhead perspective.
6. **Runing:** It controls the aircraft to fly in reality.
7. **Reset:** It resets the blocks and the SIM stage.
8. **SIM run:** It controls the SIM aircraft to SIM run on the stage.

9. **WiFi status:** You can tap it to enter the WLAN settings interface on the mobile device.
10. **More:** Daytime/night mode switch and the function of hiding gridding lines.
11. **Image transmission switch:** Turn on/off the image transmission and the parameter page of the aircraft. The recognition of blocks requires the image transmission to be turned on first.
12. **Save:** It saves your block programming works.
13. **Display/hide the blocks editing area:** After you hide this area, you can drag the stage perspective.
14. **Programming area:** This entire page is a programming area. You can drag blocks into this area for programming.

8.1.2 Basic operation of programming

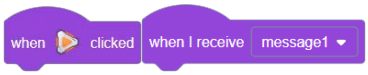
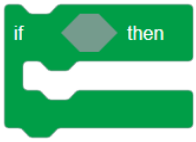


1. You can press and drag the blocks into the empty area, and the blocks will automatically snap together after you lift your fingers.
2. You can drag the blocks to the red area on the left-handed side to delete them.
3. When you complete your programming, tap the icon  to [Save]. It will be saved as My work.
4. Tap the icon  to [Run SIM]. Tap the icon  to [Pause SIM].
5. Tap the icon  to [Reset]. It will reset the programming and the stage to its initial state.
6. The aircraft has been connected. Tap and hold the icon  to [Run], and the aircraft will take off and execute the block instructions.
7. Tap the icon  to [Stop]. The aircraft will execute the landing instruction and reset the

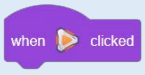
programming to its initial state.

After taking off, the aircraft will establish a relative coordinate with the takeoff point as the origin, represented as XYZ (0, 0, 0). As the minimum flight height of the aircraft is 50 cm, When the aircraft takes off from the desktop (the origin is on the desktop) and flies below the desktop, the relative altitude will be negative. At this time, TOF absolute altitude positioning is adopted. When the absolute altitude is less than 0.5 meters, there is a risk of crash.

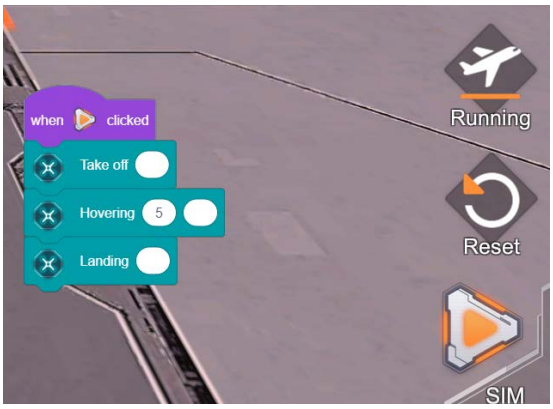
8.2 Programming block instructions

Block type	Type explanation	Block example
Execution block	Controlling Hula to execute the corresponding instructions.	
Events block	Event-triggered block. When the triggering condition is met, it will immediately interrupt the main thread and start running the program within the corresponding events block.	
Parameter block	Blocks that are used for setting the parameters of the variates or returning different types of data they got from variate values and lists.	
Conditional block	These blocks execute the corresponding instructions based on whether the conditions are met.	

8.2.1 Events

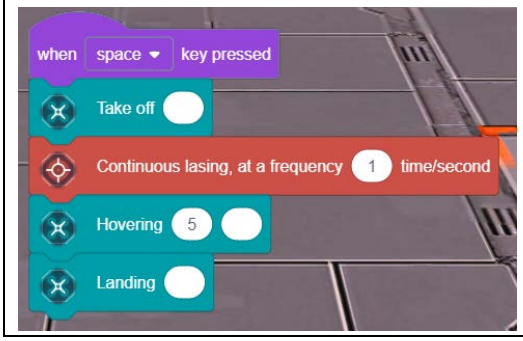


1. Name: **when**  **clicked**
2. Function: It is an events block. When  is clicked, it triggers the program to run. When  is clicked, it triggers the Hula to run.
3. Type: events block
4. Example: when  clicked, take off



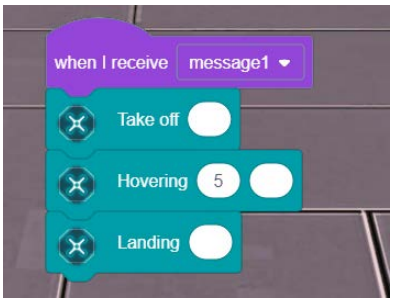


1. Name: **when [space] key pressed**
2. Function: It is an events block. When the [space] key is pressed, it triggers the program to run.
3. Type: events block
4. Example: when [space] key is pressed, lasing



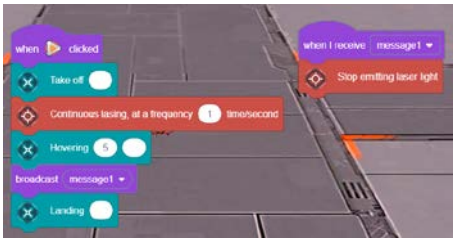


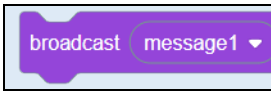
1. Name: **when I receive [message 1]**
2. Function: It is an events block. When receiving the specified message, it will execute the following script below this instruction. It is used in conjunction with the "**broadcast [message 1]**" block.
3. Type: events block
4. Example: When I receive [message 1], take off



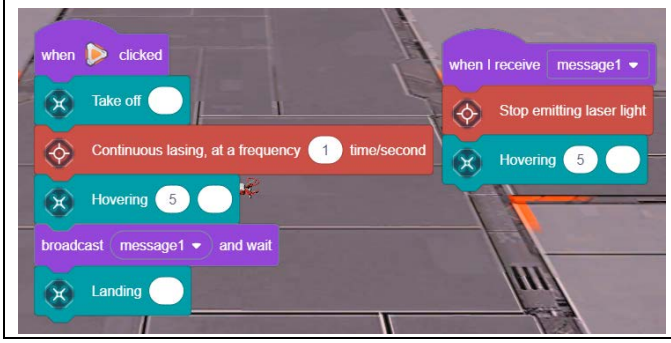


1. Name: **broadcast [message 1]**
2. Function: It broadcasts a specified message. This instruction has a drop-down parameter for specifying the message name, including defaults like "new message", "message 1", and any other newly created messages.
3. Type: execution block
4. Example: broadcast [message 1]

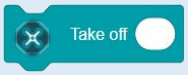




1. Name: **broadcast [message 1] and wait**
2. Function: This instruction differs from the previous broadcast message instruction in that, after broadcasting a message, it doesn't immediately execute the following program. Instead, it waits until all scripts that have received this message finish executing and then continues the following execution.
3. Type: execution block
4. Example: broadcast [message 1] and wait



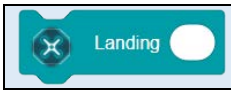
8.2.2 Flight operation



1. Name: **Take off**
2. Function: It is an instruction for taking off. This instruction, defaulting to a flight height of 100 cm, must be set when programming a flight, or it will fail to take off. The "white circle" at the end of the block can be combined with information blocks in an arc shape. It's typically used in conjunction with light blocks or lasing blocks, enabling actions like lighting up or lasing while taking off. Other blocks can be used similarly.
3. Type: execution block

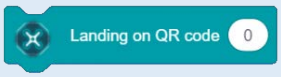
4. Example: Take off with the yellow light on





1. Name: **Landing**
2. Function: It is an instruction for landing. This instruction must be set when programming a flight, or it will fail to land after taking off. This block is usually put at the end of a flight program.
3. Type: execution block
4. Example: Landing





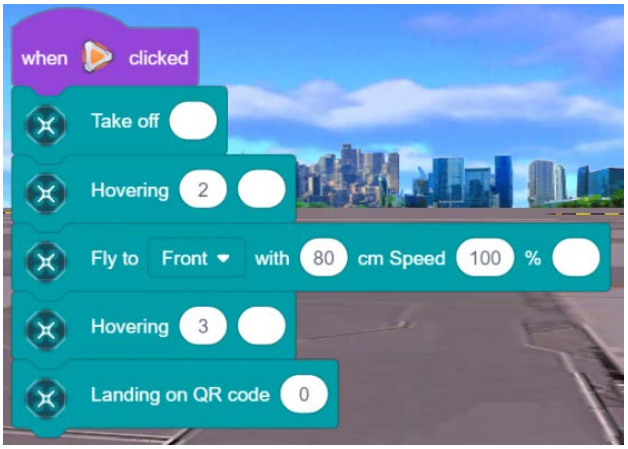
1.Name: **Landing on Qr code[0]**

2.Function: Align QR code landing instructions, Bottom (optical flow) camera, use with 0-9 QR code label card, landing instructions generally need to be placed at the end of the program.

3.Type: execution block, No simulation

4.Example: Qr code Landing

Place a zero QR code on the ground 1 meter directly in front of the drone,



Qr code landing requirements:

After indoor take-off, the lighting condition is good, the flight height is 1 meter, and a QR code of 0~9 is placed within 30*30cm below the aircraft, and the aircraft will automatically land on the QR code; If the flight altitude is 2 meters, the QR code should be placed within 60*60cm below the aircraft; The flight height is 3 meters, the placement range is within ±100*100cm, and the maximum height of the effective identification of the two-dimensional code cannot exceed 3.5 meters.



1. Name: **Hovering [n] second**
2. Function: It is an instruction for hovering. This instruction is used for hovering, and the hovering time can be set between [1~600] seconds, with a default of 1 second. Generally, after the aircraft performs flight actions, it is recommended to add 1 to 2 seconds of hovering, so that the fuselage can hover smoothly before performing the next action.
3. Type: execution block
4. Example: Hovering



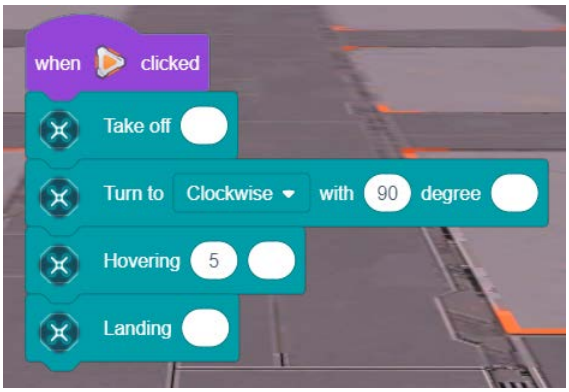


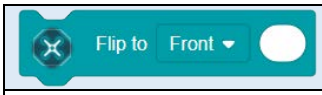
1. Name: **Fly to [Direction x] with [Distance l] cm Speed [v]% [Light]**
2. Function: It is an instruction for flying. This instruction defaults to Direction x = "top". The aircraft will fly in the direction of x = [front/back/left/right/top/bottom] for a set distance l at a set speed. The distance l can be set between [0-500] cm, with a default of 50 cm. The speed can be set between [1-100%], with a default speed of 100% (that is, 1 m/s).
3. Type: execution block
4. Example: Fly to [top] with [50] cm Speed [100]% [Light]



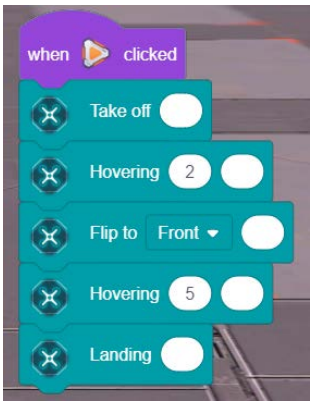


1. Name: **Turn to [Clockwise] with [n] degree**
2. Function: It is an instruction for turning. The aircraft will rotate at a certain angle [n] [Clockwise/Anticlockwise]. The angle [n] can be set between [0-360] degrees, with a default of 90 degrees.
3. Type: execution block
4. Example: Turn to [Clockwise] with [90] degree

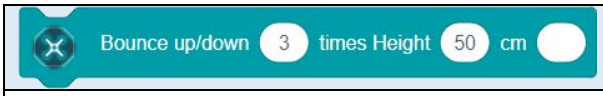




1. Name: **Flip to [Front]**
2. Function: It is an instruction for flipping. The aircraft will flip to [Front/Back/Left/Right].
3. Type: execution block
4. Example: Flip to [Front]



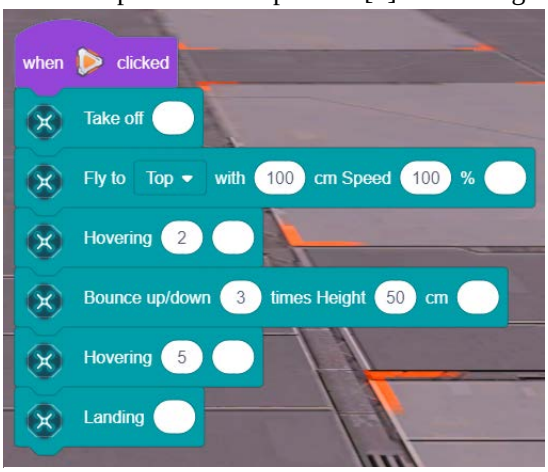
Note: Please used at indoor and far away from people， more than 30% battery， within1-3 meters.



1. Name: **Bounce up/down [n] times Height [h] cm**
2. Function: It is an instruction for bouncing. The aircraft will bounce up and down in flight at a speed of 1.5 m/s. The times of bounces, n, can be adjusted between [0~60] times, with a default of 3 times. The height of the bounce, h, can be adjusted between [50~150] cm, with a default of 50 cm. Please note that the

height of the downward bounce is represented as a negative value. This feature requires the aircraft to be at a flight height of 1 meter or higher.

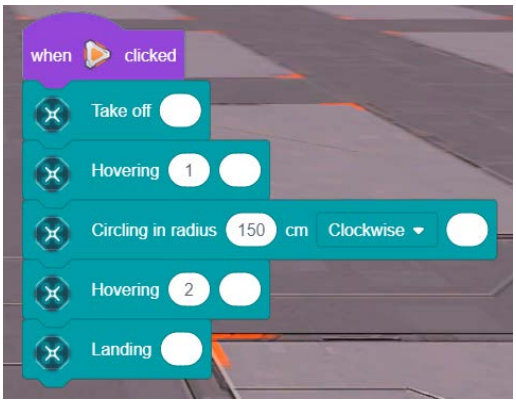
3. Type: execution block
4. Example: Bounce up/down [3] times Height [50] cm



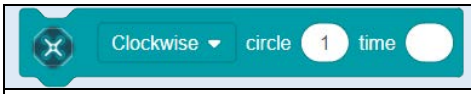
Note: Please used at indoor and far away from people， more than 30% battery， within1-3 meters.



1. Name: **Circling in radius [r] cm [Clockwise]**
2. Function: It is an instruction for circling in radius r. The aircraft will circle with the center at a distance of [r] cm in front of the aircraft nose. During circling [Clockwise/Anticlockwise], the aircraft nose will consistently face the center. The value of r can be set between [15~300 cm], with a default of r = 150 cm.
3. Type: execution block
4. Example: Circling clockwise with the center at a distance of 150 cm in front of the aircraft nose.

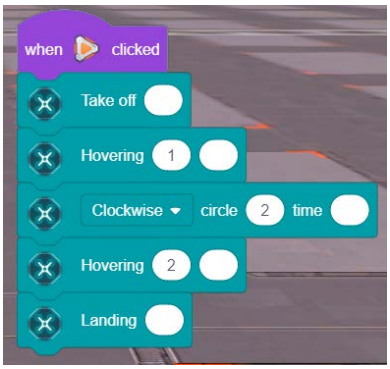


Note: Please used at indoor and far away from people，more than 30% battery，within1-3 meters.

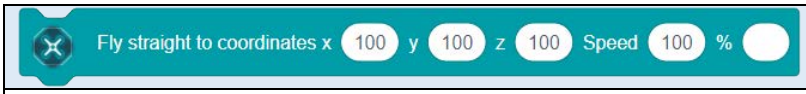


1. Name: **[Clockwise] circle [n] time**
2. Function: It is an instruction for circling for n times. The aircraft will circle [Clockwise/Anticlockwise]. n is a positive integer and can be set between [1~10] times, with a default of n = 1 time.
3. Type: execution block

4. Example: [Clockwise] circle [2] times



Note: Please used at indoor and far away from people， more than 30% battery， within1-3 meters.



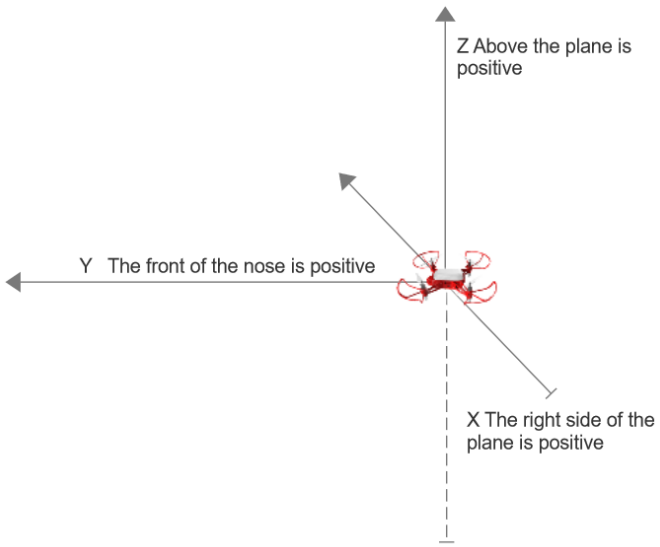
1. Name: **Fly straight to coordinates [x] [y] [z] Speed [v]%**
2. Function: It is an instruction for straight flying. The aircraft flies straight to a specific coordinate point at a certain speed. The speed is adjustable, and the light blocks can be embedded in this instruction. If not embedded, the current aircraft light will be displayed by default.
3. Type: execution block
4. Example:

What are relative and absolute coordinates?

Relative coordinate: is relative to the user coordinate system, is relative to the previous point.

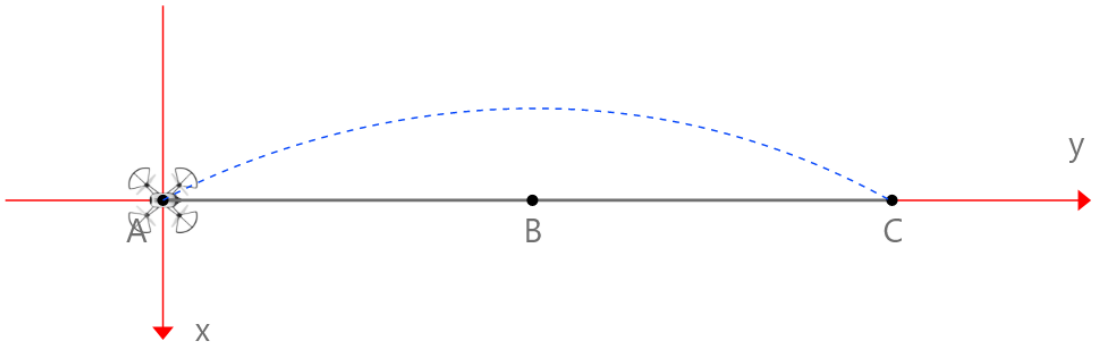
Absolute coordinates: Absolute coordinates are relative to the origin of the world coordinate system. The origin of the world coordinate system here refers to the origin of the two-dimensional code carpet or the origin of the base station.

4.1 In relative coordinates, take "Fly straight to coordinates X(100) Y(100) Z(100)" for example:

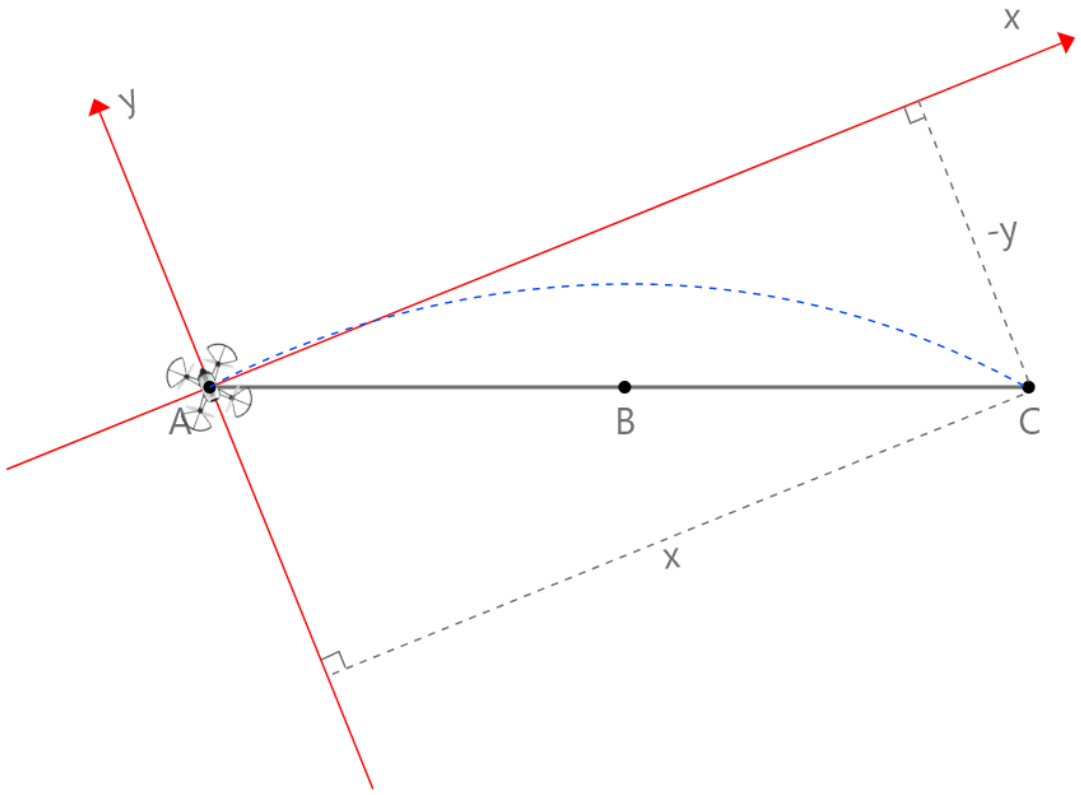


In the absence of the QR-code absolute localization or the base station absolute localization, the coordinate

points of the aircraft during flight are represented in relative coordinates (that is, incremental coordinates). The direction that the aircraft nose faces is considered the positive Y-axis, while the direction that the aircraft tail faces is the negative Y-axis direction. The left side of the aircraft nose is the negative X-axis direction, and the right side of the nose is the positive X-axis direction. The positive Z-axis is directly above the aircraft, and the negative Z-axis is directly under the aircraft.



The plane is at point A, nose direction is facing point C, fly straight to X (0) Y (AC) Z (0), can fly to point C.



The plane is at point A, the nose direction and AC have an Angle, fly straight to X (x) Y (-y) Z (0), can fly to point C.

When the aircraft needs to execute the straight flight block multiple times, it is important to note that after each straight flight, the coordinate point of the current aircraft position becomes the new origin coordinates. The input values for X, Y can be between [-1000~1500]cm, and Z can be between [-300~400] cm. A speed of 100% corresponds to 1 m/s, and the speed value can be set between 1-100%.



In relative coordinates, take "Fly straight to coordinates X(100) Y(100) Z(100)" for example:
Suppose the current hovering position of the aircraft is the origin point A: X(0) Y(0) Z(0). After executing the instruction, the aircraft reaches the origin point B, where the coordinates relative to point A are X = 100, Y = 100, Z = 100. If, on this basis, you execute "Fly straight to coordinates X(100) Y(100) Z(100)" again, the new origin point C relative to point A will have coordinates X = 200, Y = 200, Z = 200. The coordinates of point C relative to point B will be X = 100, Y = 100, Z = 100. To return to the location of the previous instruction (the origin point B), you would need to execute "Fly straight to coordinates X(-100) Y(-100) Z(-100)".

4.2 In absolute coordinates, Fly straight to coordinates X(700) Y(700) Z(800)

In the presence of the QR-code carpet positioning or the base station localization, the coordinate points of the aircraft are in absolute coordinates. In absolute coordinates, flying straight to a specific coordinate point is constrained by the following ranges: X and Y can be between [-1000~1500]cm, and Z can be between [-300~400]cm. If the input values exceed the maximum or minimum limits, they will be forcefully adjusted to fit within the specified range in order to generate valid flight maneuvers.

Please pay special attention to this: Out of safety considerations, the maximum flight distance for each action is limited to 500 cm. Under this circumstance, if the calculated distance between the input coordinates and the previous coordinates exceeds 500 cm, the input values will be forcibly adjusted to the maximum flight distance of 500 cm so as to generate valid flight maneuvers. If needed, it is recommended to complete the desired distance in two step.

For example: Activate QR code carpet positioning， For safety, the actual flight altitude should not exceed 300cm.

If the aircraft needs to fly from the coordinate point X(0) Y(0) Z(0) straight to the coordinate point X(700) Y(700) Z(200), please split it into two blocks:

Fly straight to coordinates X(500) Y(500) Z(100)

Fly straight to coordinates X(700) Y(700) Z(200)

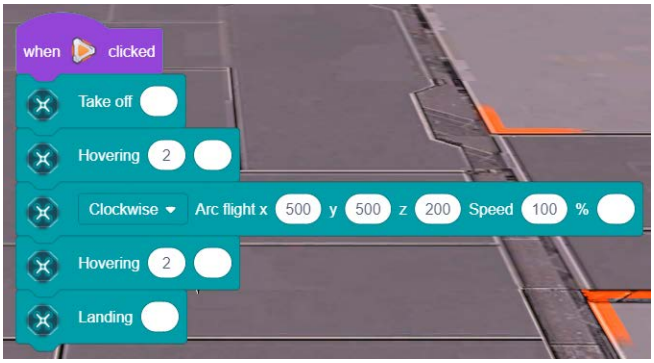


After the aircraft lands, the QR code is automatically turned off for carpet positioning.



1. Name: **[Clockwise] Arc flight [x] [y] [z] Speed [v]%**
2. Function: It is an instruction for arc flying. It will map an ellipse with the current position to the input coordinate point as the diameter, and the aircraft will fly along its arc. The input values, X and Y, can range between [-1000~1500]cm, and Z, can range between [-300~400]cm. If the input values exceed the maximum or minimum limits, they will be forcefully adjusted to fit within the specified range in order to generate valid flight maneuvers.
3. Type: execution block
4. Example:

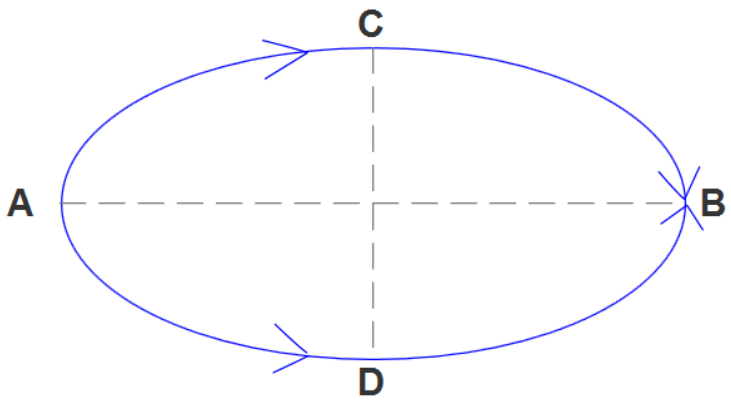
4.1 In relative coordinates, [Clockwise] Arc flight x [500] y[500] z[200] Speed [100]%



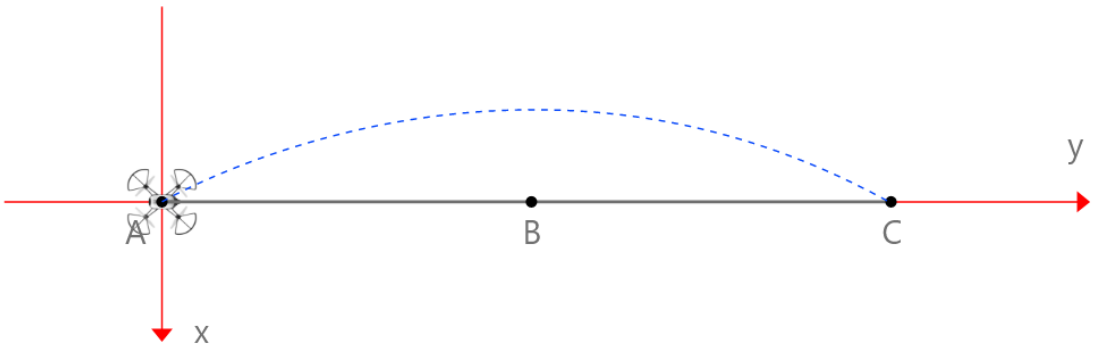
Please pay special attention to this: Out of safety considerations, the maximum flight distance for each action is limited to 500 cm. Under this circumstance, if the calculated distance between the input coordinates and the previous coordinates exceeds 500 cm, the input values will be forcibly adjusted to the maximum flight distance of 500 cm so as to generate valid flight maneuvers. The aircraft will fly in a clockwise/anticlockwise arc to the set coordinate point at a certain speed. The speed is adjustable, and the coordinates during flight are represented in relative coordinates (that is, incremental coordinates). When you need to execute the arc flight block multiple times, please note that each time you complete the flight, the current destination coordinates

will become the new origin coordinates. A speed of 100% corresponds to 1 m/s, and the speed value can be set between 1-100%. For safety, the actual flight altitude should not exceed 300cm.

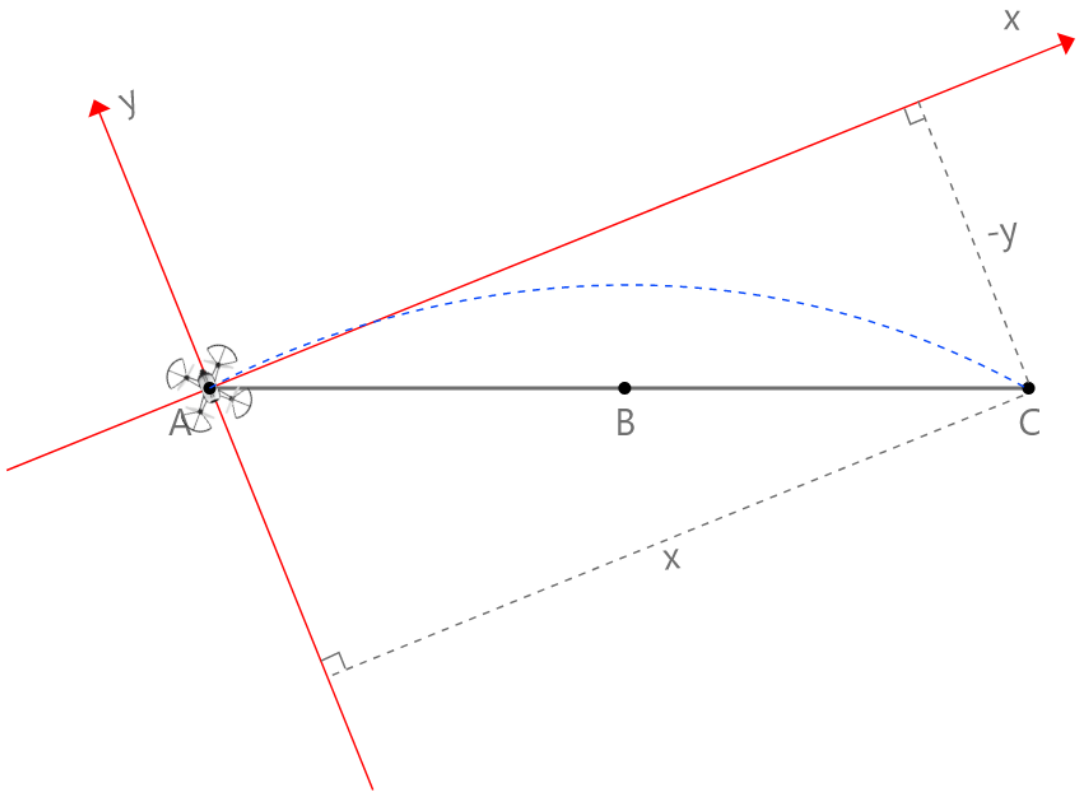
Elliptical curve: Define four elliptical vertices A, B, C, and D, with the major axis AB and the minor axis CD. Let AB = 2CD. The arc trajectory from point A to point B (clockwise/anticlockwise) is called the elliptical curve.



AB =2CD

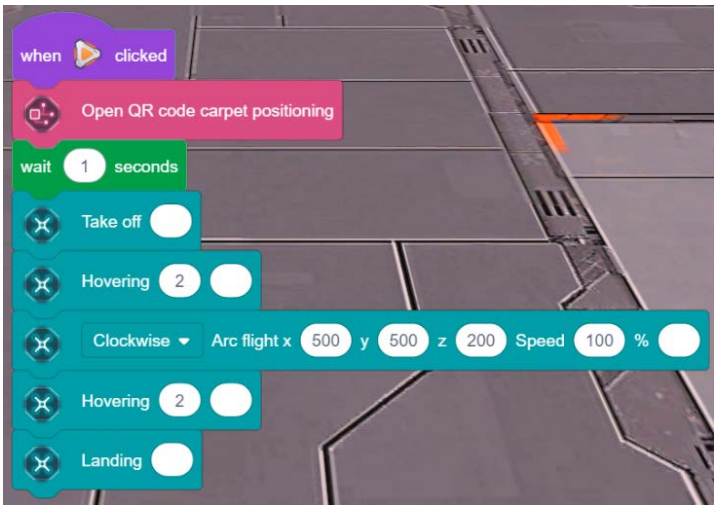


The aircraft is at point A, nose direction is facing point C, [clockwise] arc to X (0) Y (AC) Z (0), can fly to point C.

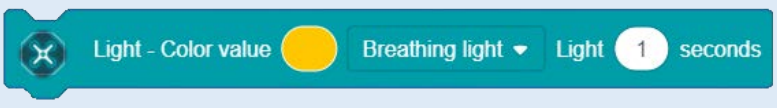


The aircraft is at point A, nose direction and AC Angle, [clockwise] arc to X (x) Y (-y) Z (0), can fly to point C.

4.2 In absolute coordinates, [Clockwise] Arc flight x[500]y [500] z[100] Speed [100]%



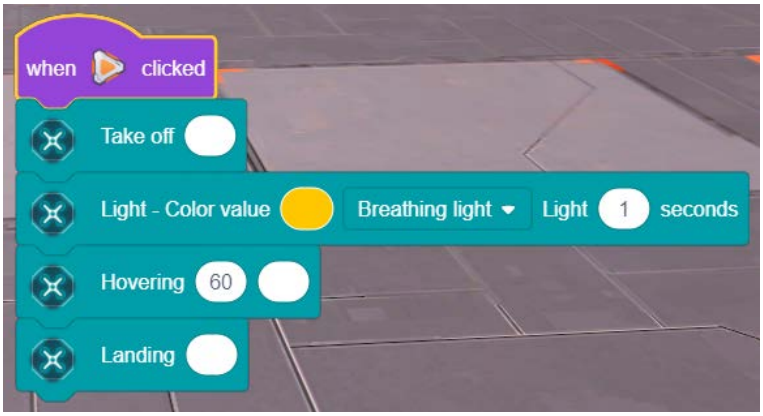
After the aircraft lands, the QR code is automatically turned off for carpet positioning.



1. Name: **Light-Color value [Breathing light] Light [n] seconds**
2. Function: It is an instruction for light. It can control the light color and the lighting time of the aircraft.

The lighting time, n, can be set between [0-50] seconds. The color values are entered using the RGB color picker. When the light of the aircraft is turned on, if there is a new light instruction, it will immediately override the existing light effect, light color, and lighting time.

3. Type: execution block
4. Example: Light-Color value [Breathing light] Light [60] seconds



Available light effects: breathing light, crush out, steady on, colorful lantern, and flash.

Colorful lantern: This light effect will automatically transition through seven colors in sequence, at a frequency of changing its color every 0.3 seconds. It is switched through a gradient method.

Flash: The light quickly flashes approximately every 0.2 seconds.

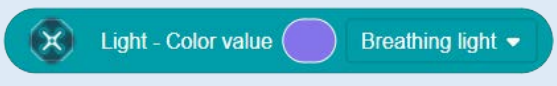
Breathing light: Its brightness gradually decreases, and its frequency matches the human breathing rate.

Steady on: The light maintains the current color and brightness until the specified time ends or a new light instruction is received.

Crush out: Turn off the light.

Note: The 3 light blocks running consecutively will be overwritten by the last command.



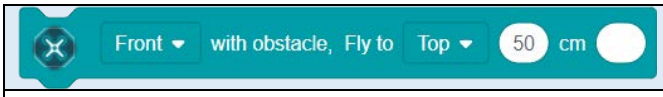


1. Name: **Embedded light [green][breathing light]**
2. Function: It is an instruction for embedding light. It is a parameter block of light RGB value and can be

embedded in the white box at the end of a flight action block. The lighting time matches the duration of the flight action. Available light effect instructions include breathing light, colorful lantern, crush out, steady on, and flash.

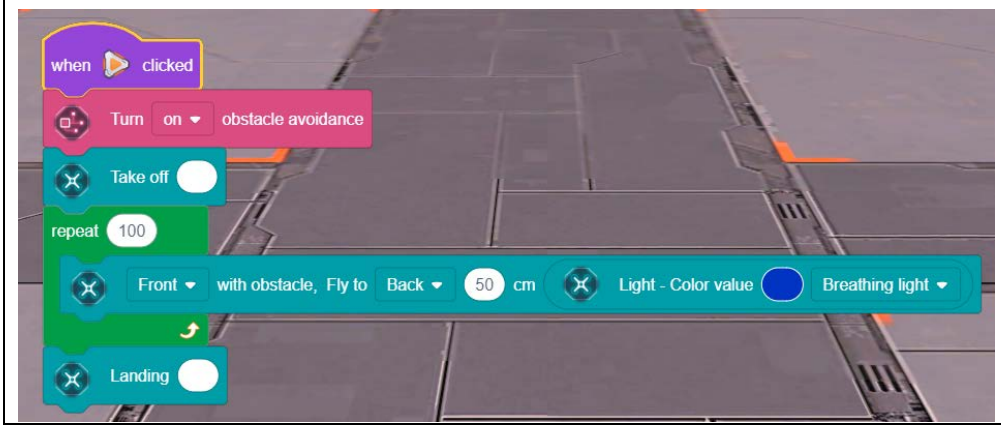
3. Type: parameter block
4. Example: Take off with the green light on, and Fly to top with the yellow light on



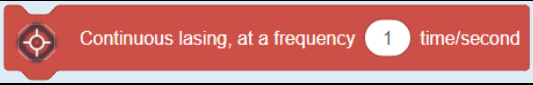


1. Name: **[Front] with obstacle, Fly to [Top] [n] cm**
2. Function: It is an instruction for obstacle avoidance. This instruction is used in conjunction with the [Turn on] obstacle avoidance block. When the obstacle detection is turned on, the aircraft assesses obstacles in [Front/Back/Left/Right] directions. If obstacles are detected, the aircraft will move [Front/Backward/Left/Right/Top/Bottom] by [n] cm, where [n] can be adjusted within the range of [0~500] cm. The default value is [50] cm. The lights might interfere with the aircraft in obstacle avoidance, so this instruction defaults to not turning on the lights. Yet this instruction can be embedded with the light block.
3. Type: execution block, No simulation
4. Example: Active obstacle avoidance.

[Front] with obstacle, Fly to [Back] [50] cm，After the drone takes off, slowly approach the front of the drone with a white board.

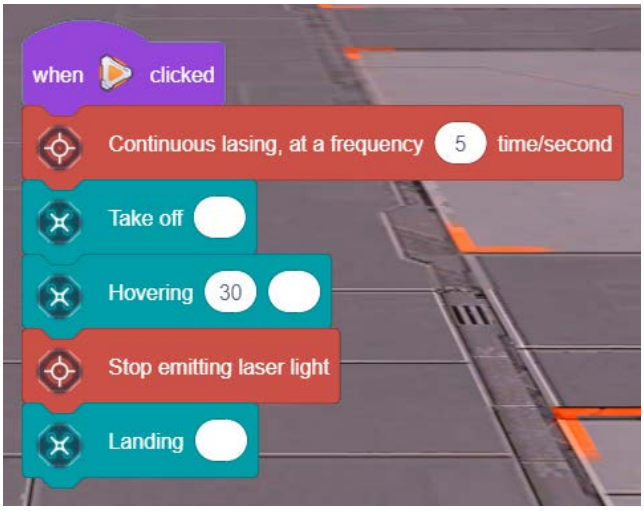


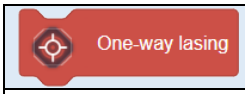
8.2.3 Laser



1. Name: **Continuous lasing, at a frequency [1] time/second**

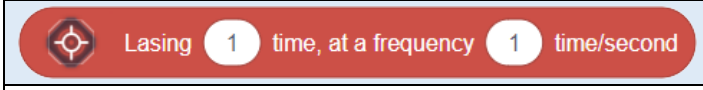
2. Function: It is an instruction for continuous actions. It controls the laser to emit laser light continuously at the frequency of f times/second, with a default frequency of f = 1. The frequency f can be adjusted between [1~5] times/second.
3. Type: execution block
4. Example: Continuous lasing, at a frequency [5] time/second





1. Name: **One-way lasing**
2. Function: It is an instruction for one-way actions. It controls the laser to emit laser light 1 time.
3. Type: execution block
4. Example: One-way lasing



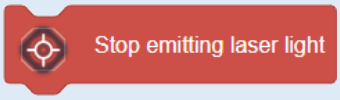


1. Name: **Lasing [n] time, at a frequency [1] time/second**
2. Function: It is an embedded instruction for continuous actions. It can be embedded into the white box of a

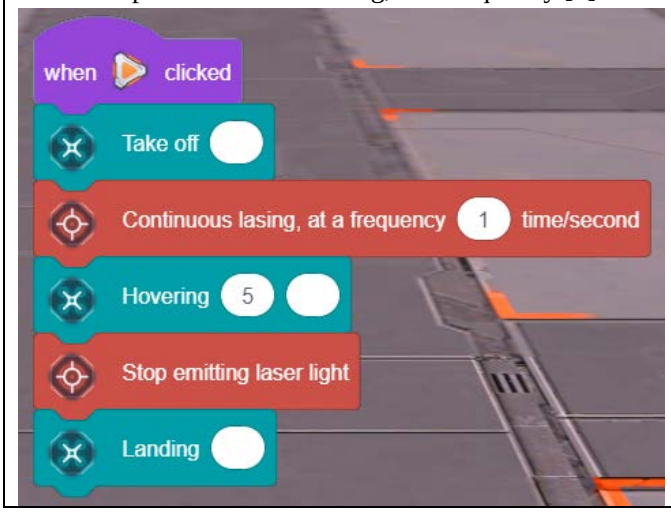
flight action block, and it will be executed simultaneously with the action block. When the action block finishes execution, the embedded lasing block also stops. This instruction controls the laser to emit laser light [n] times at a frequency of [f] time/second, with the value of [n] ranging from [1~20], and the frequency [f] ranging from [1~5] time/second.

3. Type: parameter block
4. Example: Fly to [left], Fly to [right], and Continuous lasing, at a frequency [5] time/second

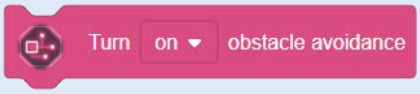




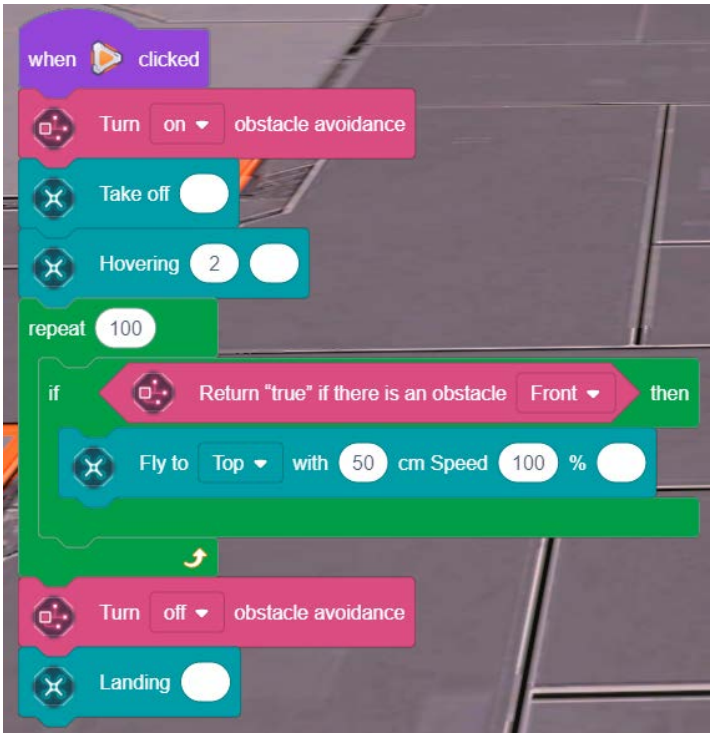
1. Name: **Stop emitting laser light**
2. Function: It is an instruction for stopping emitting laser light. During the process of continuous lasing, this instruction can stop emitting laser light.
3. Type: execution block
4. Example: Continuous lasing, at a frequency [1] time/second, and Stop emitting laser light

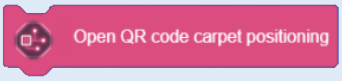


8.2.4 Sensor



1. Name: [Turn on] obstacle avoidance
2. Function: It is an instruction for turning on/off obstacle avoidance. When it is turned on, the aircraft can detect the obstacles from front, back, left, and right.
3. Type: execution block, No simulation
4. Example: [Turn on] obstacle avoidance, detecting [Front] with obstacle, Fly to [Top], then [Turn off] obstacle avoidance



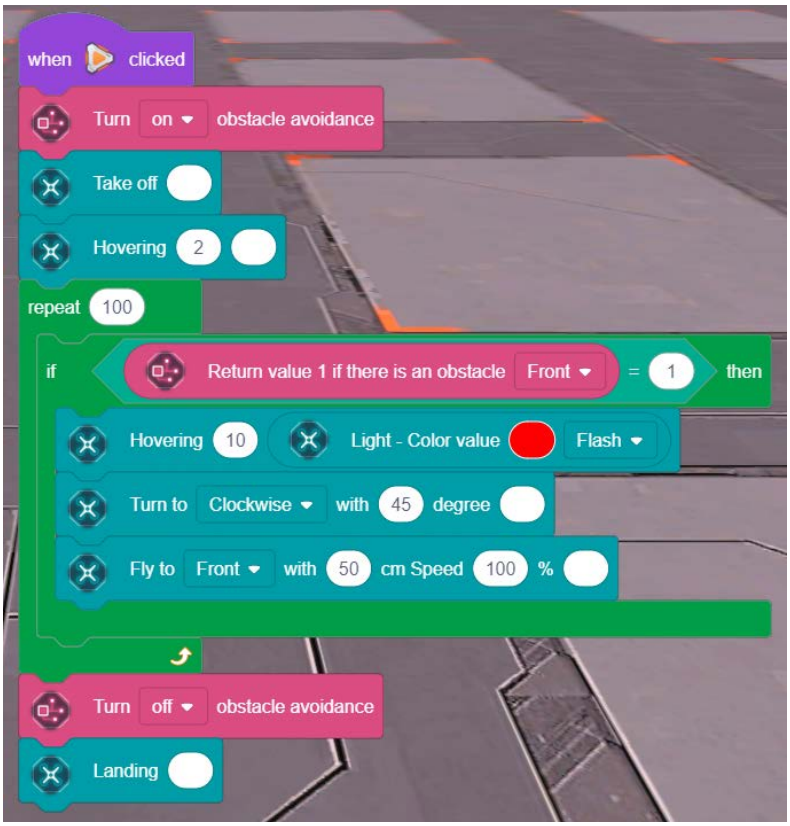


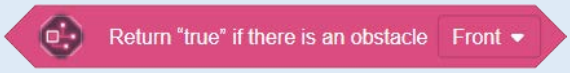
1. Name: [Turn on] QR-code carpet positioning
2. Function: It is an instruction for turning on/off the QR-code carpet positioning. When it is turned on, the aircraft will enter the QR-code carpet positioning mode and use the absolute coordinates. This feature uses optical flow camera positioning. After the aircraft lands, the QR code is automatically turned off for carpet positioning.
3. Type: execution block, No simulation
4. Example: [Turn on] QR-code carpet positioning



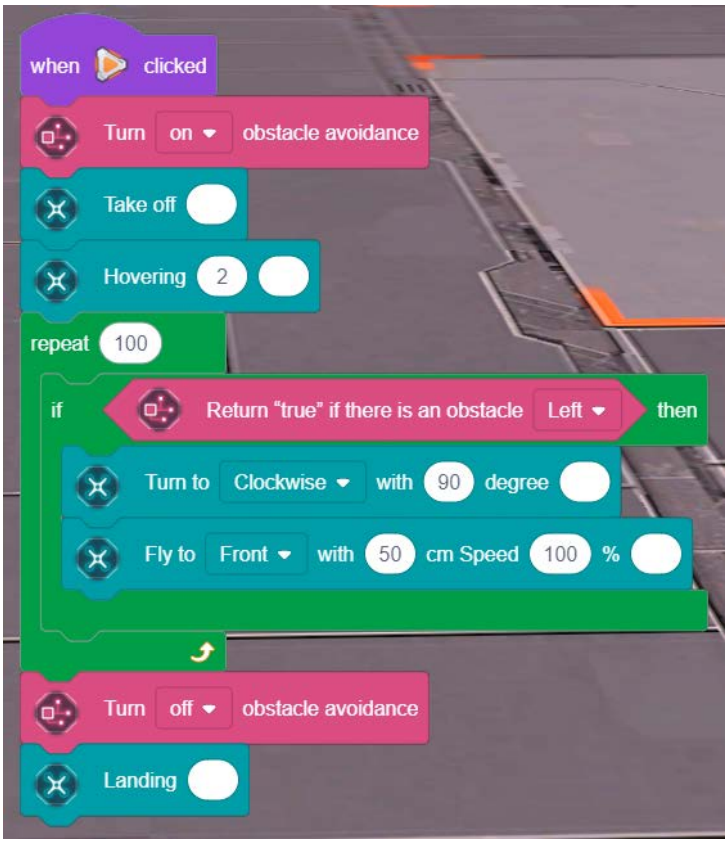
Return value 1 if there is an obstacle Front

1. Name: Return value 1 if there is an obstacle [Front]
2. Function: It is an instruction to get the return value after detecting an obstacle. [Turn on] obstacle avoidance, and return value 1 if there is an obstacle [Front/Back/Left/Right].
3. Type: parameter block, No simulation
4. Example: Detect that there is an obstacle in front, return the value 1, flash the red light, and after making a 45-degree turn, fly forward by 50 cm.

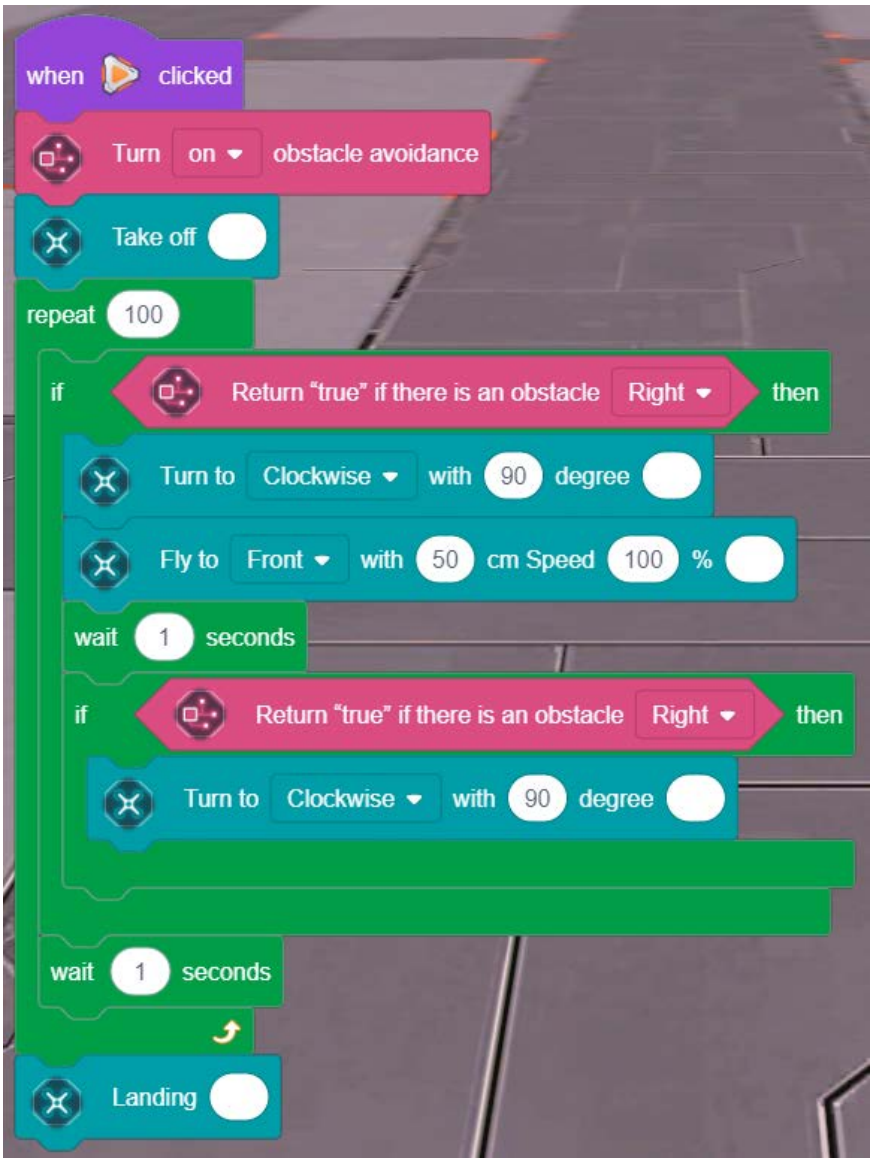




1. Name: Return value “true” if there is an obstacle [Front]
2. Function: It is an instruction for obstacle detection. [Turn on] obstacle avoidance, and return value “true” if there is an obstacle [Front/Back/Left/Right].
3. Type: parameter block, No simulation
4. Example: Detect that there is an obstacle on the left, flash the blue light, and after making a 90-degree turn, fly forward by 50 cm. Repeat this instruction 100 times.

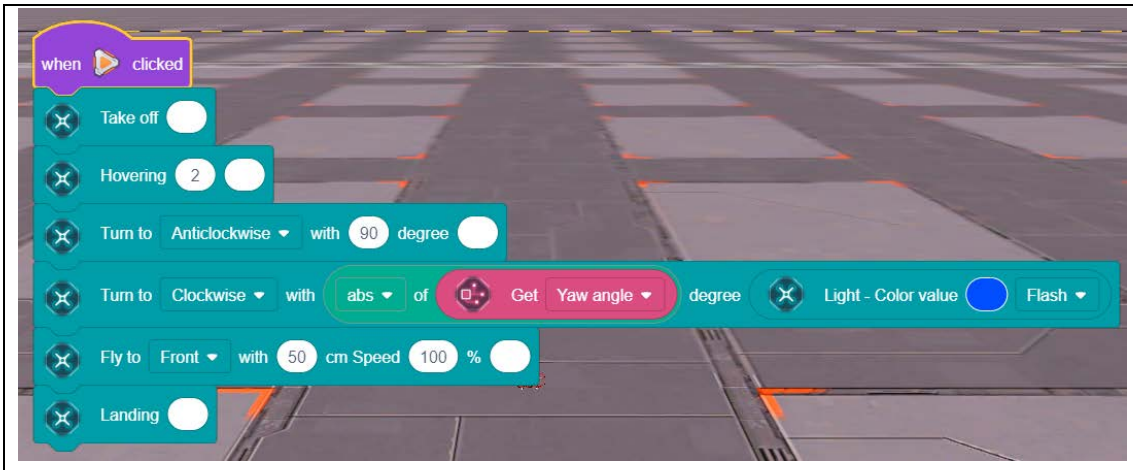


In order to avoid accidental obstacle avoidance missed detection, the waiting time can be increased by 1 second, and the detection frequency can be increased to improve the detection accuracy.

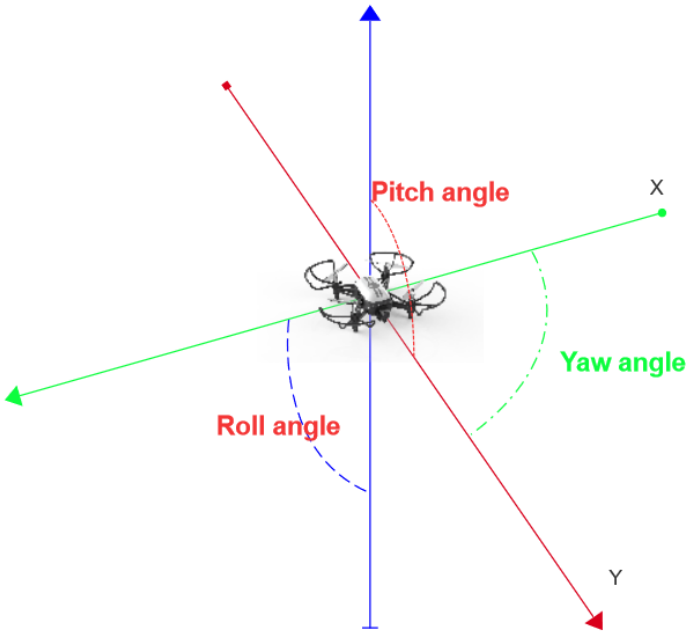


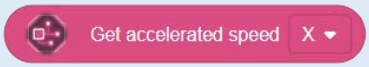


1. Name: Get [Yaw angle]
2. Function: It is an instruction to get parameters. During flight, the aircraft can get the current [Yaw angle/Pitch angle/Roll angle]. This block can be nested with the variable blocks, and variable names can be defined by yourself.
3. Type: parameter block, No simulation
4. Example: Get the [Yaw angle], flash the blue light, and after making a turn of [Yaw angle] degrees, fly forward by 50 cm.

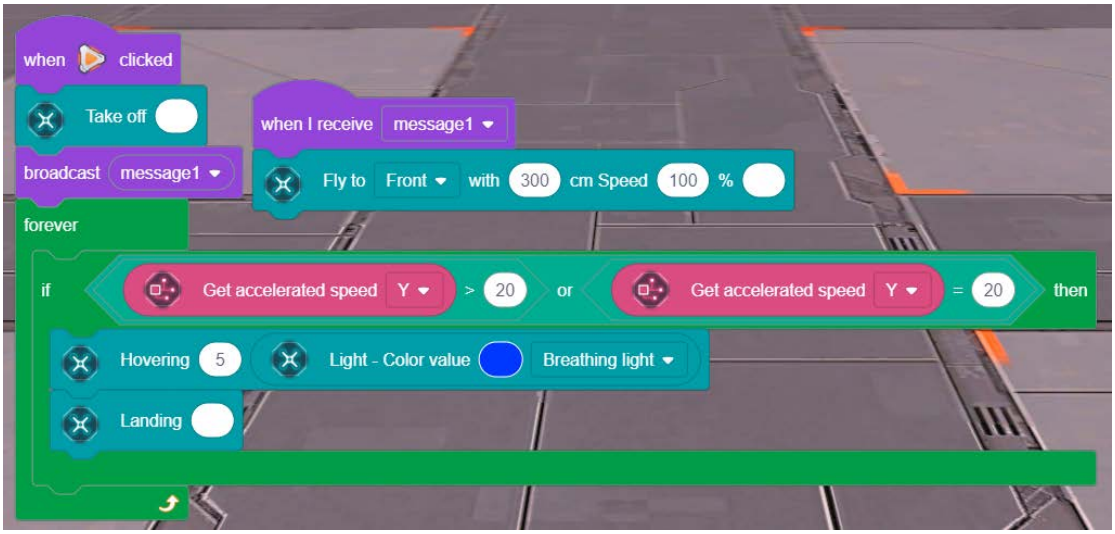


After takeoff, turn 90 degrees counterclockwise, and the yaw Angle obtained at this time is -90 degrees. The steering block cannot handle negative values, so the absolute value needs to be added.

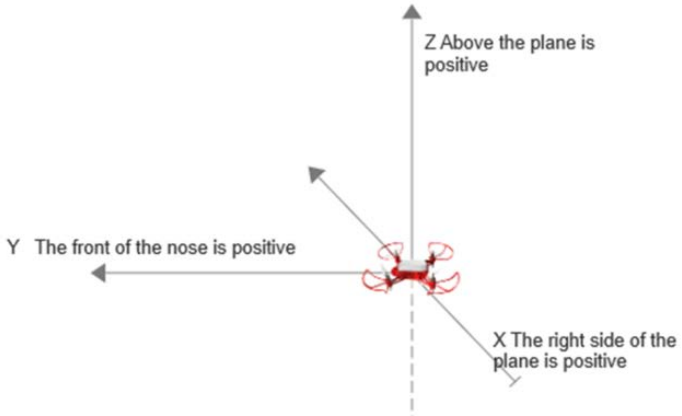


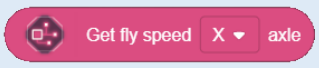


1. Name: Get accelerated speed [X]
2. Function: It is an instruction to get parameters. During flight, the aircraft can get the current accelerated speed of the single [X/Y/Z] axis. This block can be nested with the variable blocks, and variable names can be defined by yourself.
3. Type: parameter block, No simulation
4. Example: During fly 3 meters to front at 100% speed ,If the accelerated speed [Y] is greater than or equal to 20, flash the blue light and [Hovering].

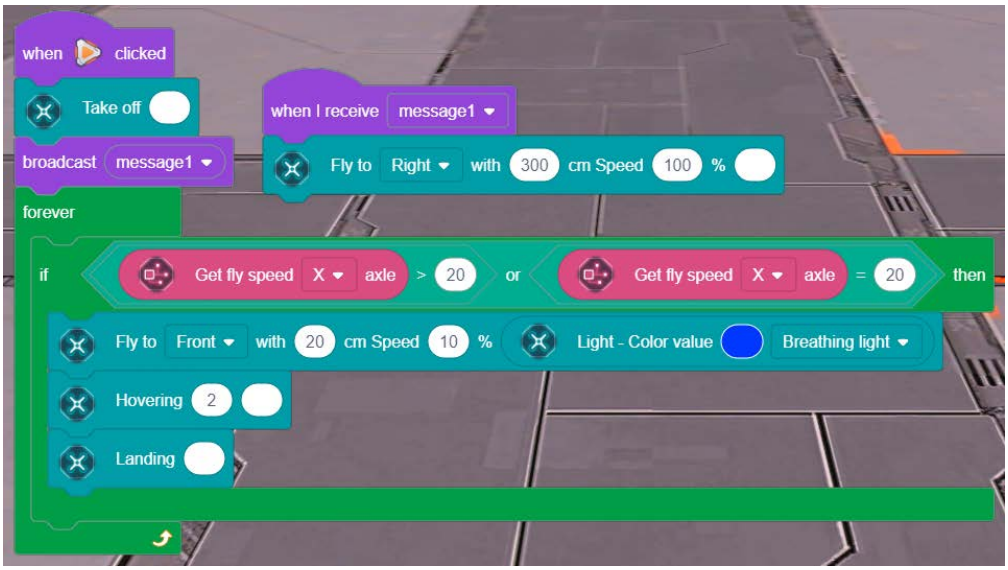


After taking off, the drone broadcasts the message 1, the drone receives the message 1 and begins to fly forward, and the acceleration is monitored while flying.



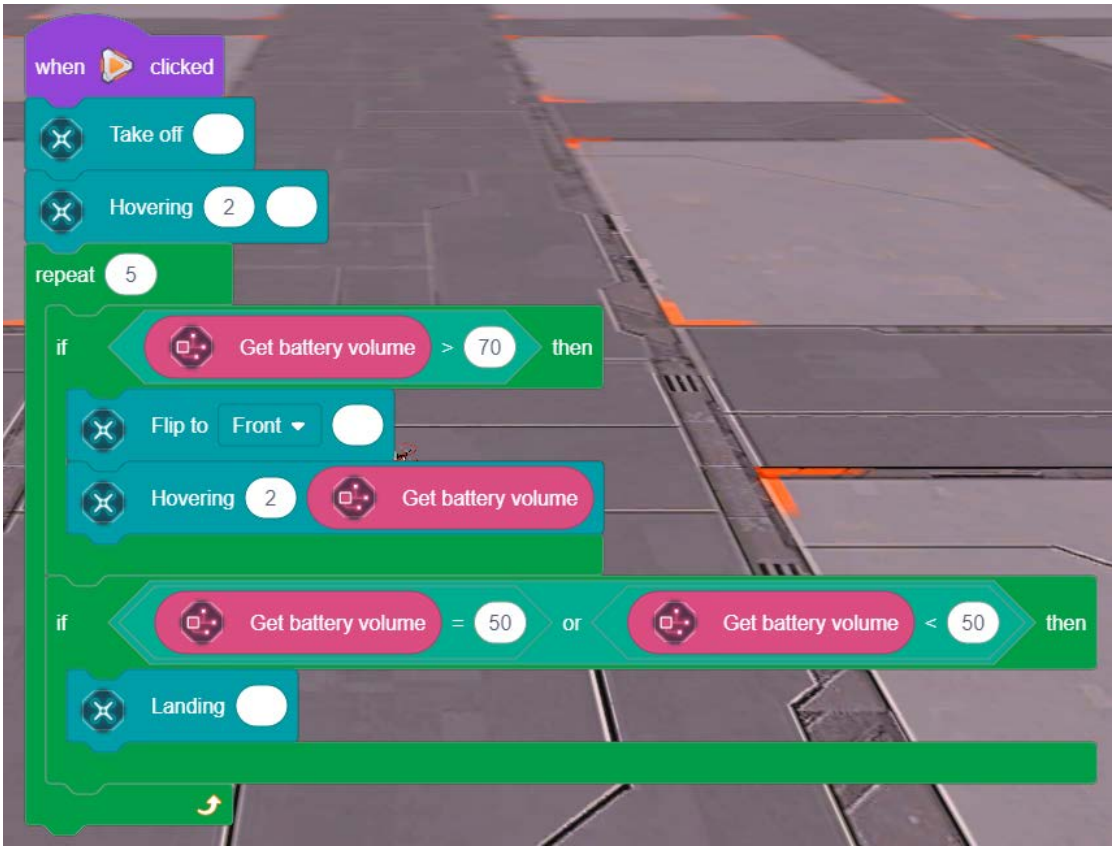


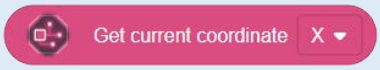
1. Name: Get fly speed [x] axle
2. Function: It is an instruction to get parameters. During flight, the aircraft can get the current speed of the single [X/Y/Z] axis. This block can be nested with the variable blocks, and variable names can be defined by yourself.
3. Type: parameter block, No simulation
4. Example: During fly 3 meters to the left at 100% speed, if flying speed [x]≥20, light blue and fly 20cm forward at 10% speed.



Get battery volume

1. Name: Get battery level
2. Function: It is an instruction to get parameters. It will get the current battery level of the aircraft. This block can be nested with the variable blocks, and variable names can be defined by yourself.
3. Type: parameter block, No simulation
4. Example: If the battery level is greater than 70, the aircraft will roll, and if the battery level is less than or equal to 50, the aircraft will land.

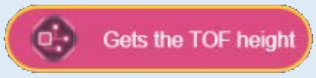




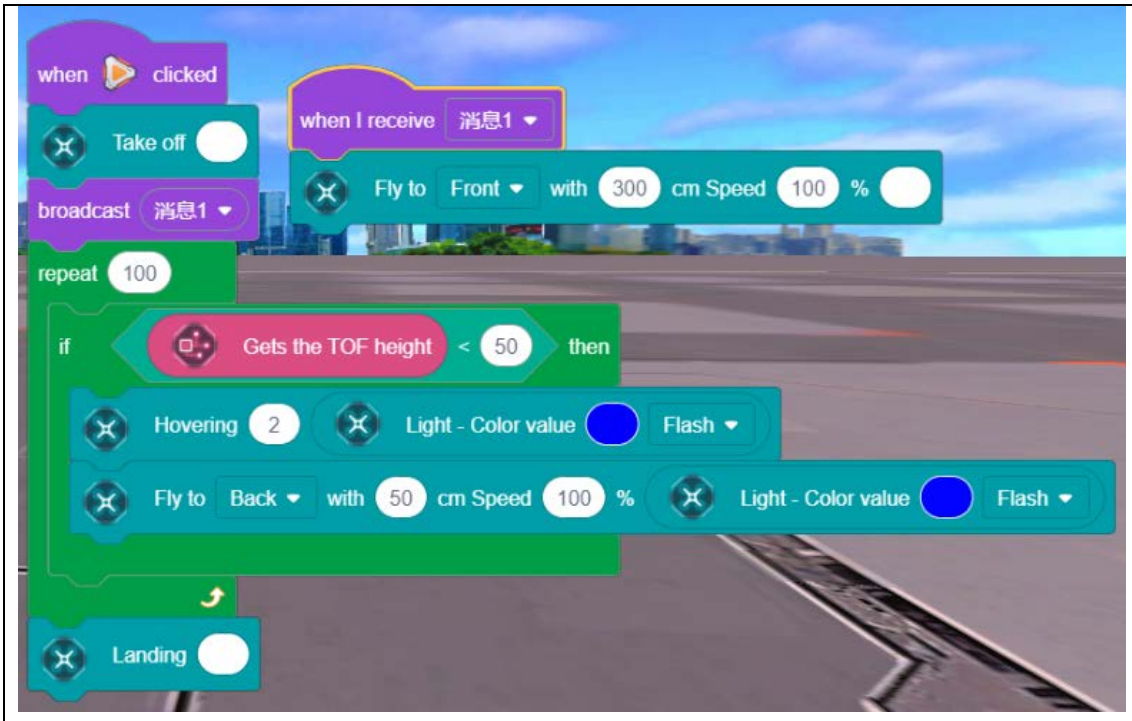
1. Name: Get current coordinate [X]
2. Function: It is an instruction to get parameters. During flight, the aircraft can get the current coordinate [X/Y/Z]. This block can be nested with the variable blocks, and variable names can be defined by yourself. The positive coordinate direction is consistent with the orientation of the aircraft's nose when the aircraft is powered on.
3. Type: parameter block, No simulation
4. Example: During fly to [front] with [300] cm, and when the current coordinate [Y] ＞ 200, [Hovering] and flash the blue light.

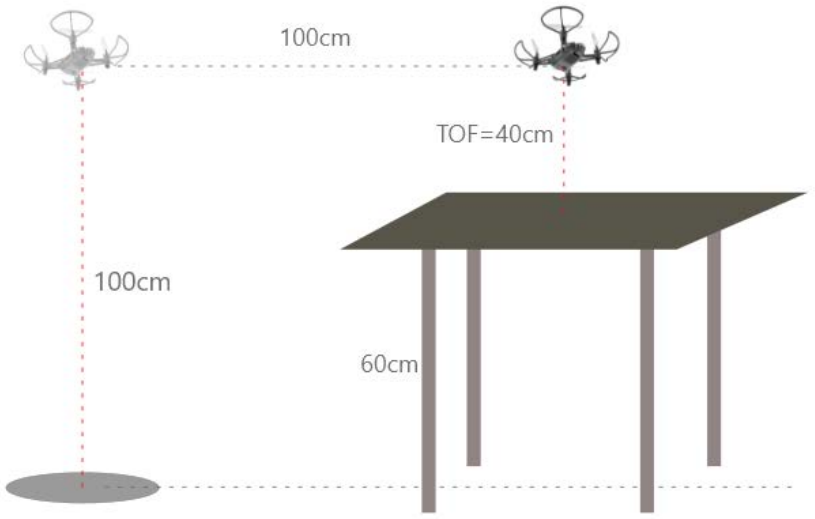


When the current coordinate [Y]=200 is replaced, due to the small time difference between program execution and aircraft communication, there is a low possibility of obtaining the current coordinate Y=200 at anytime . Please do not use "=" to make real-time flight data judgment.

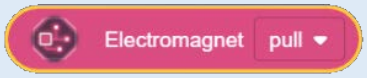


1. Name: Gets the TOF height
2. Use: It is an instruction to get TOF height, the TOF height value can be obtained during the flight of the aircraft, the unit is cm;
3. Type: Parameter block, No simulation
4. Example: In the process of flying forward 300cm, when the current TOF height value is < 50, light the blue light and retreat 50cm;

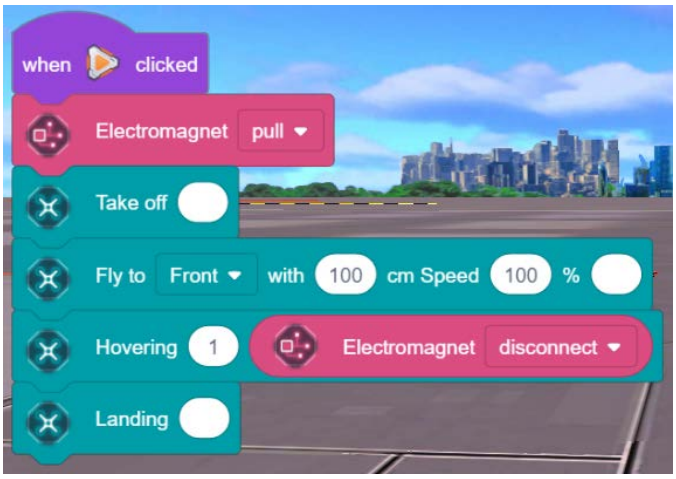




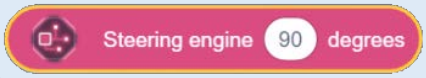
When judging TOF height =50, due to the small time difference between program execution and aircraft communication, it is very unlikely that the current coordinate TOF height =200 can be obtained at a certain time. Please do not use "=" to judge real-time flight data.



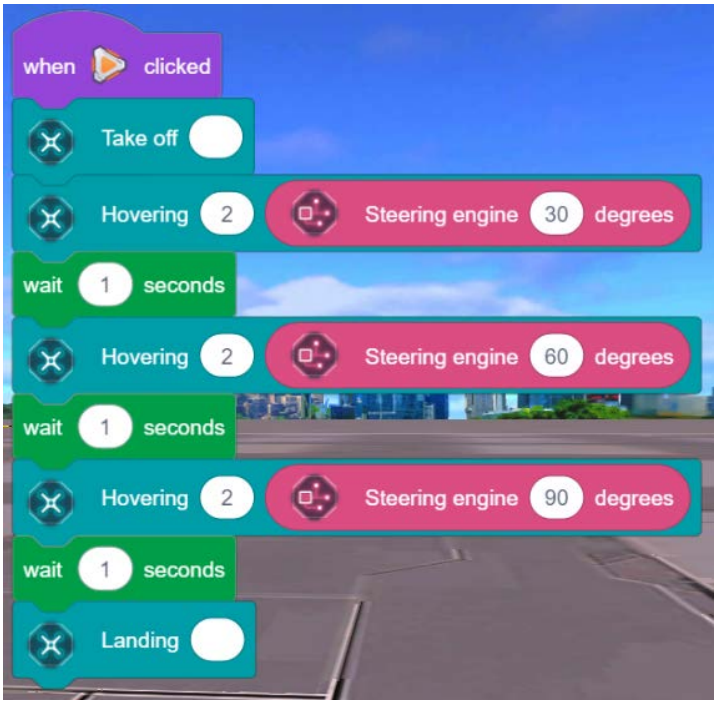
1. Name: Electromagnet [pull]
2. Use: It is an instruction to Control the mounted electromagnet pull/disconnect command, used to absorb or drop the mounted items during the flight
3. Type: Parameter block, No simulation
4. Example: Fly forward 100cm, electromagnet [disconnect], drop items



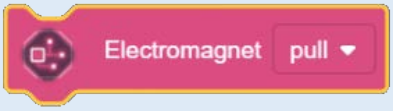
The aircraft is attached to the electromagnet kit, which attracts the items before taking off, and after flying to the target location, the electromagnet is disconnected and the items are dropped.



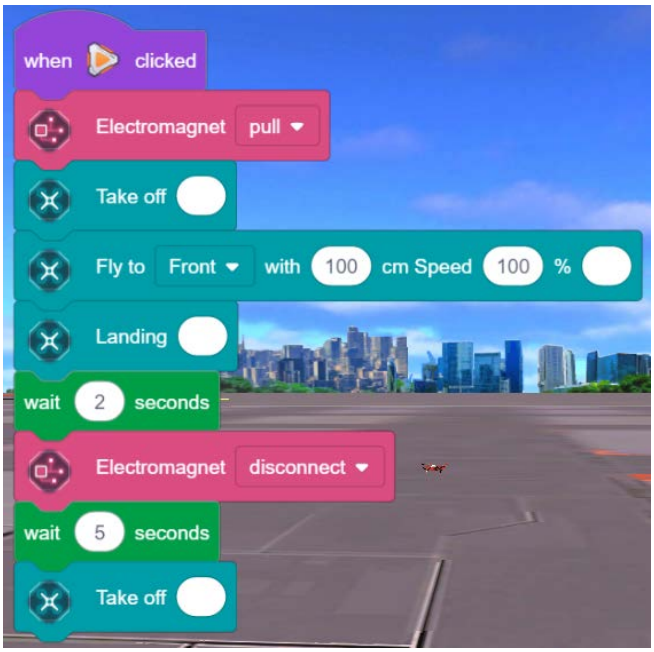
1. Name: Steering engine [90] degrees
2. Use: It is an instruction to Control steering gear Angle, during flight, used to grab or release items.
3. Type: Execution block, No simulation
4. Example: Control the steering gear Angle



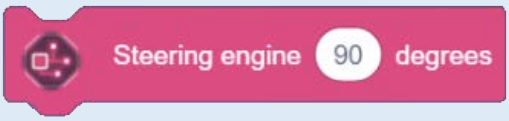
After the external steering gear is connected, the clamp can be opened and closed by controlling the steering gear Angle.



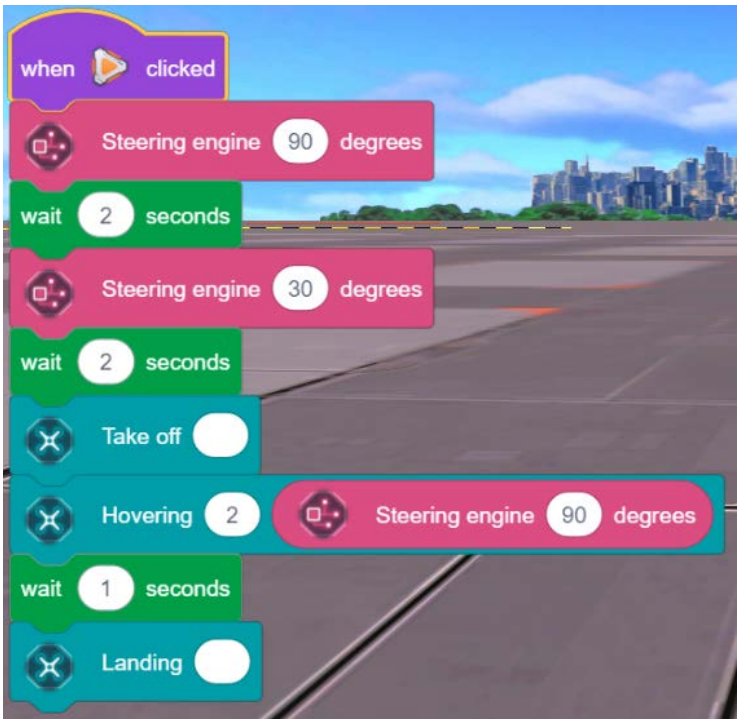
1. Name: Electromagnet [pull]
2. Use: It is an instruction to Control electromagnet pull/disconnec, The aircraft can pull or disconnec items while on the floor.
3. Type: Execution block, No simulation
4. Example: After landing, the electromagnet [disconnects] and takes off again



The aircraft is attached to the electromagnet kit, which attracts the items before taking off, and after landing at the target location, the electromagnet is disconnected and the items are placed.

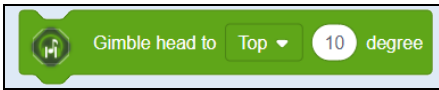


1. Name: Steering engine [90] degrees
2. Use: It is an instruction to control the steering gear Angle, used for the aircraft does not take off, can control the steering gear to grab/release items;
3. Type: Execution block, No simulation
4. Example: Control the steering gear Angle



After the external steering gear is connected, the Angle of the steering gear is controlled when it does not take off to realize the ground grab and airdrop.

8.2.5 Multimedia

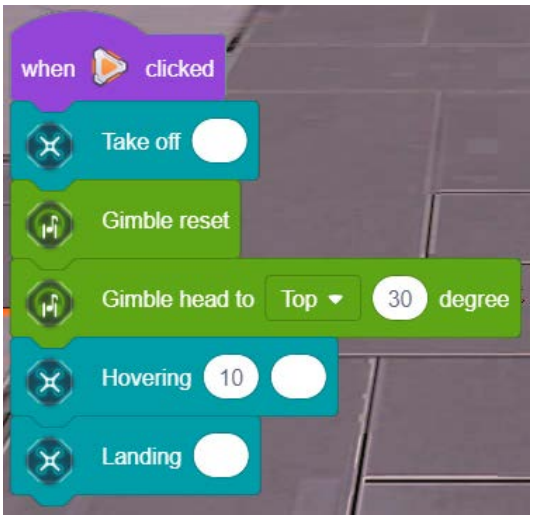


1. Name: Gimbal head to [Top/Bottom] [n] degree
2. Function: It is an instruction for adjusting the gimbal angle. This instruction controls the gimbal to rotate [Top/Bottom] [n] degree. The gimbal can be adjusted within a range of [0~130] degrees, with a minimum adjustment of 1 degree. It can be used in conjunction with the reset blocks. First, reset the gimbal to its default horizontal position, and then adjust the corresponding angle [up/down].
3. Type: execution block, No simulation
4. Example: Gimbal reset, and Gimbal head to [Bottom] [90] degree



Gimble reset

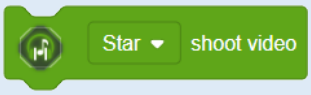
1. Name: Gimbal reset
2. Function: It is an instruction for resetting the gimbal. It will reset the gimbal to its default horizontal position.
3. Type: execution block, No simulation
4. Example: Gimbal reset, and gimbal head to [Top] [30] degree



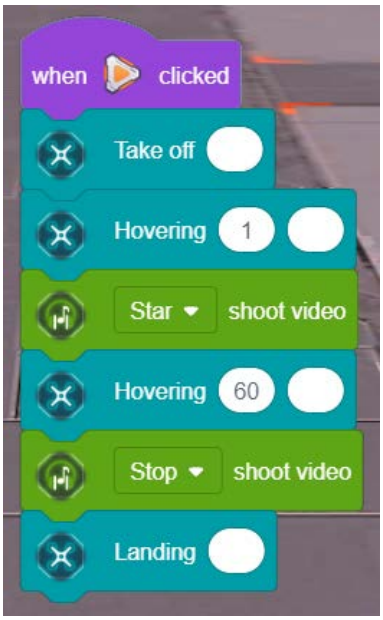
Shoot

1. Name: Shoot
2. Function: It is an instruction for shooting a picture.
   This function operates on the aircraft motherboard without the need to turn on the image transmission.
3. Type: execution block, No simulation
4. Example: Shoot 5 pictures. The pictures can be viewed on the home page - Album.

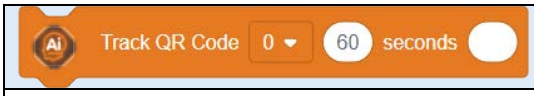




1. Name: [Start] shoot video
2. Function: It is an instruction for shooting videos. [Start/Stop] shoot video. After the [Start] shoot video block, you need the [Stop] shoot video block, or the aircraft will continue shooting videos. Currently, video shooting is limited to a maximum of 8 minutes per session.
   This function operates on the aircraft motherboard without the need to turn on the image transmission.
3. Type: execution block, No simulation
4. Example: [Start] shoot video, and after 1 minute, [Stop] shoot video. The video can be viewed on the home page - Album.



8.2.6 AI-Apply



1. Name: Track QR Code [0] [60] seconds
2. Function: It is an instruction for tracking QR codes. The aircraft uses the **front camera to recognize QR codes,** with the option to choose QR codes numbered from 0 to 9. The time can be set between [0~400] seconds, and can be embedded with the light blocks. This function operates on the aircraft motherboard without the need to turn on the image transmission. Tap  to turn on the image transmission, to see if the target is out of the image transmission range and cannot be recognized.
3. Type: execution block, No simulation
4. Example: Track QR Code [0] [60] seconds



QR code tracking requirements: With the image transmission function enabled, it requires good indoor lighting and no backlight. Also, the flight height of the aircraft should be between [1~2.2] meters. The QR code should maintain a vertical distance of [1~2.2] meters from the front camera for optimal recognition results.

Note: Please used at indoor and far away from people， within0.5-3 meters.

Start tracking: The aircraft uses the **front camera** to recognize the QR code. The aircraft will dynamically maintain its distance from the QR code to track it in 1.5m.

Following: You can hold the QR code and slowly move it back and forth and left to right to ensure stable QR code recognition. Please note that the QR code should remain within the aircraft's screen area. The optimal following effect is achieved with a QR code movement speed of less than or equal to 0.3 m/s horizontally and less than or equal to 0.5 m/s vertically.

Search timeout: When the aircraft performs QR code tracking, if no QR code is recognized within 30 seconds, the tracking function will be turned off, and a prompt "Search timeout, exiting tracking" will be displayed.

Stop tracking: The aircraft maintains the distance at which it initially recognized the QR code and continues to dynamically track the QR code until the tracking time is up. Only after the tracking instruction is stopped will it proceed to execute the other blocks below.

Cautions: During the tracking, the image transmission defaults to 360 px. If you record videos or take photos during tracking, the resolution will be 360 px.

When holding the QR code, do not block the outer white edge of the QR code with your finger, which will lead to failure to identify it.

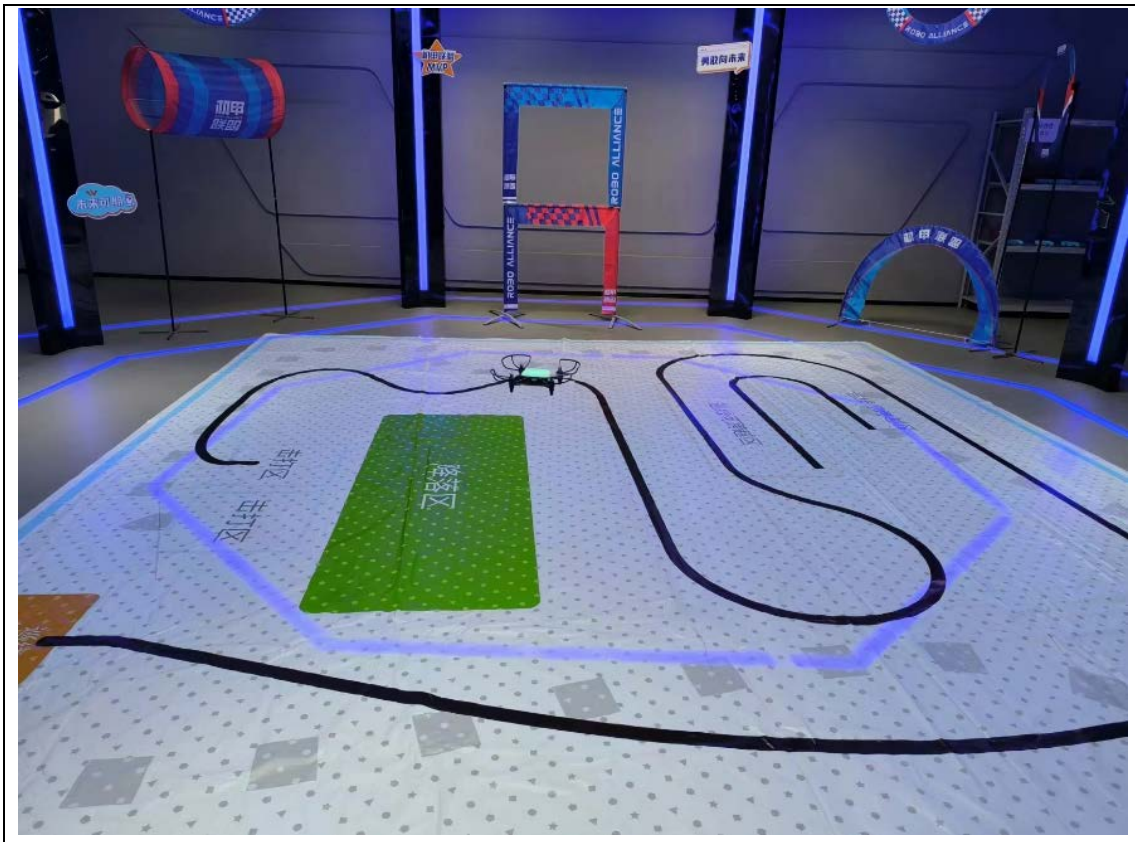




1. Name: Line [White] patrol [60] cm
2. Function: It is an instruction for line patrol. The aircraft uses the downward-facing camera to recognize a solid line with a certain width, which can be either Dark line on Light background (black line) or Light line on Dark background (white line) . The length of line patrol can be set between [0~600] cm, with an actual margin of error of around ±10 cm. This block can also be embedded with the light blocks.
   This function operates on the aircraft motherboard without the need to turn on the image transmission, and the proactive image transmission is used to see if the target is out of the image transmission range and cannot be recognized.
3. Type: execution block, No simulation
4. Example: Line [Black] patrol [200] cm

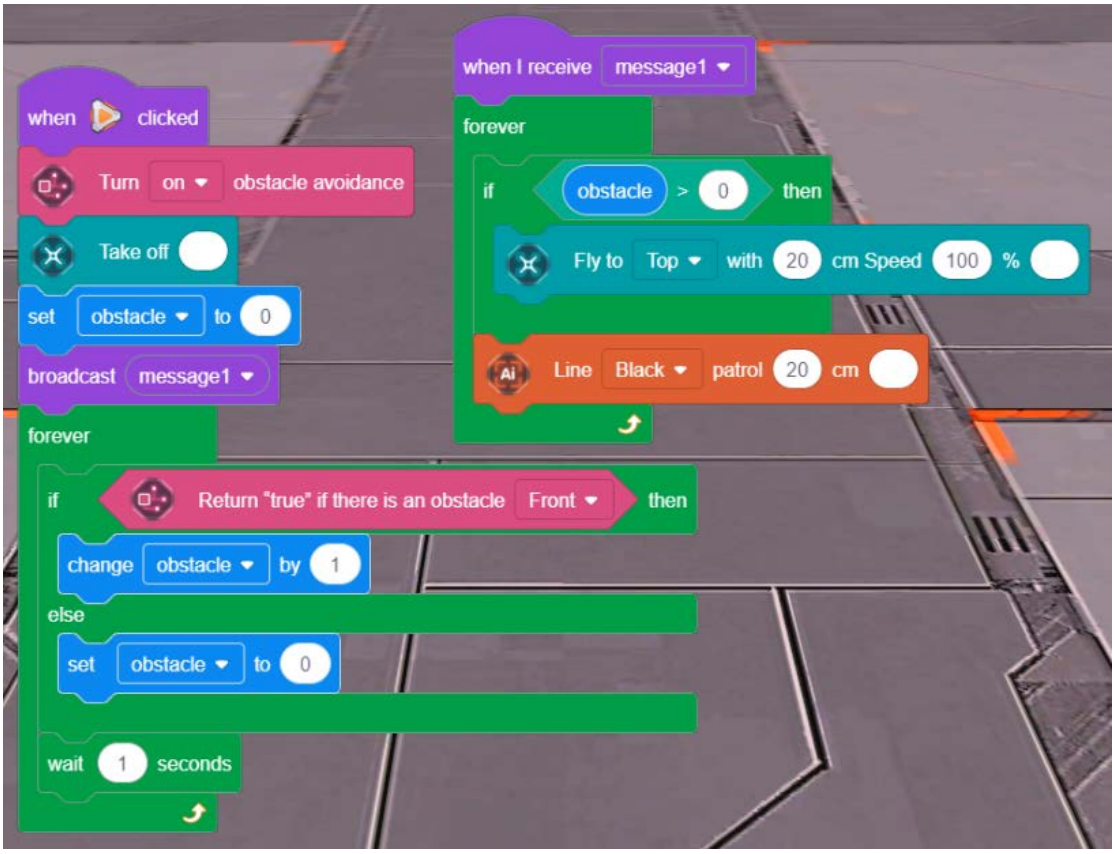


Line patrol requirements: The optimal flight height of the aircraft is [1~2.2] meters. The best visual effect will be achieved by Dark line on Light background (black line) or Light line on Dark background (white line) , and for other colors, the greater the contrast between the line and the background, the better the effect. The line width is recommended to be more than 5cm, The angle between lines at turns should be greater than 135°, meaning the turning angle of the aircraft nose at each turn cannot exceed 45°. Flying over indoor surfaces with good lighting and non-reflective textures produces the best flight results.
Note: Please used at indoor and far away from people， within0.5-3 meters.

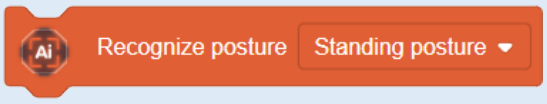


Start line patrol: After recognizing the line, the aircraft will automatically perform a line patrol flight.
Line patrol timeout: When the aircraft searches the line, and the line on the ground is not recognized within 30 seconds, the line patrol function will be turned off, and a prompt "Line patrol timeout, exited" will be displayed.
Stop line patrol: The aircraft will automatically stop line patrol after covering the input distance, with an actual margin of error of around ±10 cm.

Cautions: During the line patrol, the image transmission defaults to 360 px. If you record videos or take photos during the line patrol, the resolution will be 360 px. Avoid using this function in environments with poor lighting, where the line color is close to the background, or where the ground reflects light strongly, as it may affect the optical flow positioning function of the aircraft.
When patrolling the line, detect obstacles ahead and fly upward.



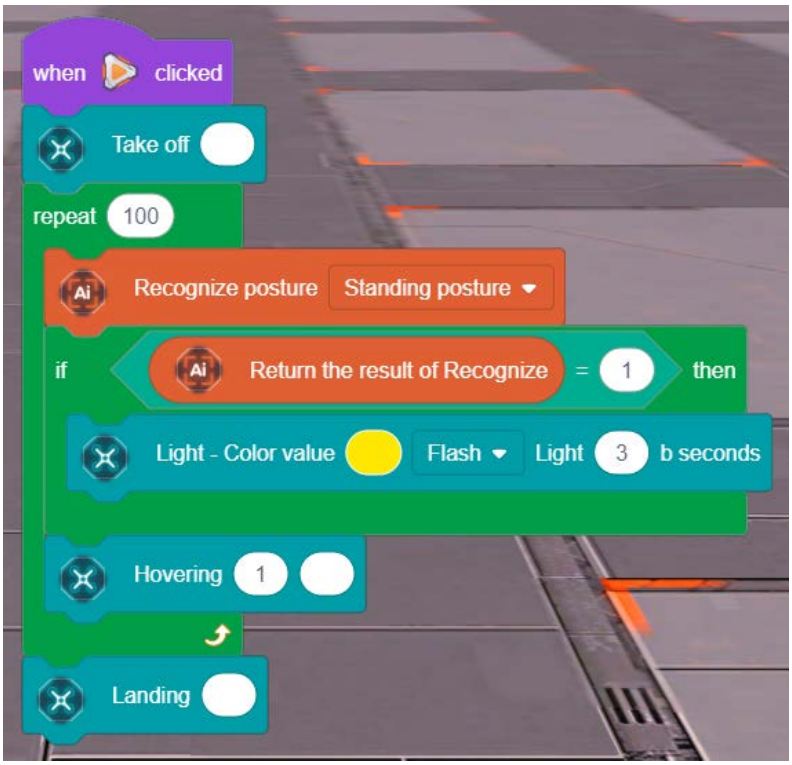
8.2.7 AI-Recognition



1. Name: Recognize posture [Standing posture]
5. Function: It is an instruction for recognizing posture. Available postures are [Sitting posture, Standing posture, Ultraman ray, Extend right hand, and Extend left hand]. Before recognizing, you need to tap  to open the image transmission. It can be used in conjunction with the "Return the result of Recognize" block. It can recognize individual targets or multiple targets of the same posture. The returned results and coordinates are based on the largest recognized target. This function runs on the APP and can only be used if you enable the image transmission.

2. Type: execution block, No simulation
3. Example: Recognize posture [Standing posture]

Recognition requirements: It requires the aircraft to hover at a height of [1~2] meters or remain grounded, the image transmission to be enabled, adequate indoor lighting, a distance from the target within [1~2.5] meters, clear visibility of the complete target, absence of backlighting, and a distraction-free solid-color background. Under these conditions, the recognition success rate is high.



Start recognition: When the recognition block is executed, it initiates the recognition of targets within the image transmission.

Stop recognition: The recognition block runs for 5 seconds, and it stops recognition upon receiving the result of "Recognized successfully" or "Recognition failed".

Block mutual exclusion: The aircraft does not support multi-threaded parallel execution of multiple types of recognition blocks. When there are parallel execution recognition blocks, only the last recognition instruction will be executed.

Standing posture：Hands should be below the waist and feet should be within range of the picture



Sitting posture：Feet need to be within range of the image




Ultraman ray:






Extend left hand:



Extend right hand:



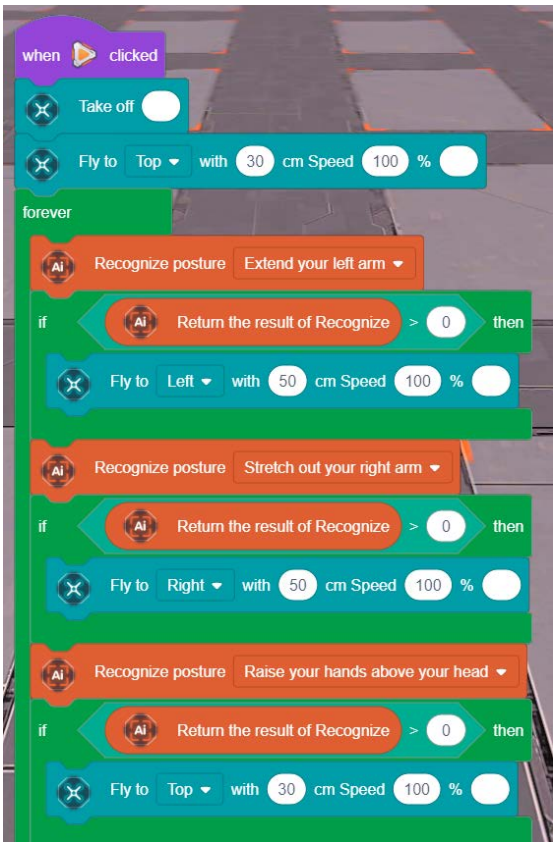
Hands over your head:

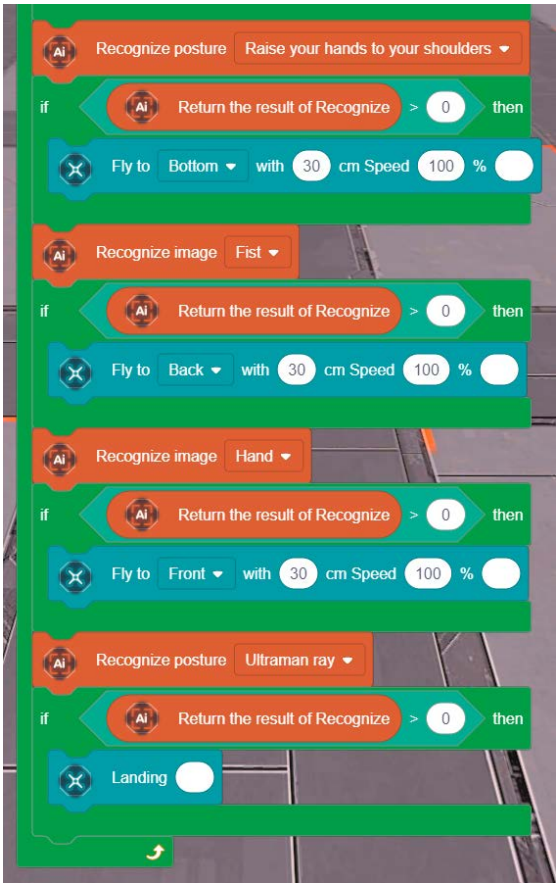


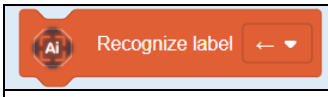

Hands up to your shoulders:




Posture controlled flight:

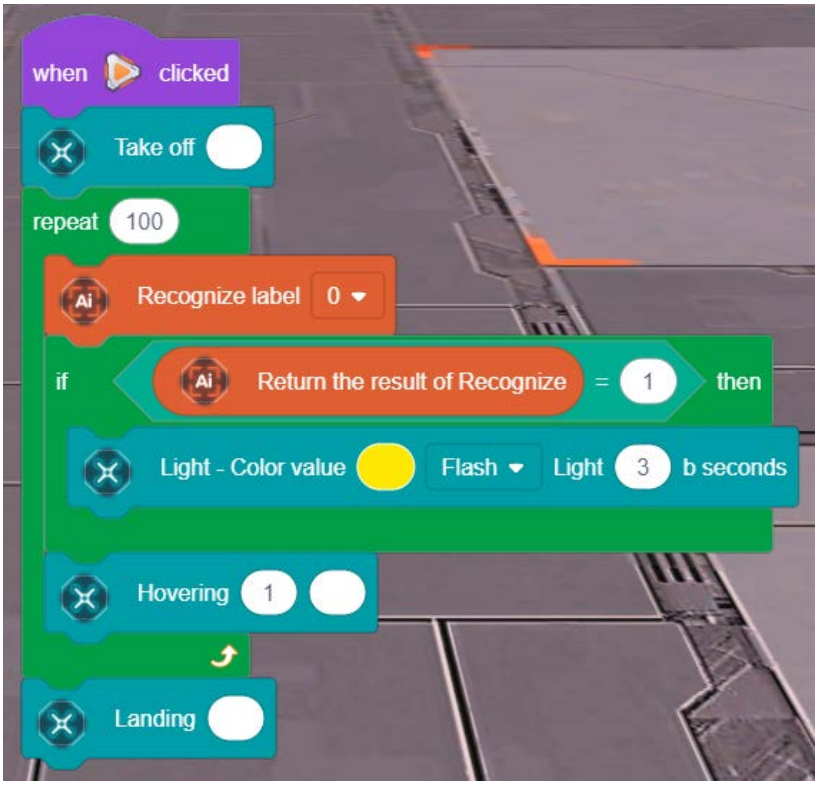






1. Name: Recognize label [0]
2. Function: It is an instruction for recognizing the label. Available labels are [←, →, ↑, ↓, and 0~9]. It can be used in conjunction with the "Return the result of Recognize" block. It can recognize individual targets or multiple targets of the same label. The returned results and coordinates are based on the largest recognized target.
   This function operates on the aircraft motherboard without the need to turn on the image transmission, and the proactive image transmission is used to see if the target is out of the image transmission range and cannot be recognized.
3. Type: execution block, No simulation
4. Example: Recognize label [0]

Recognition requirements: It requires the aircraft to hover at a height of [1~2] meters or remain grounded, the image transmission to be enabled, adequate indoor lighting, a distance from the target within [1~2.5] meters, clear visibility of the complete target, absence of backlighting, and a distraction-free solid-color background. Under these conditions, the recognition success rate is high.



Start recognition: When the recognition block is executed, it initiates the recognition of targets within the image transmission.

Stop recognition: The recognition block runs for 5 seconds, and it stops recognition upon receiving the result of "Recognized successfully" or "Recognition failed".

Block mutual exclusion: The aircraft does not support multi-threaded parallel execution of multiple types of recognition blocks. When there are parallel execution recognition blocks, only the last recognition instruction will be executed.

Hand label specification: do not block the white or black side of the label with your fingers, which will lead to recognition.





Arrow tag cards, 0-9 tag cards, and 0-9 QR codes are all tag cards that can be used in combination to flexibly design event goals/tasks.

Arrow tag card: can be used to program through the direction guide design;

0~9 digital tag card: It can be used to obtain game items or to design the designated task route;

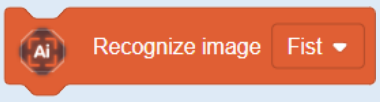
0-9 QR code: can be used for alignment, tracking, ranging, flight correction, can also be used for props, route design;

Event case: To identify tag # 3, take Route # 3 - > Identify tag # 5, get 5 bullets - > Identify tag # 9, hit QR

code # 9 (target) within 5 attacks. During the execution of the mission, if the opponent hit, you can identify the 7 tag to revive or recover HP.

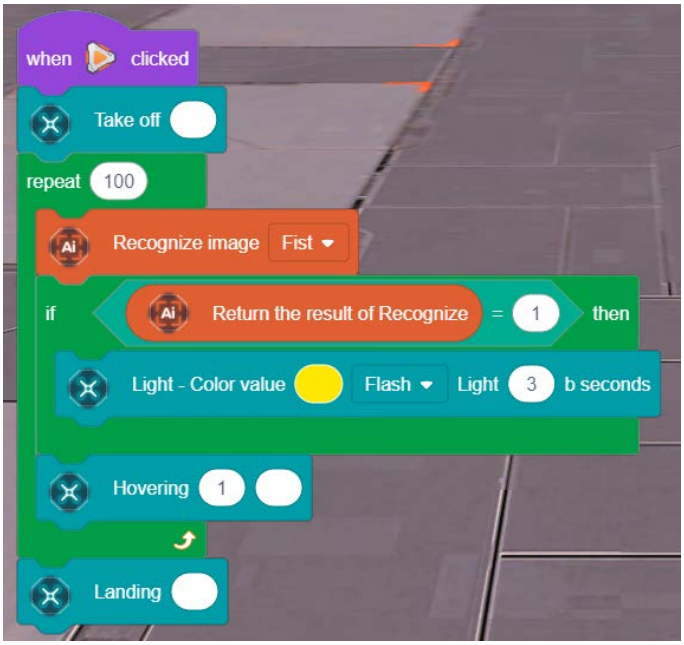
Collaborative Crossing Case:





1. Name: Recognize image [Fist]
2. Function: It is an instruction for recognizing the image. Available images are [Fist, Hand, Pedestrian, Hula, and Book]. Before recognizing, you need to tap  to open the image transmission. It can be used in conjunction with the "Return the result of Recognize" block. It can recognize individual targets or multiple targets of the same image. The returned results and coordinates are based on the largest recognized target. This function runs on the APP and can only be used if you enable the image transmission.
3. Type: execution block, No simulation
4. Example: Recognize image [Fist]

Recognition requirements: It requires the aircraft to hover at a height of [1~2] meters or remain grounded, the image transmission to be enabled, adequate indoor lighting, a distance from the target within [1~2.5] meters, clear visibility of the complete target, absence of backlighting, and a distraction-free solid-color background. Under these conditions, the recognition success rate is high.



Start recognition: When the recognition block is executed, it initiates the recognition of targets within the image transmission.

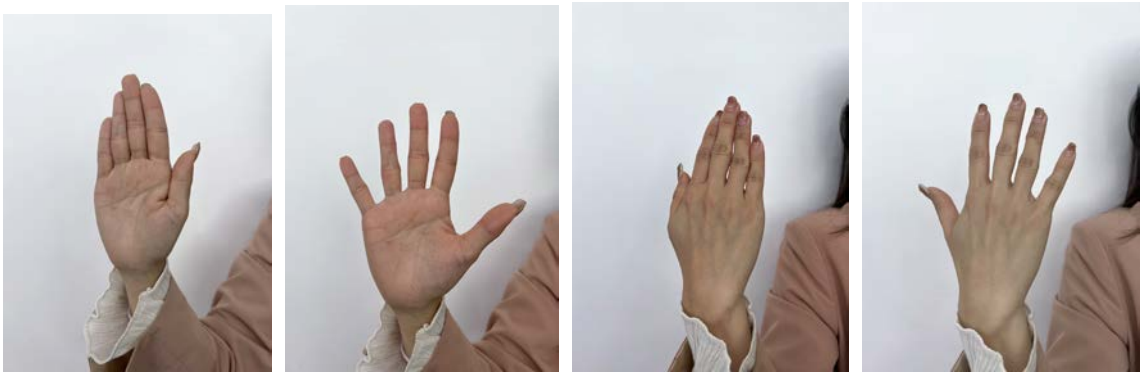
Stop recognition: The recognition block runs for 5 seconds, and it stops recognition upon receiving the result of "Recognized successfully" or "Recognition failed".

Block mutual exclusion: The aircraft does not support multi-threaded parallel execution of multiple types of recognition blocks. When there are parallel execution recognition blocks, only the last recognition instruction will be executed.

Fist：



Hand：

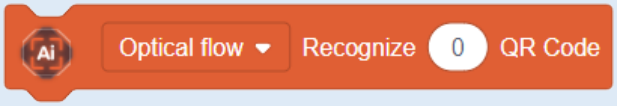


Hula: The recognition rate of the oblique side is higher




Pedestrian：





1. Name: [Front camera/Optical flow] Recognize [0] QR code
2. Function: It is an instruction for recognizing the QR code. Available options are "[Front camera/Optical flow] Recognize QR codes [0~9]". It can be used in conjunction with the "Return the result of Recognize" block. It can recognize individual targets or multiple targets of the same QR code. The returned results and coordinates are based on the largest recognized target.
   [Optical flow] refers to using the camera underneath the aircraft for recognition, while [Front camera] refers to using the camera in front of the aircraft for recognition.
   This function operates on the aircraft motherboard without the need to turn on the image transmission, and the proactive image transmission is used to see if the target is out of the image transmission range and cannot be recognized.
3. Type: execution block, No simulation
4. Example: [Front camera] Recognize image [Fist]

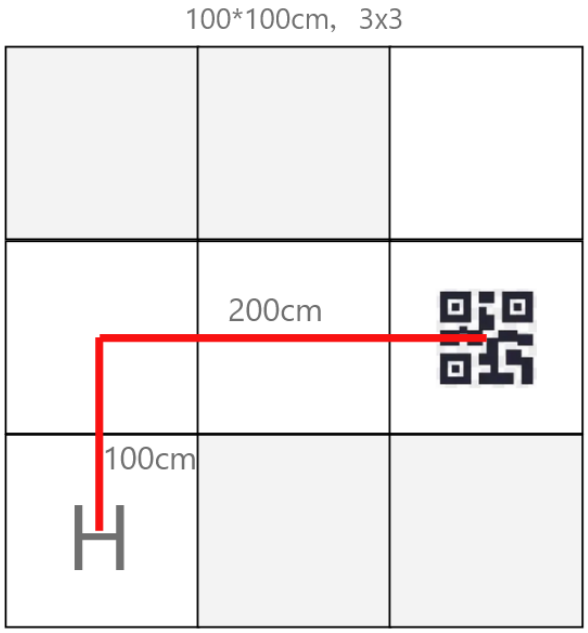
Recognition requirements: It requires the aircraft to hover at a height of [1~2] meters or remain grounded, the image transmission to be enabled, adequate indoor lighting, a distance from the target within [1~2.2] meters, clear visibility of the complete target, absence of backlighting, and a distraction-free solid-color background. Under these conditions, the recognition success rate is high.



Start recognition: When the recognition block is executed, it initiates the recognition of targets within the image transmission.

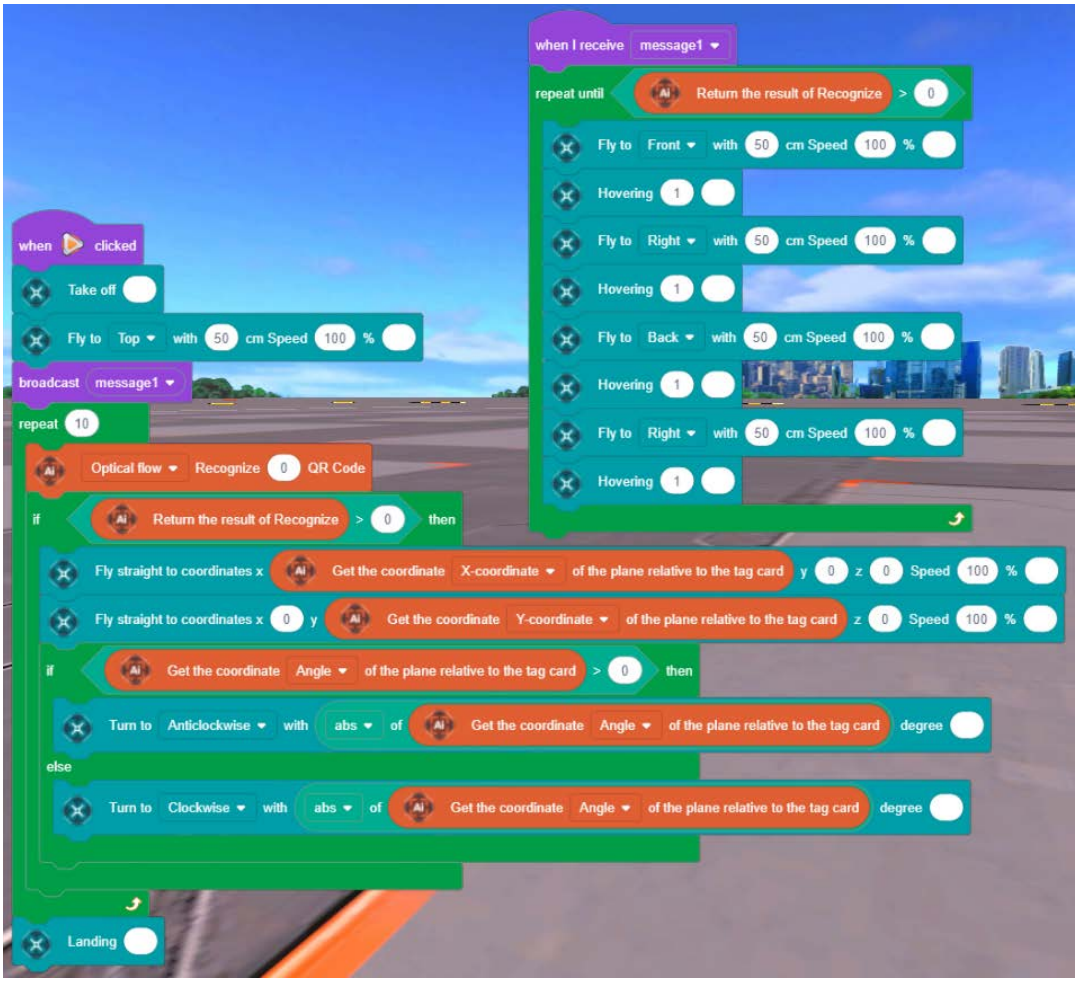
Stop recognition: The recognition block runs for 5 seconds, and it stops recognition upon receiving the result of "Recognized successfully" or "Recognition failed".

Block mutual exclusion: The aircraft does not support multi-threaded parallel execution of multiple types of recognition blocks. When there are parallel execution recognition blocks, only the last recognition instruction will be executed.

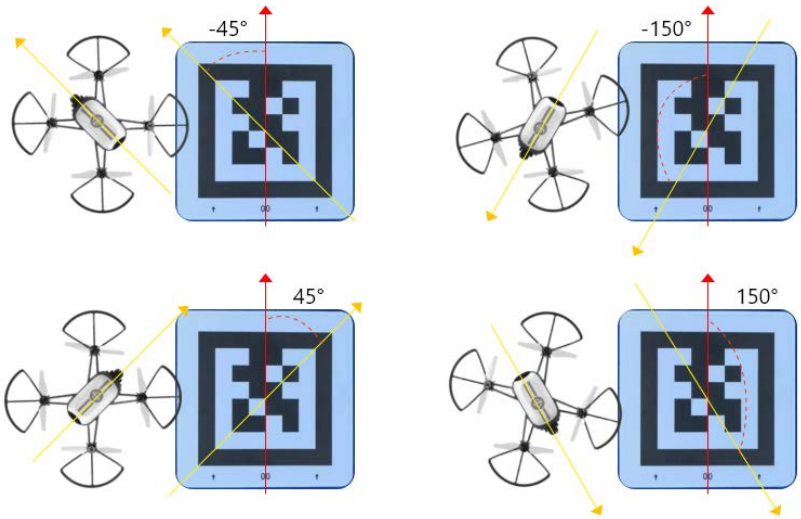


As shown in the figure, the two-dimensional code is placed on the ground, takes off from H, searches for the

two-dimensional code, aligns it with optical flow, and lands on the two-dimensional code.

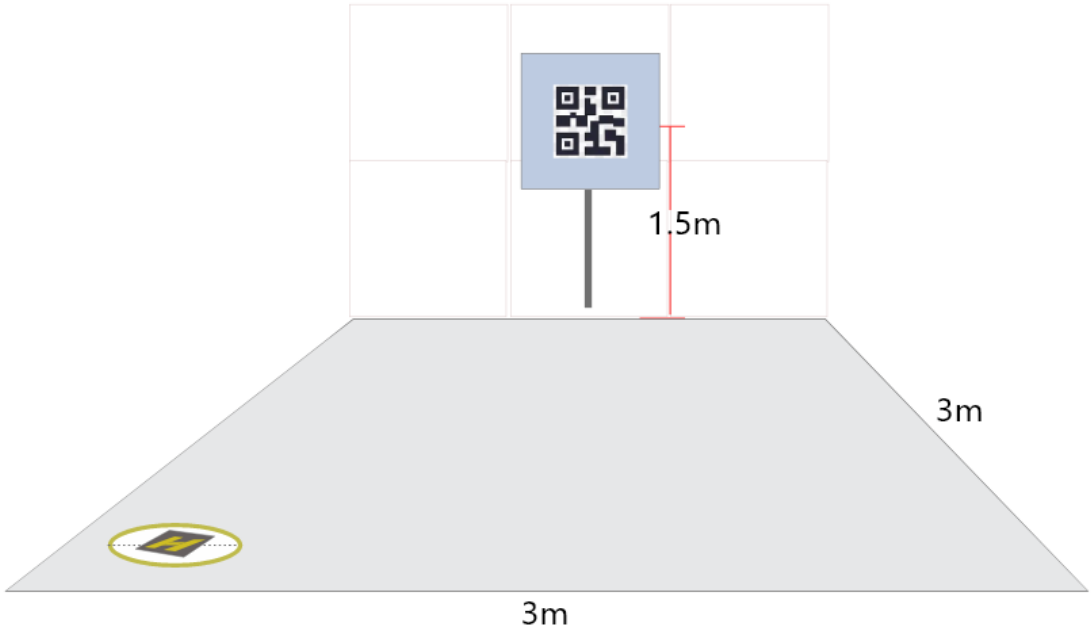


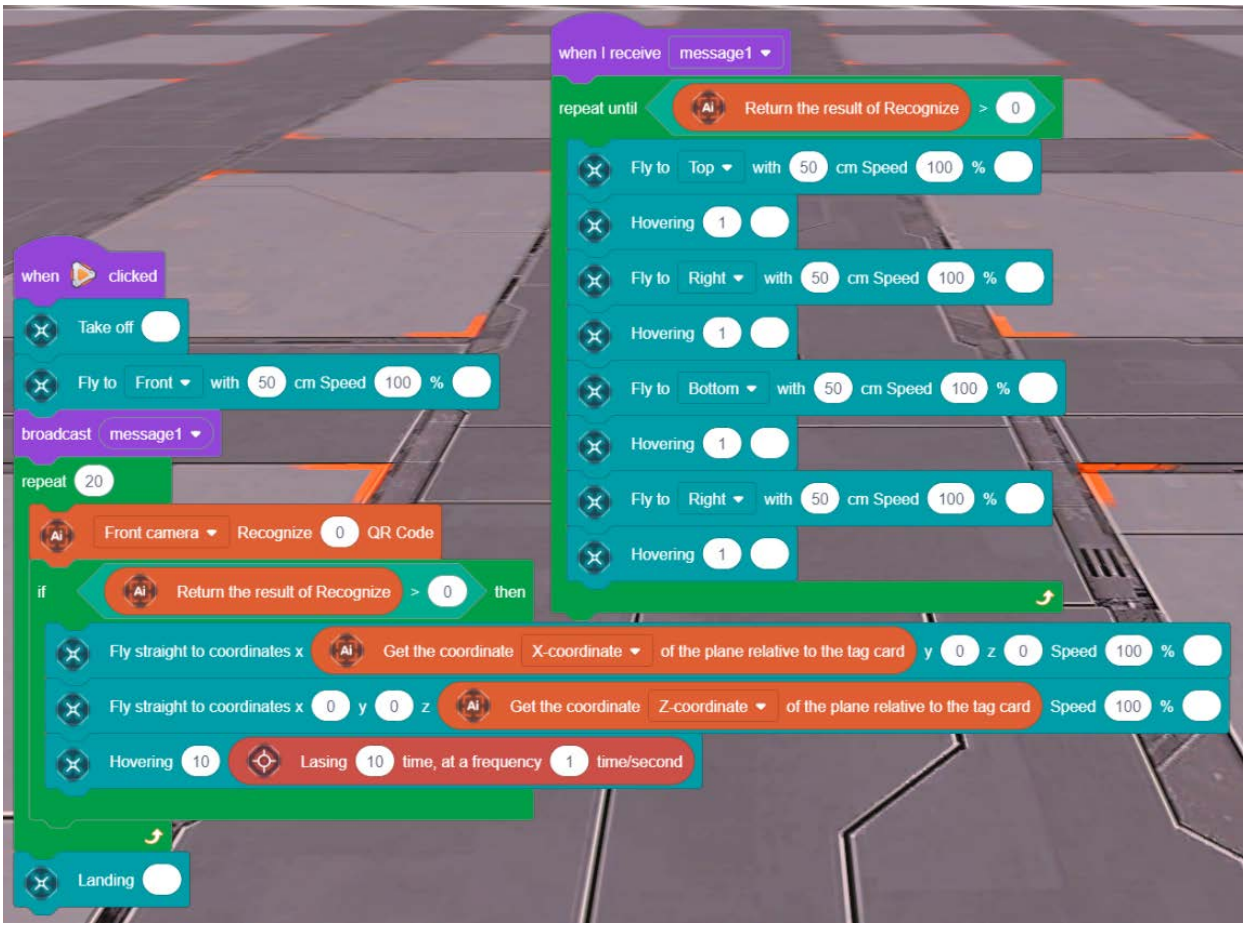
Take off, broadcast message 1, trigger the patrol thread, when the two-dimensional code is recognized, the aircraft executes the straight flight command. The patrol thread flight action will be interrupted because the new flight command overwrites the old flight command.



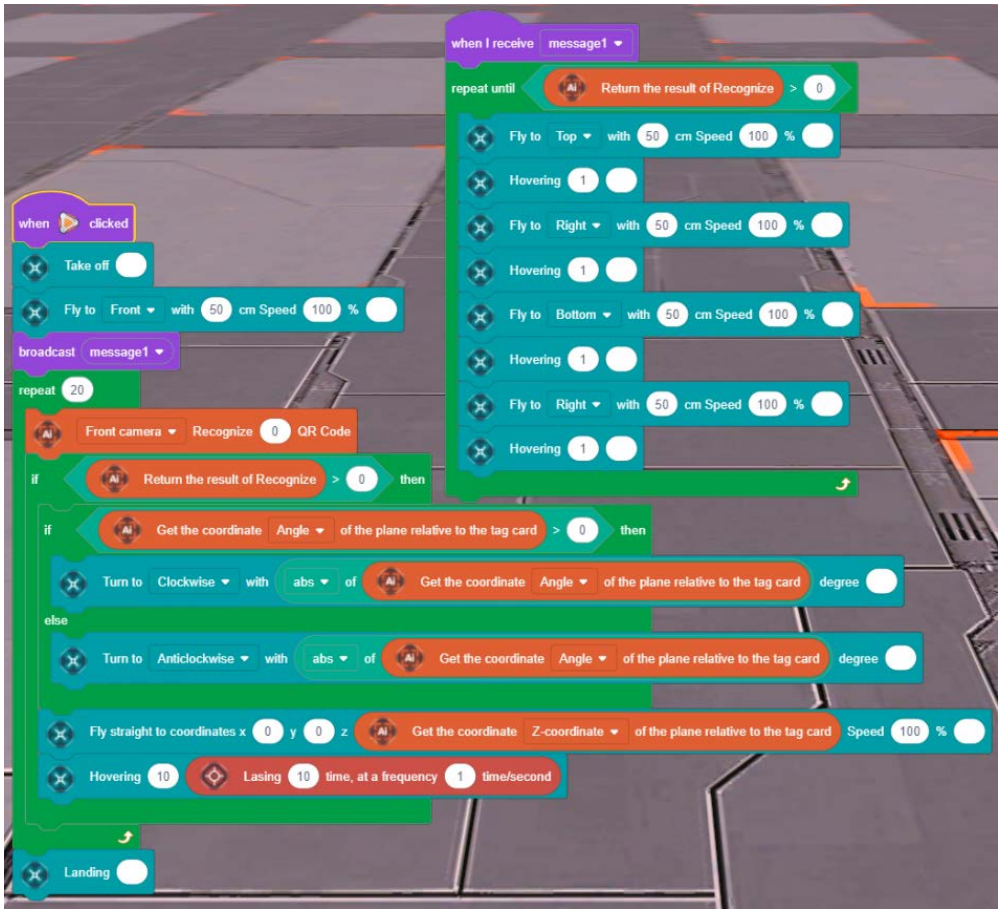
Angle between the head and the positive direction of the QR code- [Optical flow] Recognize

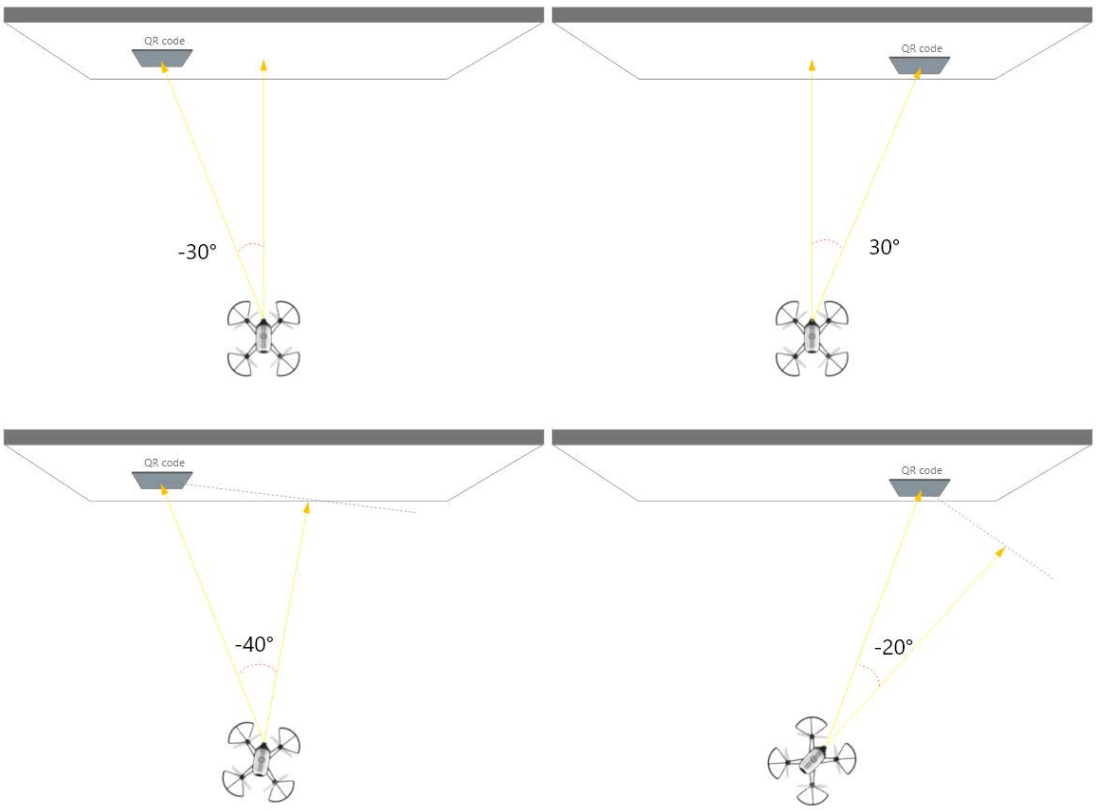
[Front camera] Recognize QR code [0] and align and fire:





[Front camera] Recognize QR code [0] and aim and fire:





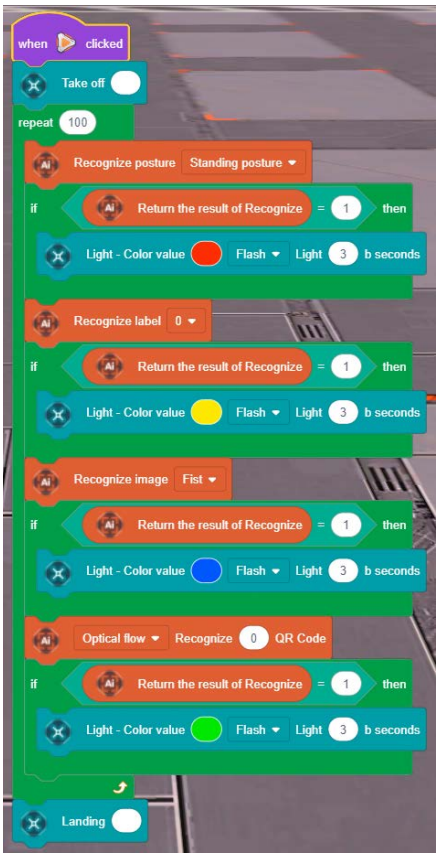
Angle between the head and the QR code-[Front camera] Recognize

When holding the QR code, do not block the outer white edge of the QR code with your finger, which will lead to failure to identify it.

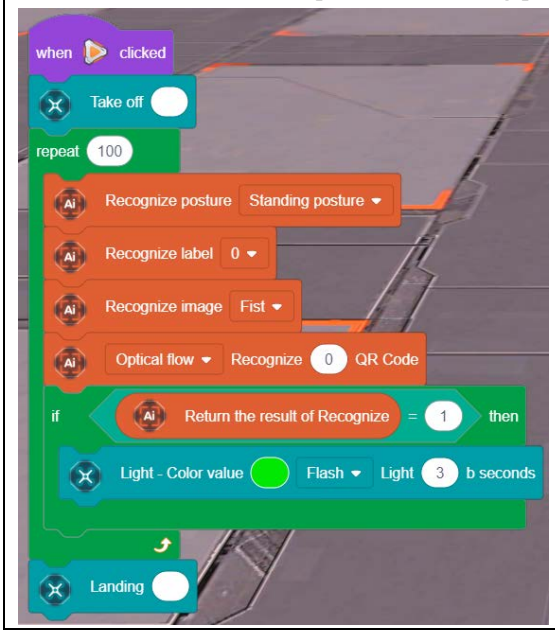


Return the result of Recognize

1. Name: Return the result of Recognize
2. Function: It is an instruction for returning the result of recognition. It can return the recognition results of labels, images, postures, and QR codes. If they are recognized successfully, it returns the value "1". If no recognition occurs or recognition fails, it returns the value "0". It is used in conjunction with the variable blocks and the conditional blocks.
3. Type: parameter block, No simulation
4. Example: Get the recognition results of labels, images, postures, and QR codes respectively, and based on the results, flash different colored lights.



Return results. For each recognition block, there is a corresponding return result. They are typically used in pairs. If multiple recognition blocks are executed sequentially, only the recognition result from the last block will be returned. For example, the following program only returns the result of identifying QR code number 0.



Return a RGB of Color Recognition

1. Name: Return a RGB of Color Recognition
2. Function: It is an instruction for returning a RGB of color recognition. It will activate the front camera to get the RGB value of the largest colored area within the forward image transmission, excluding black, white, and gray regions. This is used for color recognition and can be used in conjunction with the variable instructions or the light instructions. RGB values are arrays and cannot be used for operational programming. This function operates on the aircraft motherboard without the need to turn on the image transmission, and the proactive image transmission is used to see if the target is out of the image transmission range and cannot be recognized.
3. Type: parameter block, No simulation
4. Example: Recognize the color and flash the light of that color.

Color recognition requirements: It requires adequate indoor lighting, a significant contrast between the target color and its background, and the absence of other large areas of interference colors. Targets with high color saturation, non-reflective material and non-luminous targets have high recognition accuracy.

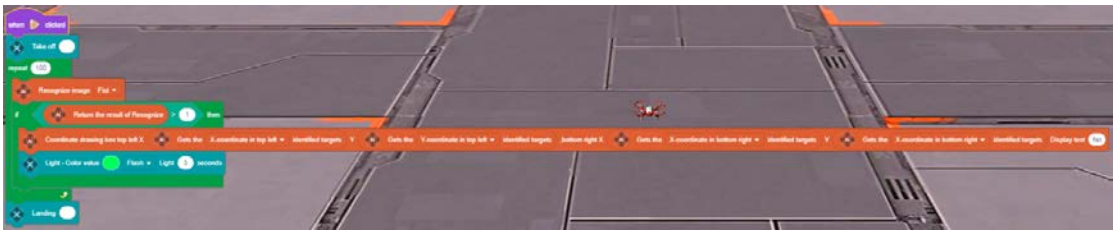


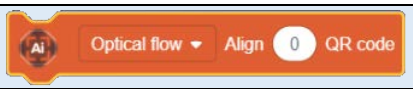
Note: The second light command, upon recognition, is immediately skipped and overwritten by the first light

command. If you want the second command to light up for a few more seconds, you can add waiting or hovering blocks.



1. Name: Gets the [X-coordinate in top left] identified targets
2. Function: Before doing identification, we need to open the image transmission,It is an instruction for getting the image transmission [coordinates] of identified targets, including images. Available coordinates include [X-coordinate in top left/Y-coordinate in top left/X-coordinate in bottom right/Y-coordinate in bottom right]. After obtaining the coordinates, with the image drawing frame building blocks, you can select the area of recognition target in the image transmission frame . If there are multiple targets recognized, return the coordinates of the target with the largest absolute values of X and Y. It is usually used in conjunction with the coordinate drawing box block.
3. Type: parameter block, No simulation
4. Example: Get the [top-left X/Y coordinates] [bottom-right X/Y coordinates] of the identified target and draw a box to show "stance"



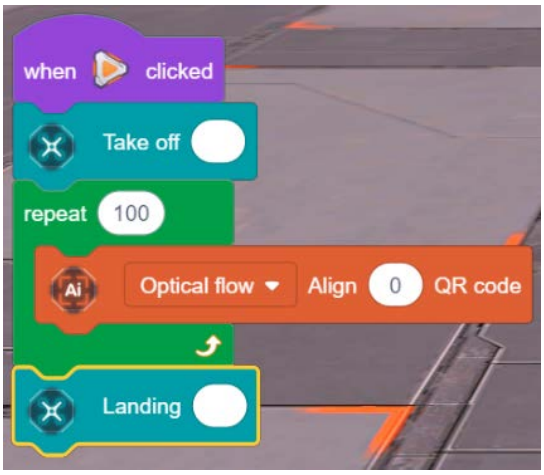


1. Name: [Optical flow] Align [0] QR code
2. Function: It is an instruction for aligning the QR code. [Optical flow] refers to aligning with QR codes using the optical flow camera underneath the aircraft. It will make the aircraft hover above the QR code and align the aircraft's orientation with the QR code's orientation. [Forward shooting] alignment refers to using the front camera of the aircraft to align the two-dimensional code, automatically adjust the aircraft head, and slowly fly left or right to display the two-dimensional code in the aircraft image transmission center. The running time is 300ms, that is, whether the identification is successful or failed within 0.3 seconds, it is skipped, and it is generally used in cycles. This function operates on the aircraft motherboard without the need to turn on the image transmission, and the proactive image transmission is used to see if the target is out of the image transmission range and cannot be recognized.
3. Type: execution block
4. Example:

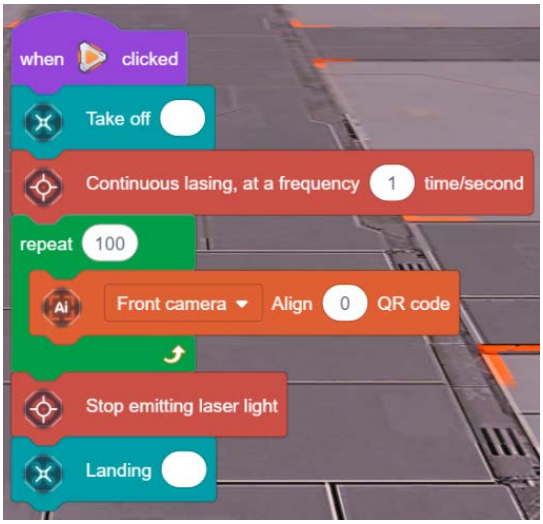
Recognition requirements: It requires the aircraft to hover at a height of [1~2] meters, the image transmission to be enabled, adequate indoor lighting, a distance from the target within [1~2.2] meters, clear visibility of the complete target, absence of backlighting, and a distraction-free solid-color background. Under these conditions, the recognition success rate is high.

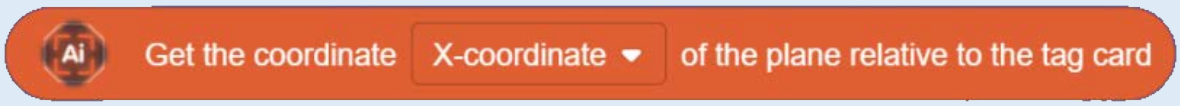
Optional alignment of 0-9 QR codes for single recognition. To ensure recognition efficiency, please do not have more than 3 QR codes in the aircraft visual range at the same time.

Optical flow Align QR code， and landing.



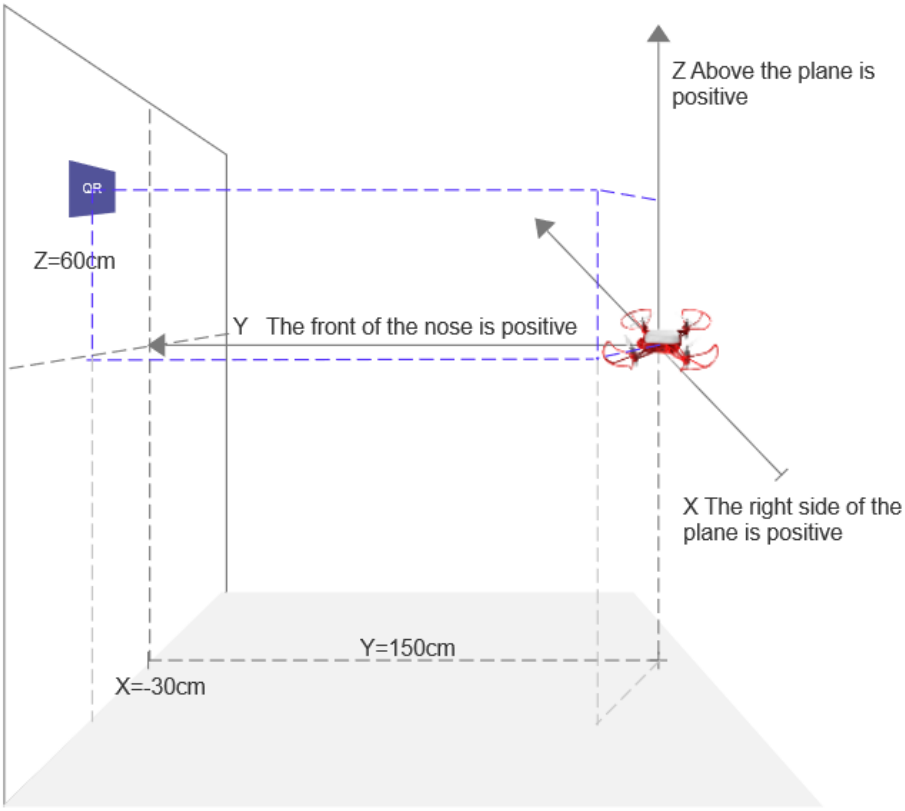
[Front shot] Align the QR code in front and fire the laser



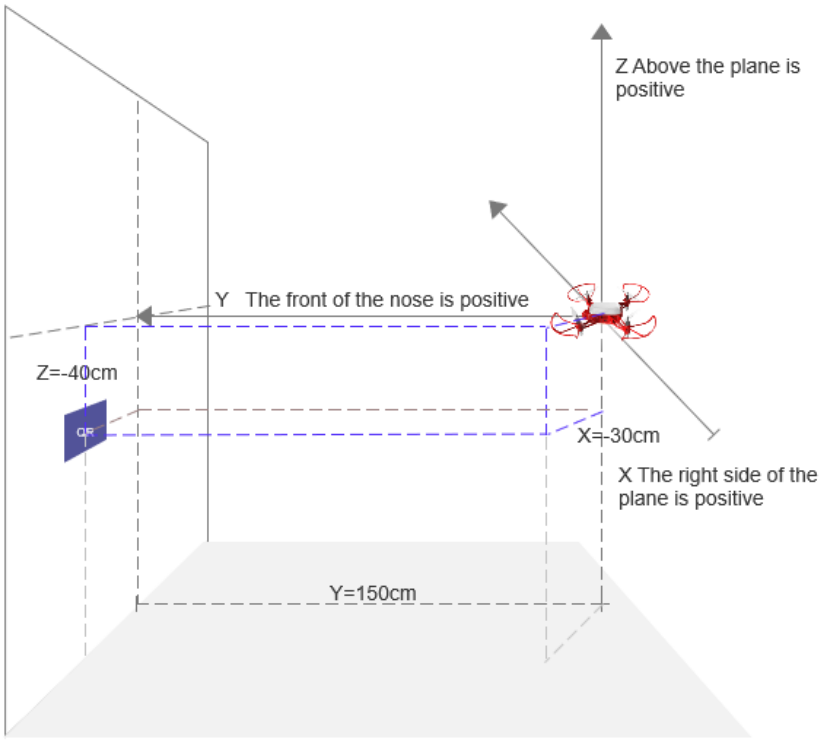


1. Name: Get the [coordinate X] of the drone relative to the tag card
2. Function: It is an instruction for getting the [coordinates] of the tag card in relation to the aircraft. It is used in conjunction with the tag card recognition blocks. With the aircraft as the origin, it can get the [X-coordinate, Y-coordinate, Z-coordinate, angle] of the tag card in relation to the aircraft. Parameters of tag cards on the ground are obtained through the [Optical flow], and parameters of tag card on the wall are obtained through the [Front camera]. This function runs on the aircraft motherboard and can be used without opening the image transmission.
   Tag card Including: QR code, Arrows, Numbers
3. Type: parameter block, No simulation
4. Example:

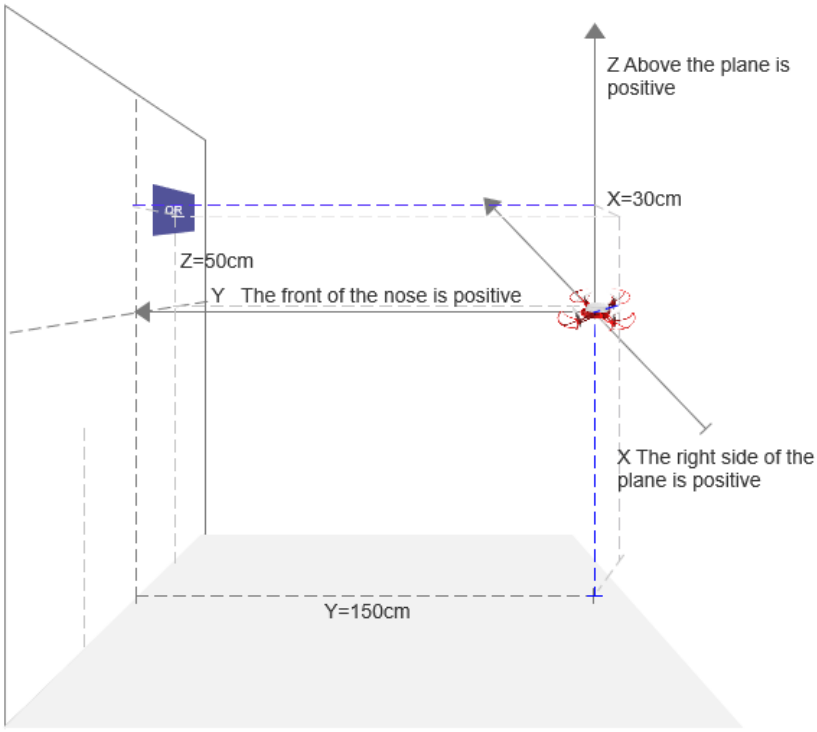
[Front shot] Identification of QRcode, aircraft relative QRcode diagram:



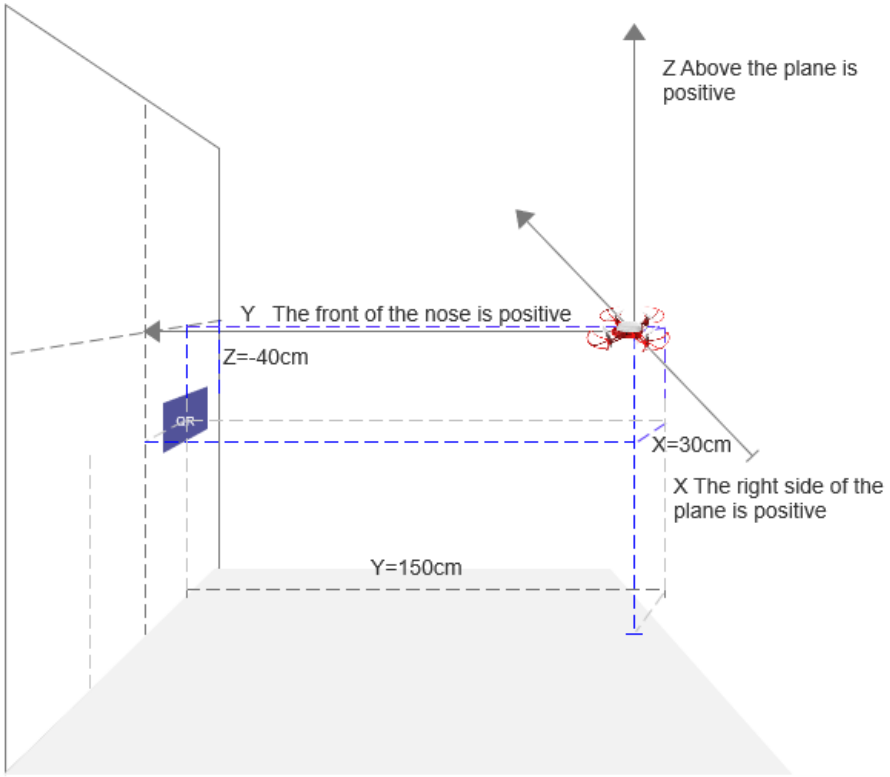
The QR code is at the top left of the plane
xyz (-30,150,60)



The QR code is at the bottom left of the plane
xyz (-30,150,-40)



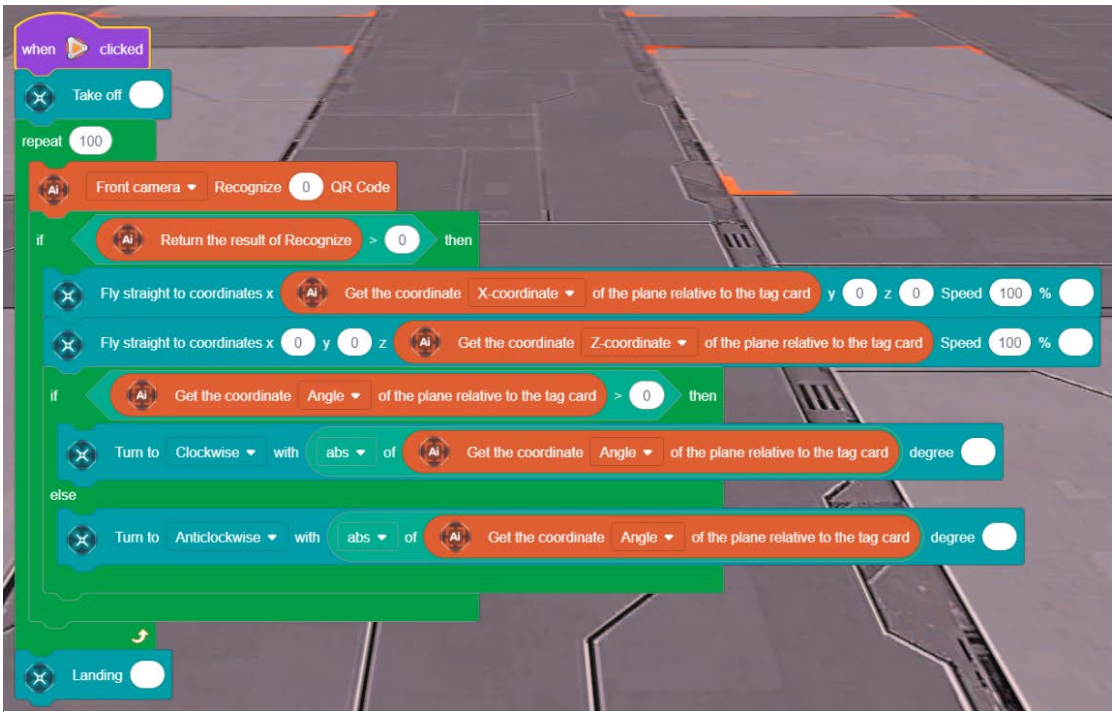
The QR code is at the top right of the plane
xyz (30,150,50)



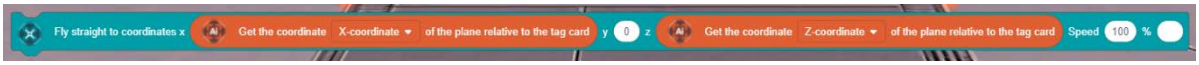
The QR code is at the bottom right of the plane
xyz (30,150,-40)

[Front camera] Recognize QR code [0], and get the [X-coordinate, Y-coordinate, Z-coordinate, angle] of the tag card in relation to the aircraft, and make the aircraft align with the QR code.

In relative coordinate system:

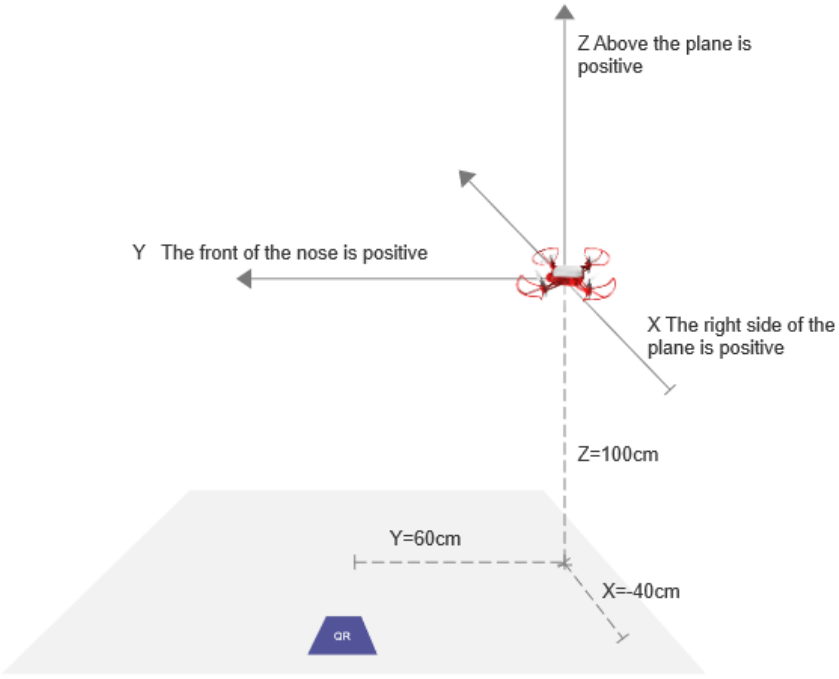


Compare the two programming options for flying straight to xyz to fully understand the relative coordinate system.

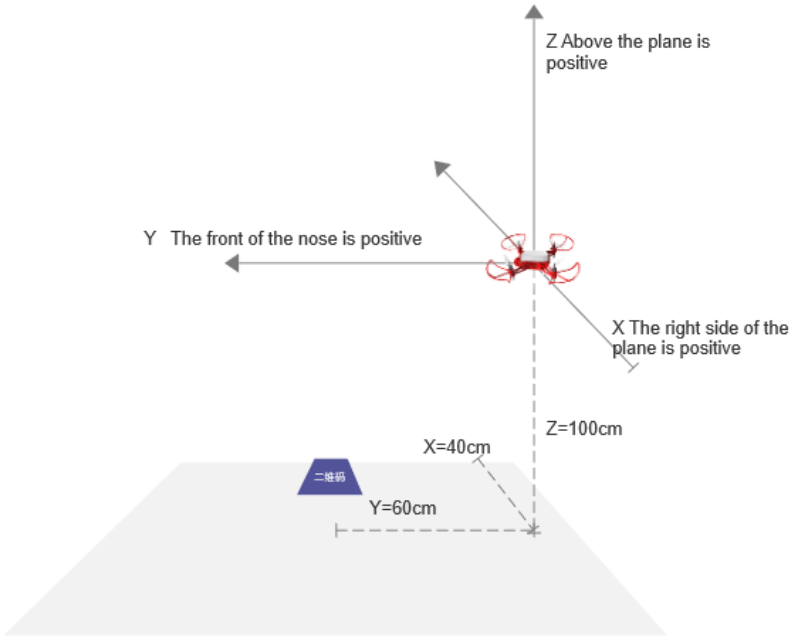


Absolute coordinate system, two-dimensional code carpet positioning does not support QR code alignment function.

[optical flow] Identification of QRcode, aircraft relative to QRcode coordinate diagram:



The QR code is on the front left side of the plane

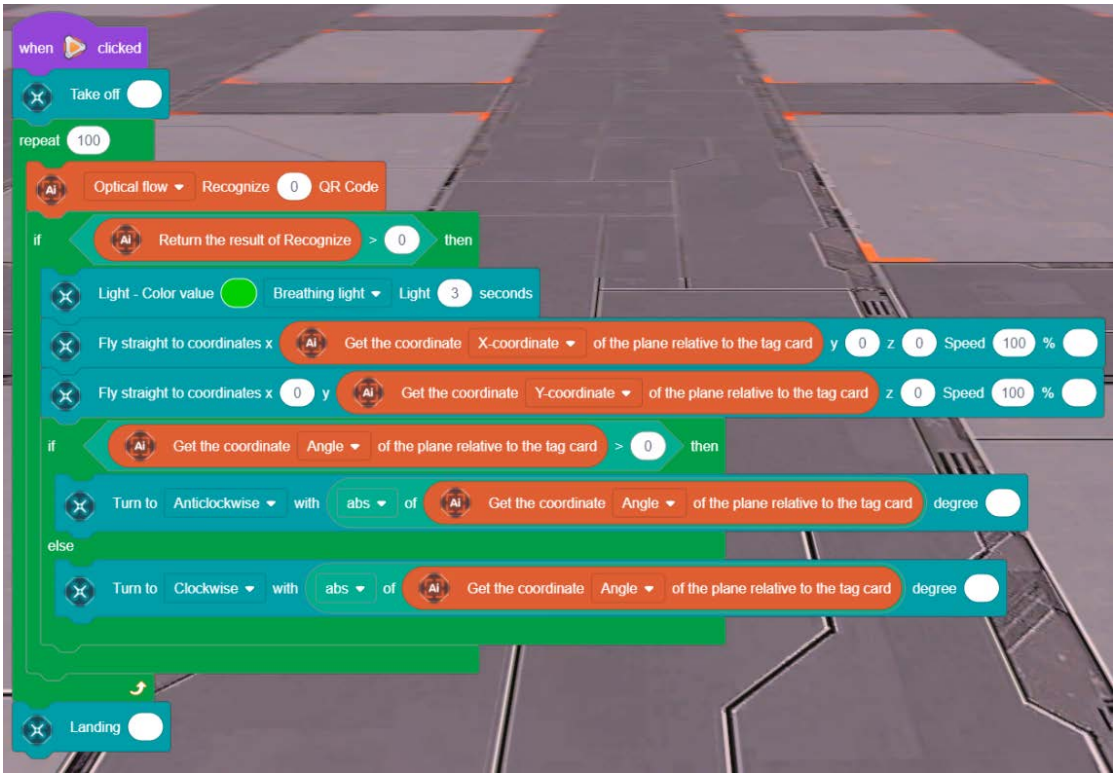


The QR code is right front of the plane

In the above figure is 60cm away from the aircraft, and the optical flow camera will not see the QR code. It is necessary to place the QR code near the aircraft within 30cm or raise the aircraft to 2 meters.

[optical flow] identifies the QR code [0], obtains the tag card and the [X coordinates, Y coordinates, Angle] of the aircraft, and aligns the QR code with the aircraft.

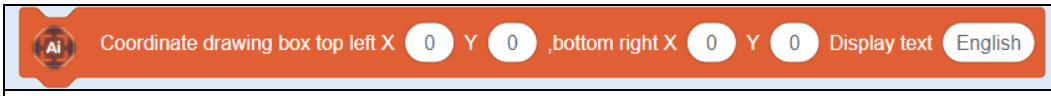
In relative coordinate system:



Compare the two programming options for flying straight to xyz to fully understand the relative coordinate system.



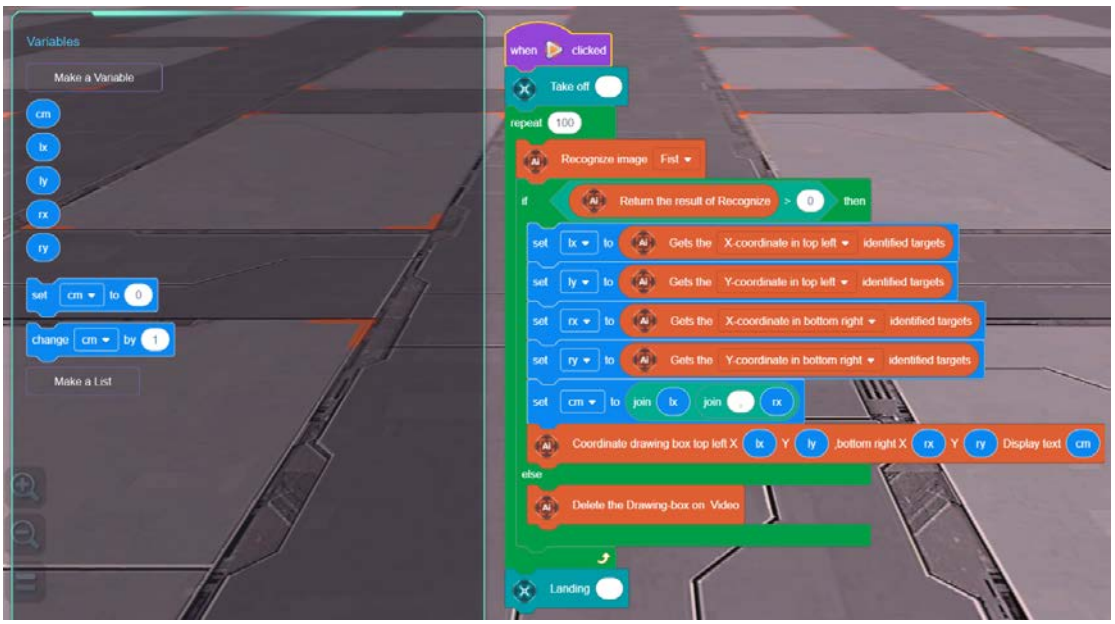
Absolute coordinate system, two-dimensional code carpet positioning does not support QR code alignment function.



1. Name: Coordinate drawing box top left [X] [Y], bottom right [X] [Y] Display text [En]
2. Function: It is an instruction to draw a box and display the text for the recognized target. Enter the coordinates, and at the corresponding position on the transmitted image, a box will be drawn along with the text displayed. After getting the coordinates of the images, it can draw a box and display the text for the recognized target. This function runs in the APP, you need to open the image transmission to use.
3. Type: execution block, No simulation
4. Example: Identify fist and frame it



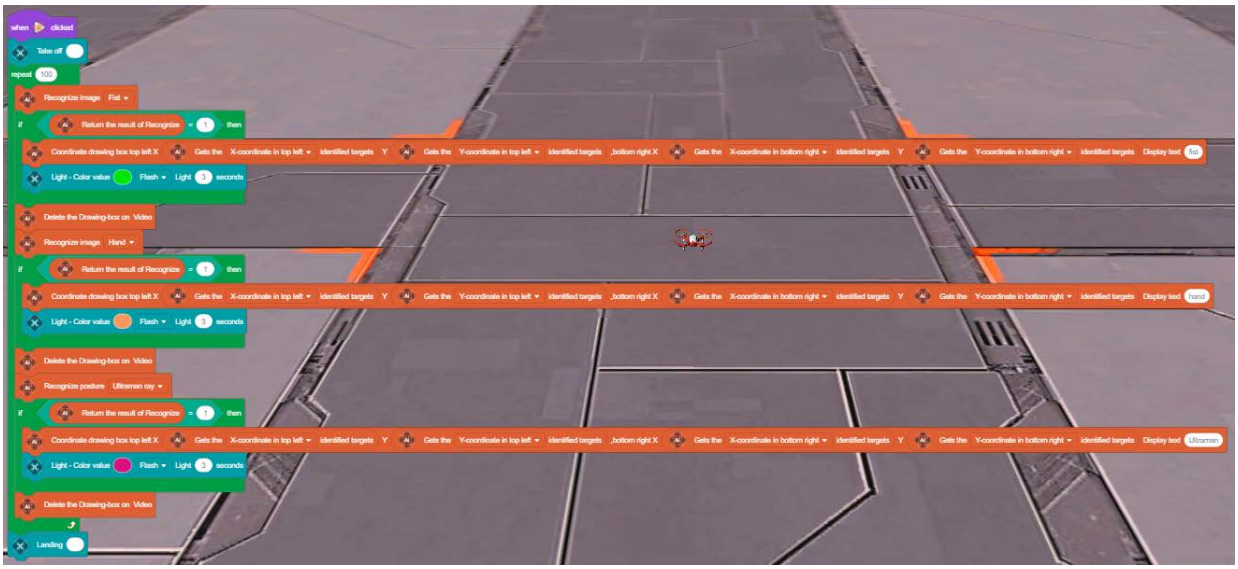
Draw a box for the target and show the x-coordinate of the upper left corner and the X-coordinate of the lower right corner of the target .





Delete the Drawing-box on Video

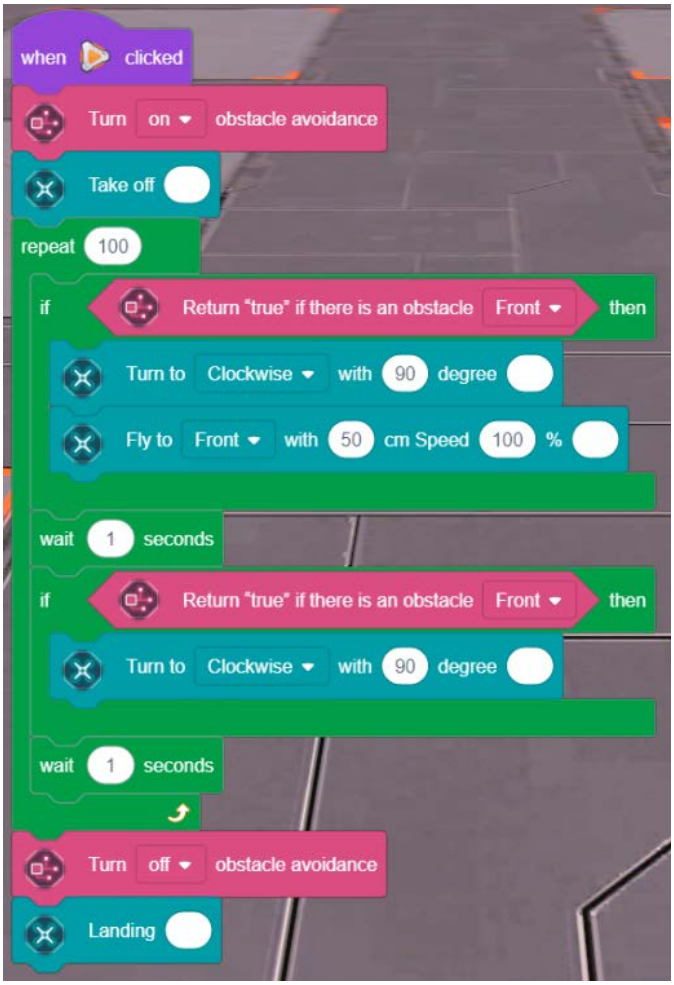
1. Name: Delete the Drawing-box on Video
2. Function: It is an instruction for deleting the drawing box on video. It will delete the drawing box of the corresponding coordinates on the video image. This function runs in the APP, you need to open the image transmission to use.
3. Type: execution block, No simulation
4. Example: Identify different targets, delete the drawing frame of different targets, and draw a new drawing frame.



8.2.8 Control



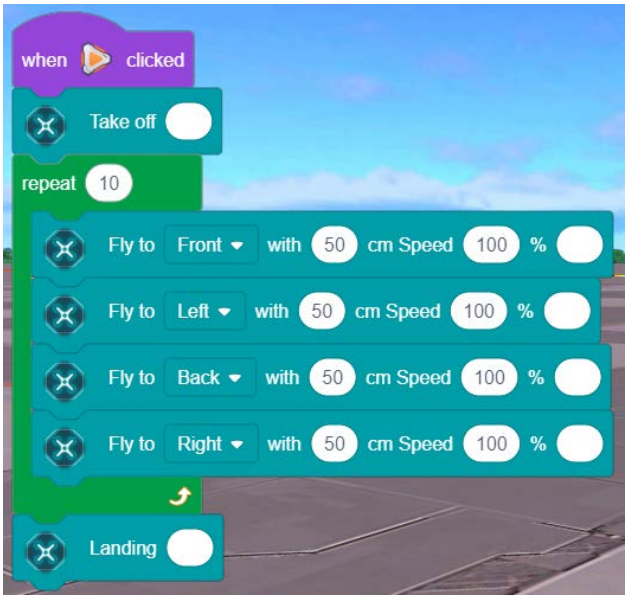
1. Name: wait [1] seconds
2. Function: It is an instruction for waiting. Wait for the specified number of seconds before executing the next instruction.
3. Type: execution block
4. Example: Wait until obstacle avoidance feedback is complete.

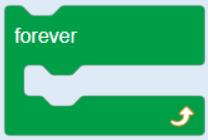




1. Name: repeat [10]
2. Function: It is an instruction of a finite loop. It will repeat the internal program a certain number of times.
3. Type: execution block

4. Example: Repeat square flight 10 times.



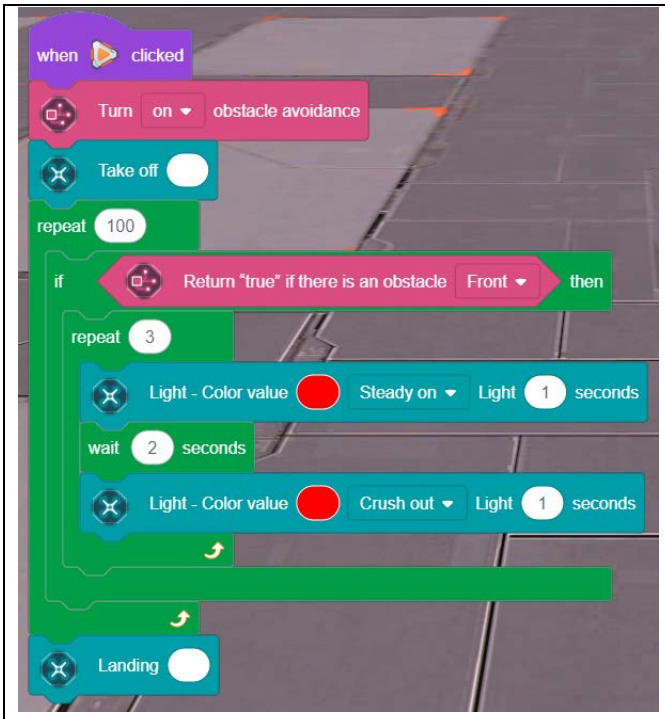


1. Name: Forever
2. Function: It is an instruction of an infinite loop. It will repeat the internal program forever.
3. Type: execution block
4. Example: forever One-way lasing

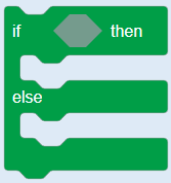




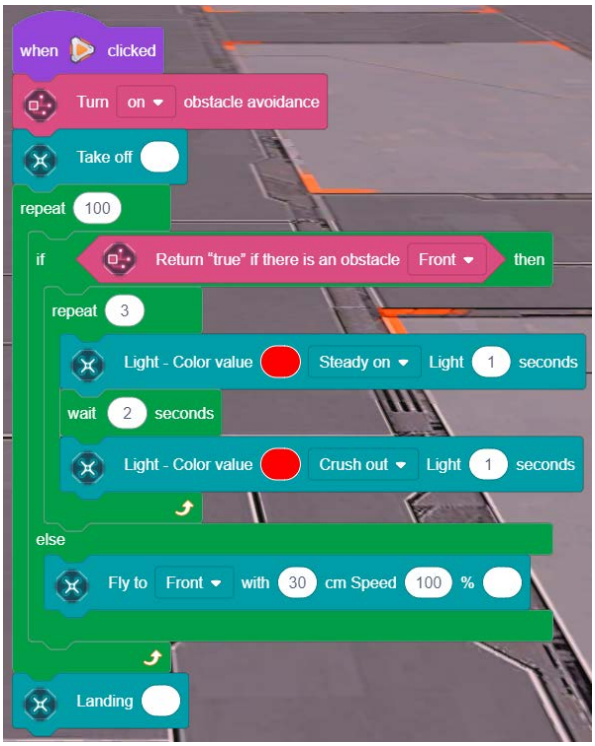
1. Name: if [] then
2. Function: It is an instruction to determine whether the condition is met. If it is, then execute the internal program.
3. Type: conditional block
4. Example: If there is an obstacle in front, then flash the red light three times.



Note: The light on command will be quickly overwritten by the light off command, and the aircraft will appear to be fast 3 times. If you need to extend the light on time, you need to add waiting or hovering before the light off command.

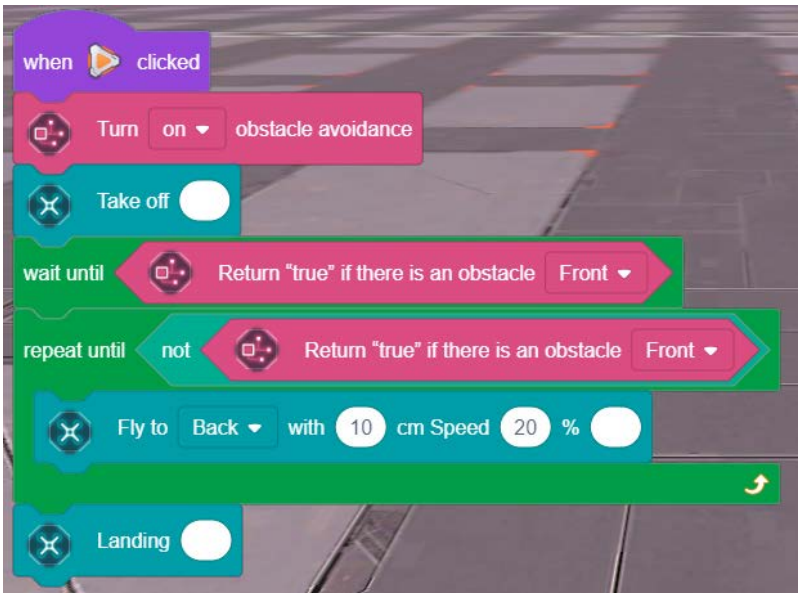


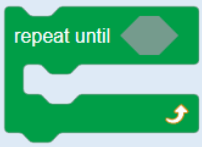
1. Name: if [] then, else
2. Function: It is an instruction to determine whether the condition is met. If it is, execute the internal program of the "then". If not, execute the internal program of the "else".
3. Type: conditional block
4. Example: If there is an obstacle in front, then flash the light three times, else fly forward by 30 cm.



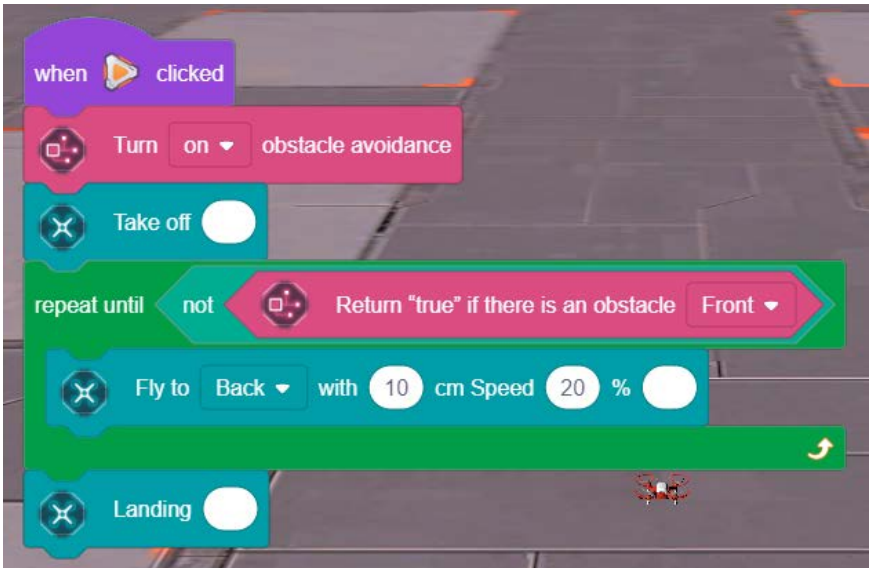
wait until

1. Name: Wait until [the condition is met]
2. Function: It is an instruction to determine whether the condition is met. It means the program will pause and wait for the specified condition to become true. Once the condition is met, the program will continue executing the subsequent instructions.
3. Type: conditional block
4. Example: Wait until an obstacle is detected ahead and continue with the procedure below.





1. Name: repeat until []
2. Function: It is an instruction to determine whether the condition is met. It will repeatedly execute the internal program until the condition is met, then exit the loop and proceed to the next instruction.
3. Type: conditional block
4. Example: Fly back 10cm at 20% speed until no obstacle is detected.





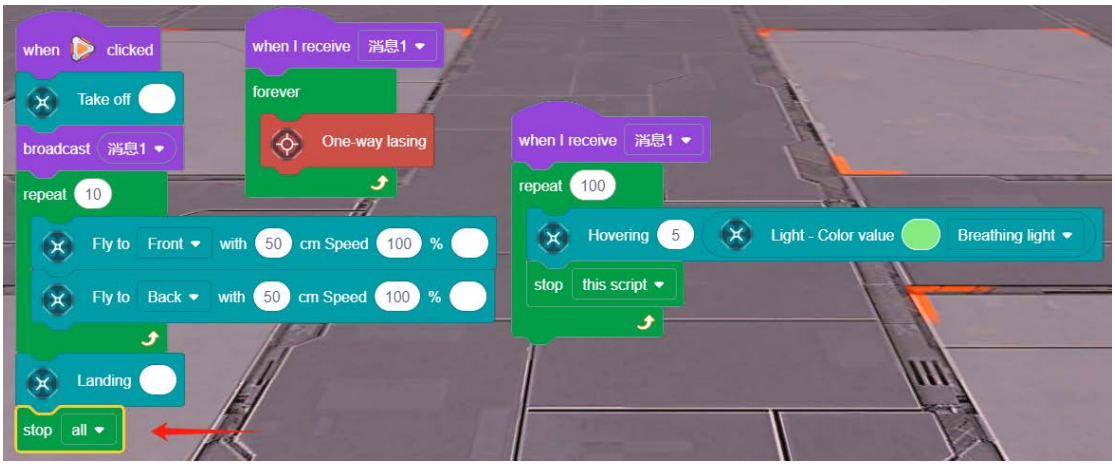
1. Name: stop [all]
2. Function: It is an instruction to stop all running programs. It will stop all the programs that are running. Available options are stopping the current script, all scripts, or other scripts associated with the character.

Stop [all scripts] : terminates the entire program execution and completely exits the program;
Stop [this script] : terminates the execution of the current code group corresponding to this role (the code group where this block resides) without affecting the execution of other code groups of the current role;
Stop [other scripts of this role] : Terminates the execution of other code groups of this role without affecting the execution of the current code group (the code group in which the block resides).

3. Type: execution block
4. Example: Broadcast message 1, trigger the multithreaded execution program, stop hovering for 5 seconds and light the breath light script .



In the above case, hover and light the green breath light, run only once and be stopped [this script] command is terminated, unlimited single laser will continue to run, if you need to stop unlimited laser emission, you can use stop [all scripts] to stop all threads.



8.2.9 Operators



1. Name: Sum of two numbers
2. Function: It is an instruction for adding two numbers, resulting in a sum as the variable.
3. Type: parameter block
4. Example: Arithmetic operations.





1. Name: Difference of two numbers
2. Function: It is an instruction for subtracting two numbers, resulting in a difference as the variable.
3. Type: parameter block
4. Example: Arithmetic operations.





1. Name: Product of two numbers
2. Function: It is an instruction for multiplying two numbers, resulting in a product as the variable.
3. Type: parameter block
4. Example: Arithmetic operations.





1. Name: Quotient of two numbers
2. Function: It is an instruction for dividing two numbers, resulting in a quotient as the variable.
3. Type: parameter block
4. Example: Arithmetic operations.





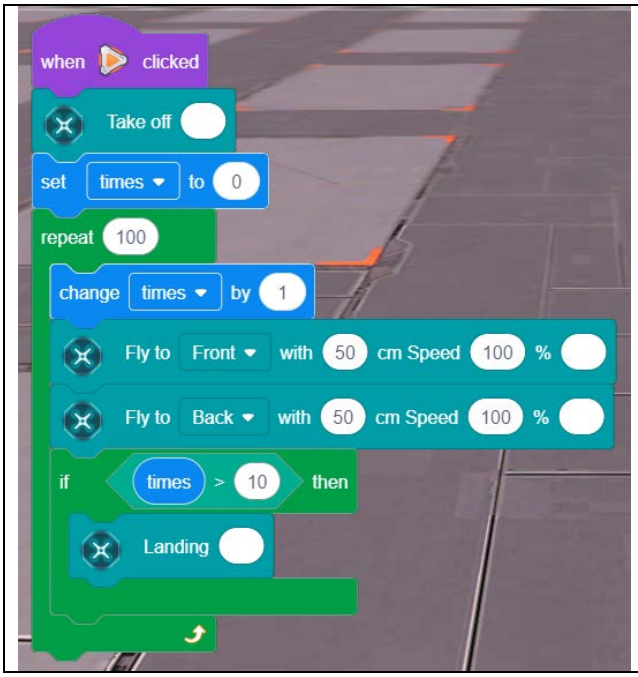
1. Name: Random picker
2. Function: It is an instruction for picking a random number within the range of two variable values.
3. Type: parameter block
4. Example: Arithmetic operations.

Example: If both the input endpoint values are integers, the output values will be integers as well.
If either of the endpoint values is a decimal (regardless of whether it is the starting or ending value), the output values will also be decimals.



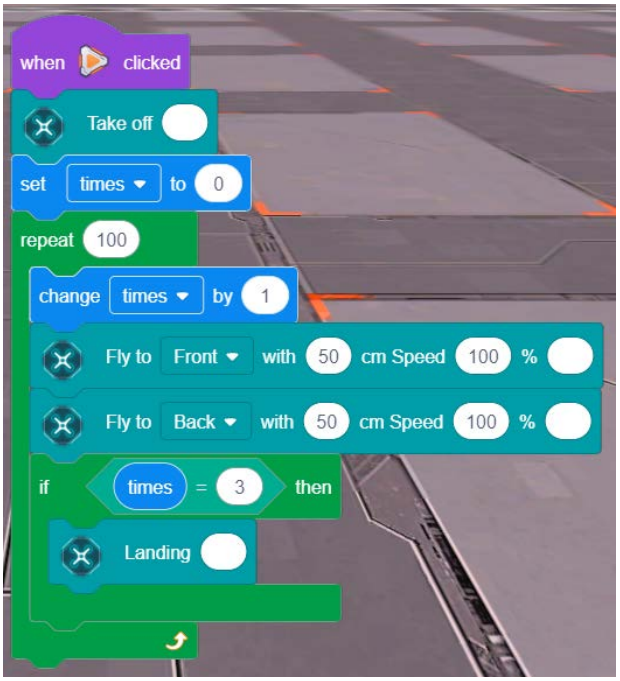


1. Name: Greater than
2. Function: It is an instruction used to determine whether the first value is greater than the second value. If it is, it will return "True". If not, it will return "False".
3. Type: conditional block
4. Example: If the number of round-trip flights is greater than 10, then the aircraft will land.



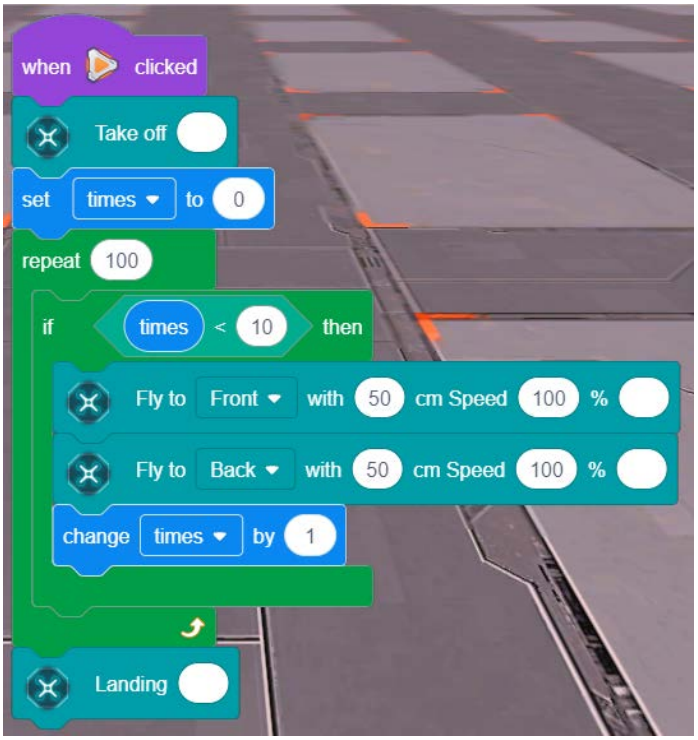


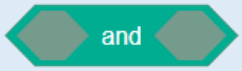
1. Name: Equal to
2. Function: It is an instruction used to determine whether the first value is equal to the second value. If it is, it will return "True". If not, it will return "False".
3. Type: conditional block
4. Example: If the number of round-trip flights is equal to 3, then the aircraft will land.





1. Name: Less than
2. Function: It is an instruction used to determine whether the first value is less than the second value. If it is, it will return "True". If not, it will return "False".
3. Type: conditional block
4. Example: If the number of round-trip flights is less than 10, then the aircraft will continue flying.



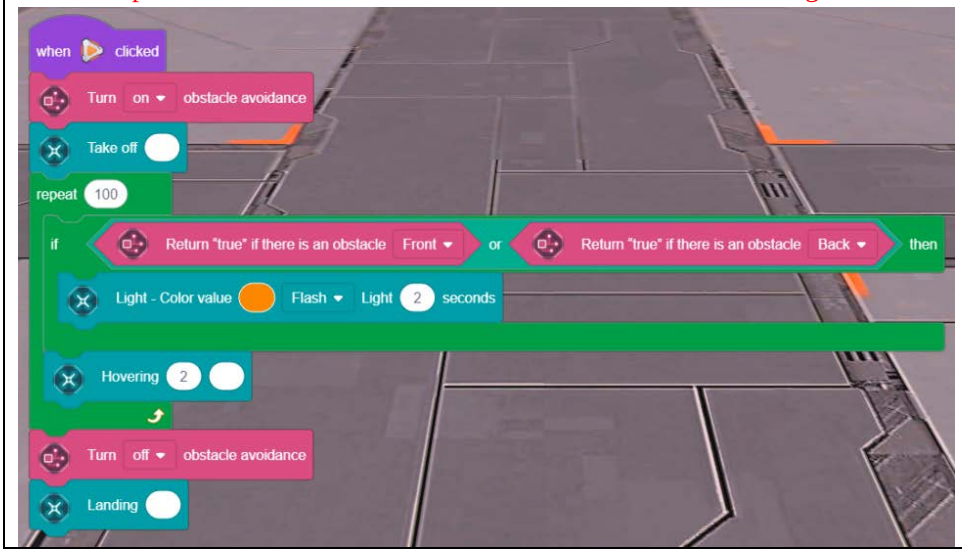


1. Name: and
2. Function: It is an instruction used to determine whether both conditions are met. If they are, it will return "True". If not, it will return "False".
3. Type: conditional block
4. Example: If there are obstacles both in front and behind, then flash the light.



or

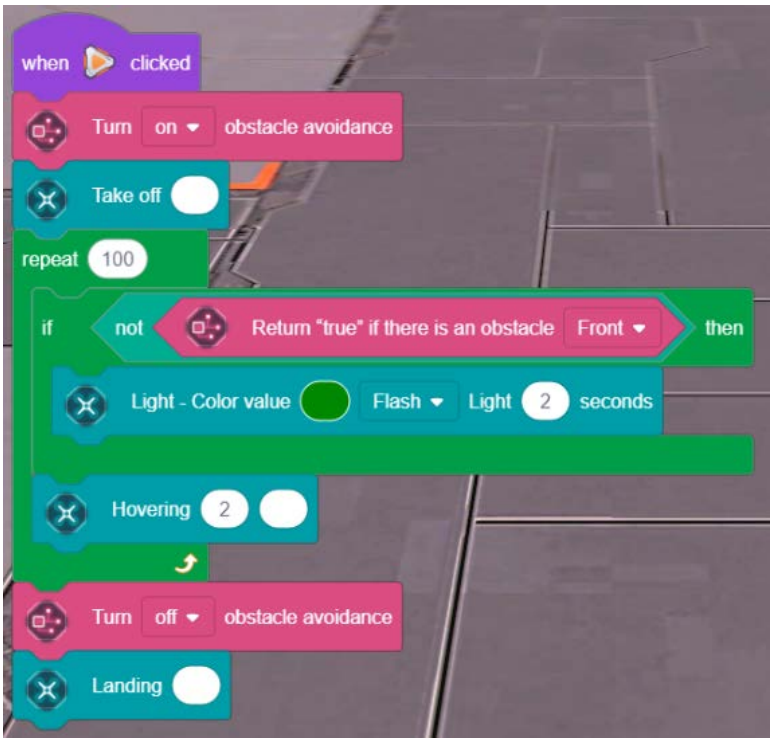
1. Name: or
2. Function: It is an instruction to determine whether the conditions are met. It returns "True" if either of the conditions is met, and returns "False" if neither of the conditions is met.
3. Type: conditional block
4. Example: If there are obstacles in front or behind, then flash the light.



not

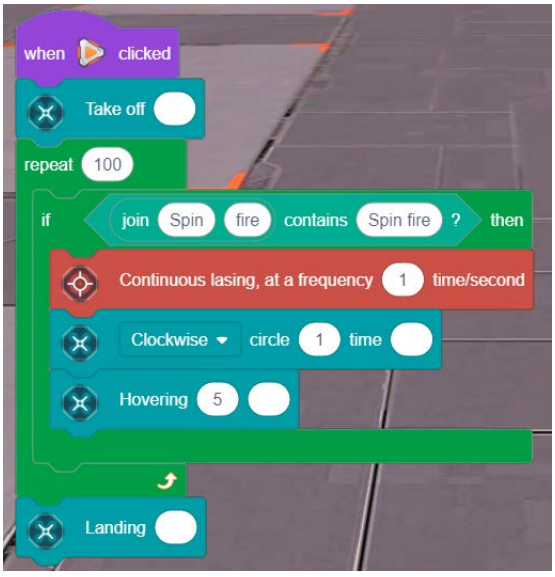
1. Name: not
2. Function: It is an instruction to determine whether the condition is met. It returns "False" if the condition is met, and returns "True" if the condition is not met.
3. Type: conditional block
4. Example: There is not an obstacle in front, flash the light.

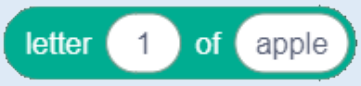
Note: AND, OR, and NOT are logical operators that will return a Boolean value that is "True" or "False".



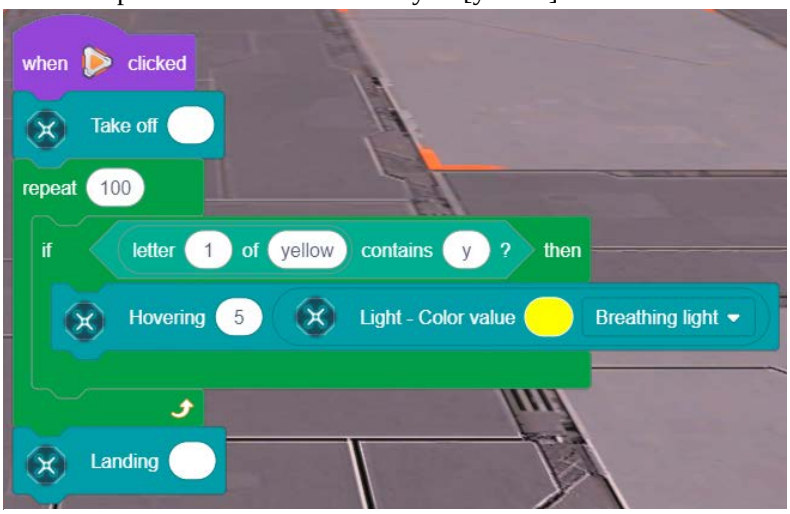


1. Name: join [letter A] and [letter B]
2. Function: It is an instruction for joining letters. It can get AB by joining [letter A] and [letter B].
3. Type: parameter block
4. Example: Join [Spin] and [fire] to get "Spin fire".



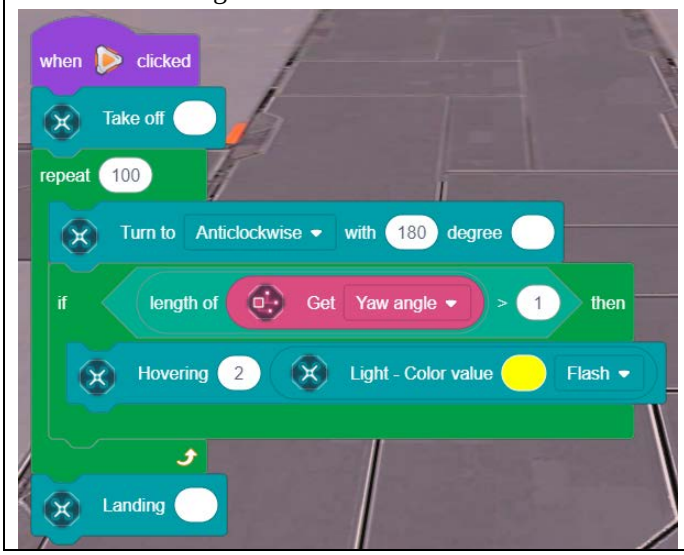


1. Name: letter 1 of [AB]
2. Function: It is an instruction for extracting the first letter A in [AB].
3. Type: parameter block
4. Example: Extract the first letter y of [yellow].



length of apple

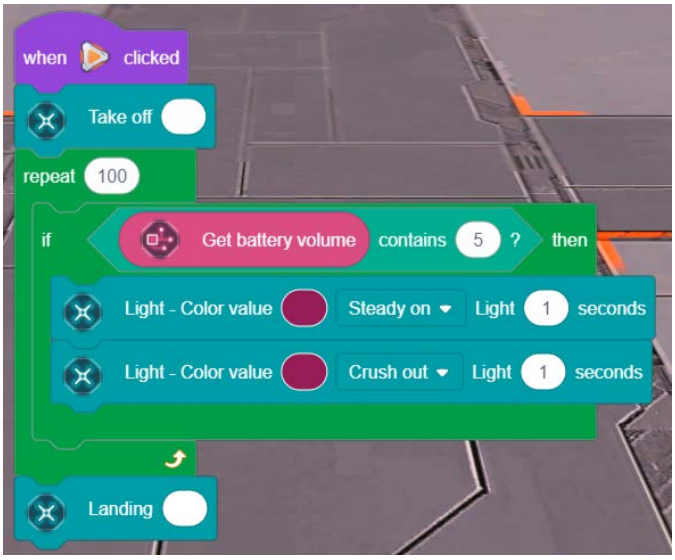
1. Name: Length of [AB]
2. Function: It is an instruction for calculating the number of letters. It calculates the total number of letters in [AB].
3. Type: parameter block
4. Example: Turn 180 degrees counterclockwise, the yaw Angle character is "-180", and the number of characters is greater than 1.



apple contains a ?

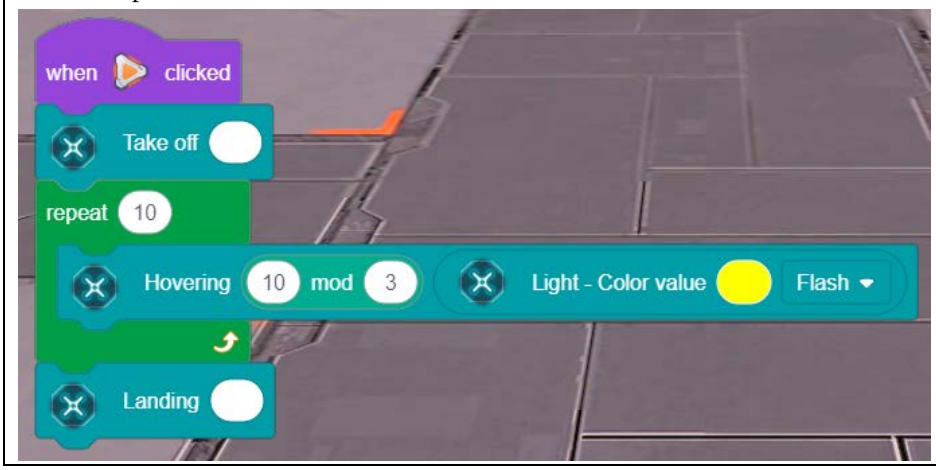
1. Name: [apple] contains [a]?
2. Function: It is an instruction used to determine whether some letter is contained. If it is, return "True".

3. Type: parameter block
4. Example: Judge the power value character, including 5 lights once?



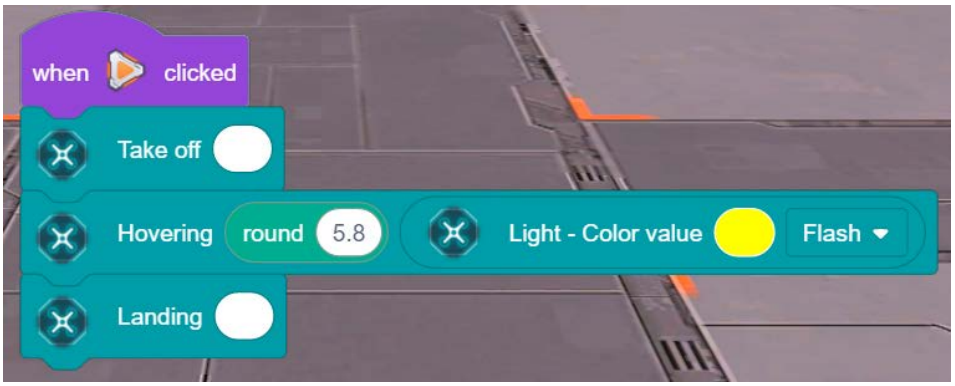
mod

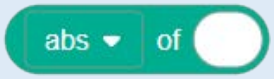
1. Name: mod operation
2. Function: It is an instruction for modulo operation. It is used to get the remainder when one number is divided by another.
3. Type: parameter block
4. Example: [10] mod [3] is "1".



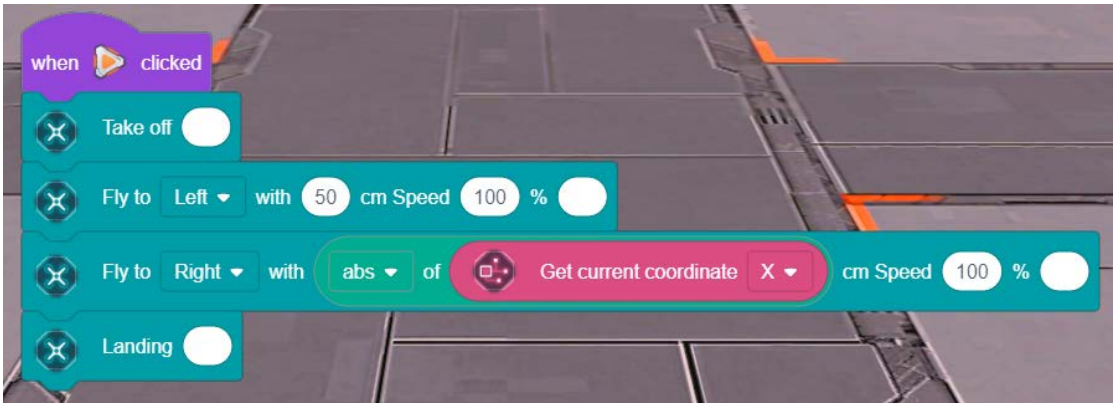
round

1. Name: rounding operation
2. Function: It is an instruction for rounding operations. Round [variable] to get its nearest integer.
3. Type: parameter block
4. Example: Round [5.8].





1. Name: [abs] operation
2. Function: It is an instruction for algebraic computation. It calculates the absolute value, the largest integer less than or equal to the real number, the smallest integer greater than or equal to the real number, the square root, the sine value, etc.
3. Type: parameter block
4. Example: [abs] of [X- coordinates]

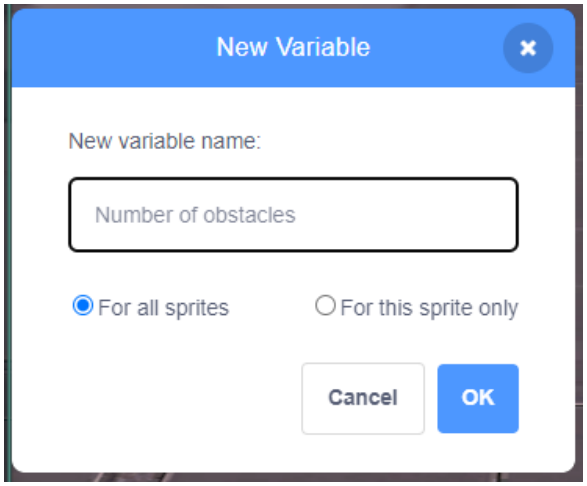


8.2.10 Variables

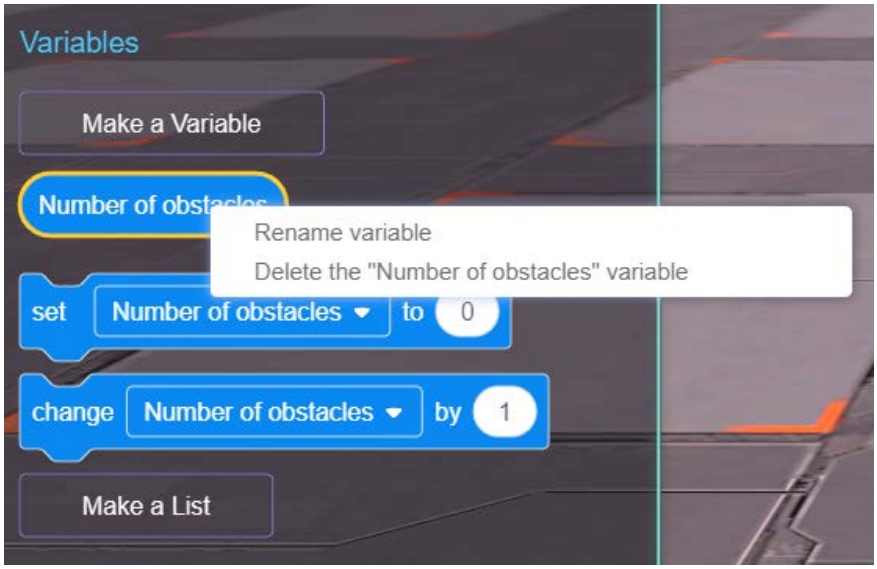
8.2.10.1 Create a variable



Make a Variable

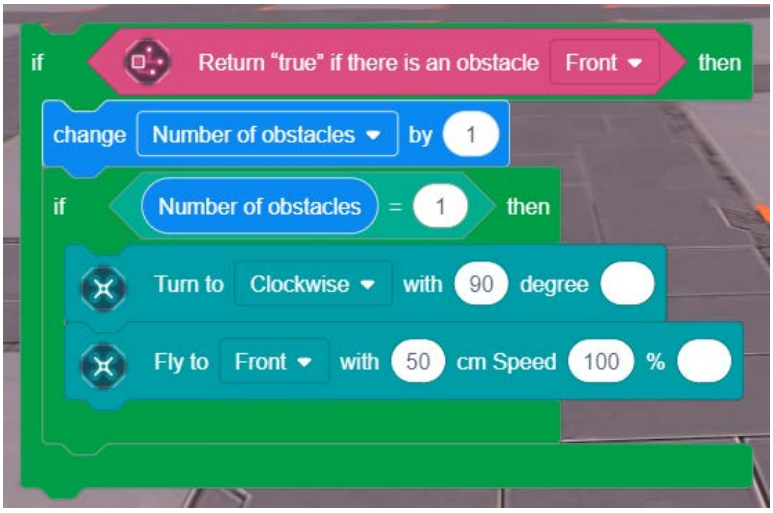


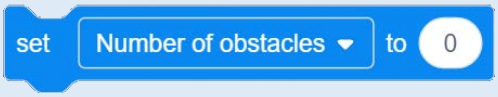
Variable names must be unique. In order to be self-explanatory, they are generally named based on the variable's function. For example, create a variable named "Number of obstacles". It is used to determine how the aircraft will respond when it comes to obstacles. After you create a variable, three related blocks will appear, respectively responsible for variable calling, assignment, and increment/decrement.



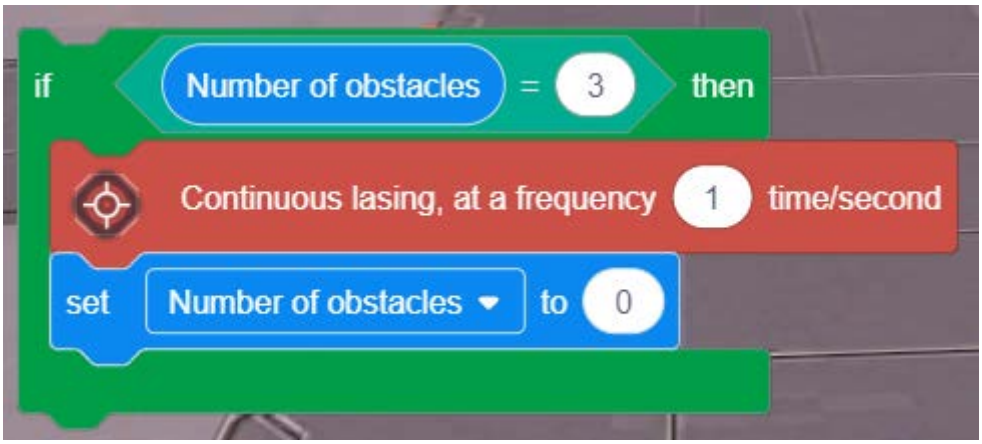
On mobile devices, tap and hold the variable name; on PC, right-click on the variable name to delete or rename it.

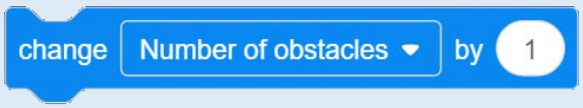
Number of obstacles
1. Name: Number of obstacles 2. Function: It is an instruction for getting the value of a variable. 3. Type: parameter block 4. Example: When the number of obstacles is 1, the drone turns 90° and. fly to front





1. Name: Set [Number of obstacles] to [0]
2. Function: It is an instruction for setting the variable to a specified value. This instruction assigns a specified value to the variable, and the variable stores the input value.
3. Type: execution block
4. Example: Recalculate the number of obstacles in front and set [Number of obstacles] to [0].

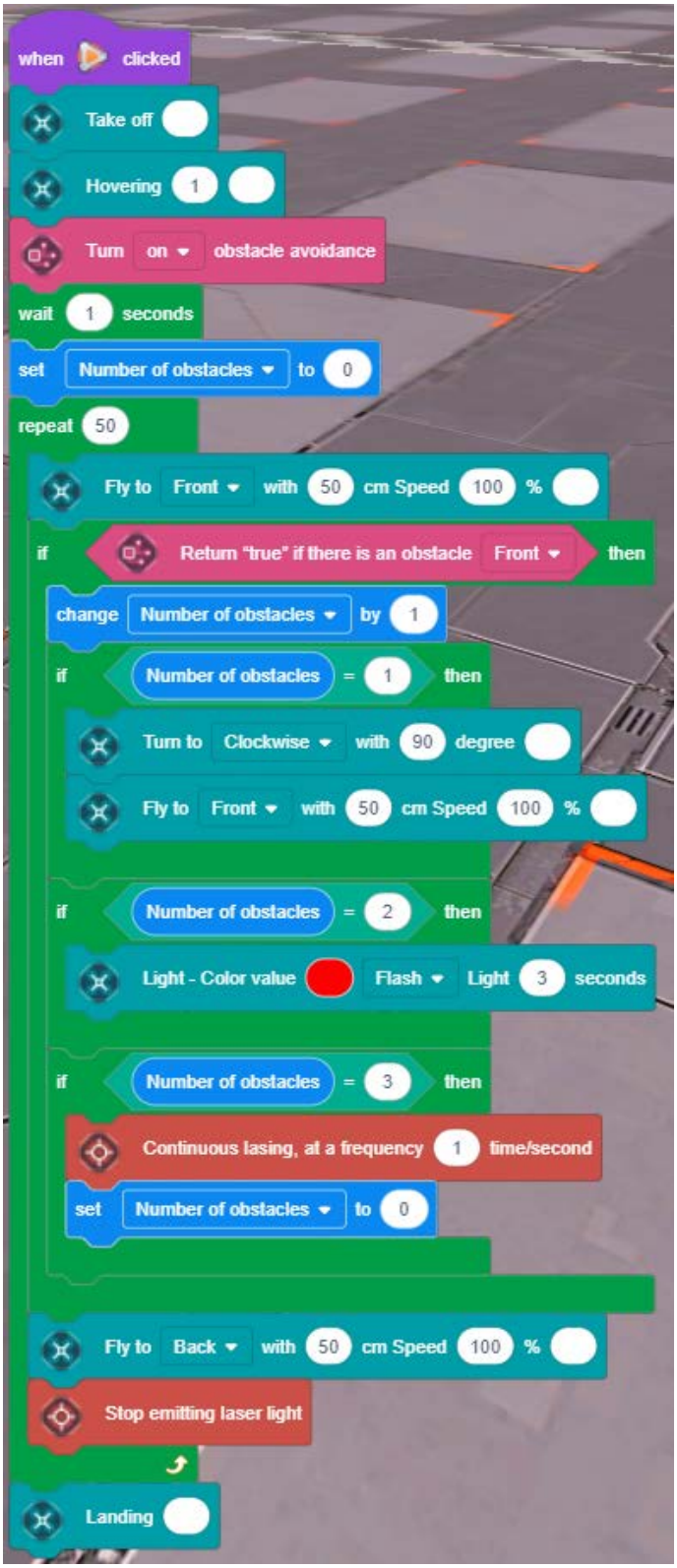




1. Name: change [Number of obstacles] by [1]
2. Function: It is an instruction for changing the value of the current variable. It changes the value of the current variable by an integer. Positive integer means increment, while negative integer means decrement.
3. Type: execution block
4. Example: Drone patrol

The drone executes this patrol instruction 50 times. During this period, you can bring obstacles close to the front of the drone.

When the drone encounters an obstacle for the first time, it turns around; for the second time, it flashes the light as a warning; for the third time, it opens fire.

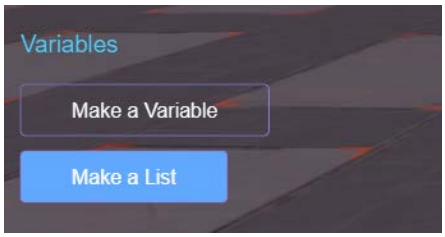


Note:

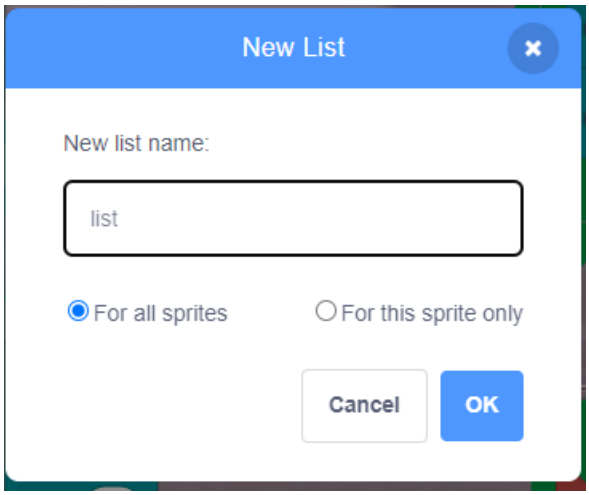
1) A variable stores a single data, while a list stores a series of data. They cannot be used interchangeably.

2) For variables, input values can be numbers, variables, or variable-type values, but not lists or list-type values.

8.2.10.2 Create a list



Make a list and fill in the name "list".



After you create a list, multiple related blocks will appear, respectively responsible for list calling, assignment, addition, deletion, and more.

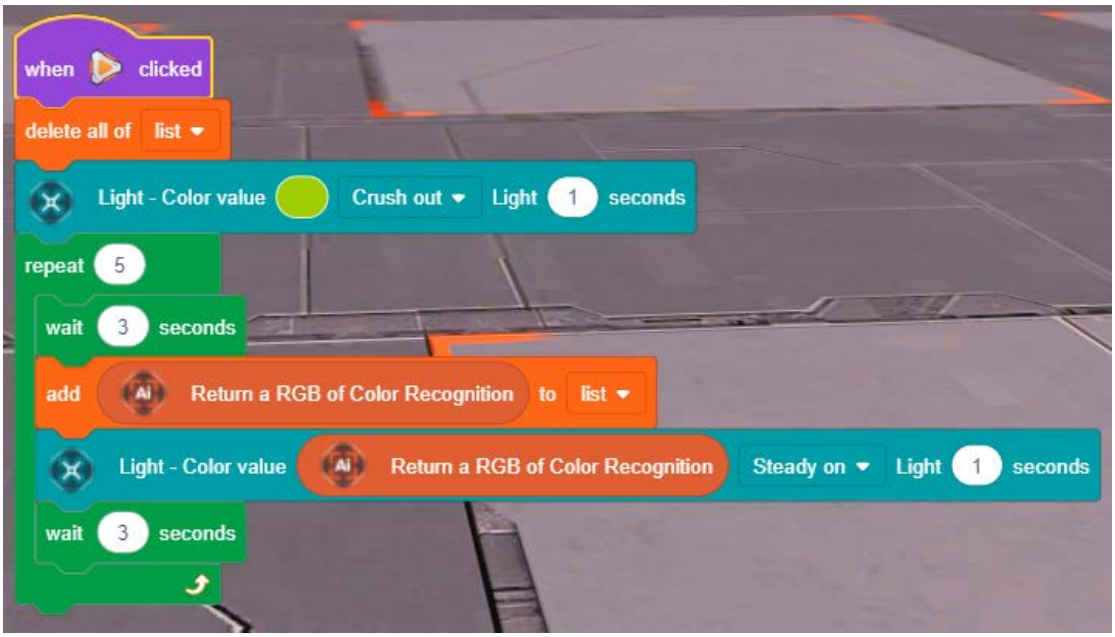


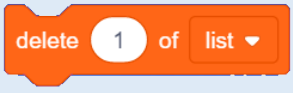
On mobile devices, tap and hold the list name; on PC, right-click on the list name to delete or rename it.

list
1. Name: list 2. Function: Invokes the whole list data directive, which can be used to copy the list. 3. Type: parameter block (list) 4. Example: Source scratch programming does not support the copy list, there is no case.

add thing to list
1. Name: add [thing] to [list] 2. Function: It is an instruction for adding an item to the end of a list. 3. Type: execution block (list) 4. Example: Add RGB values that identify colors to the list. Programming requires thinking about the environment and the feasibility of programming. Repeat this step 5 times to add 5 color RGB values. Before adding, clear the list list to know the order of adding colors. Turn off

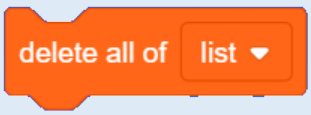
the light to reduce the interference of ambient light on color recognition.





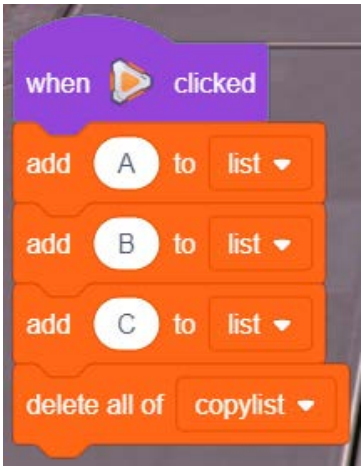
1. Name: delete [1] of [list]
2. Function: It is an instruction for deleting an item at the position indicated by the number [1] in the given list.
3. Type: execution block (list)
4. Example: Add A, B, and C to item 1/2/3 of the list, respectively, and delete item 1 “A” from the list. Remaining B and C.





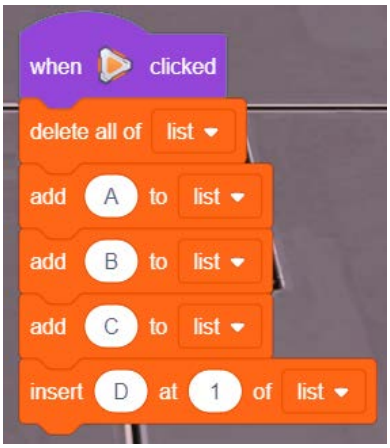
1. Name: Delete all of [list]

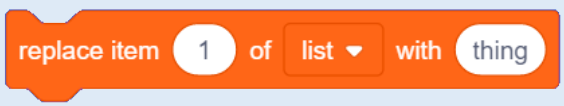
2. Function: It is an instruction for deleting an item at the position indicated by the number [1] in the given list.
3. Type: execution block (list)
4. Example: Delete all the items of this list.





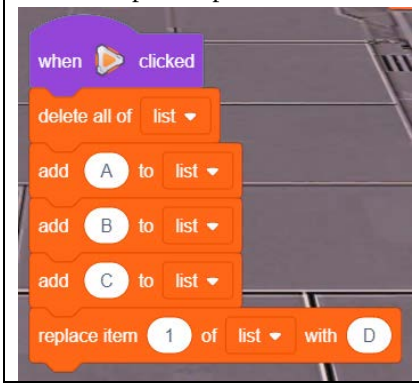
1. Name: insert [thing] at [1] of [list]
2. Function: It is an instruction for inserting an item into the given list at the position indicated by the number [1]. The item initially at that position, and all subsequent items are moved one position further down in the list.
3. Type: execution block (list)
4. Example: Insert D into the first item to get the list sorted as D, A, B, and C.





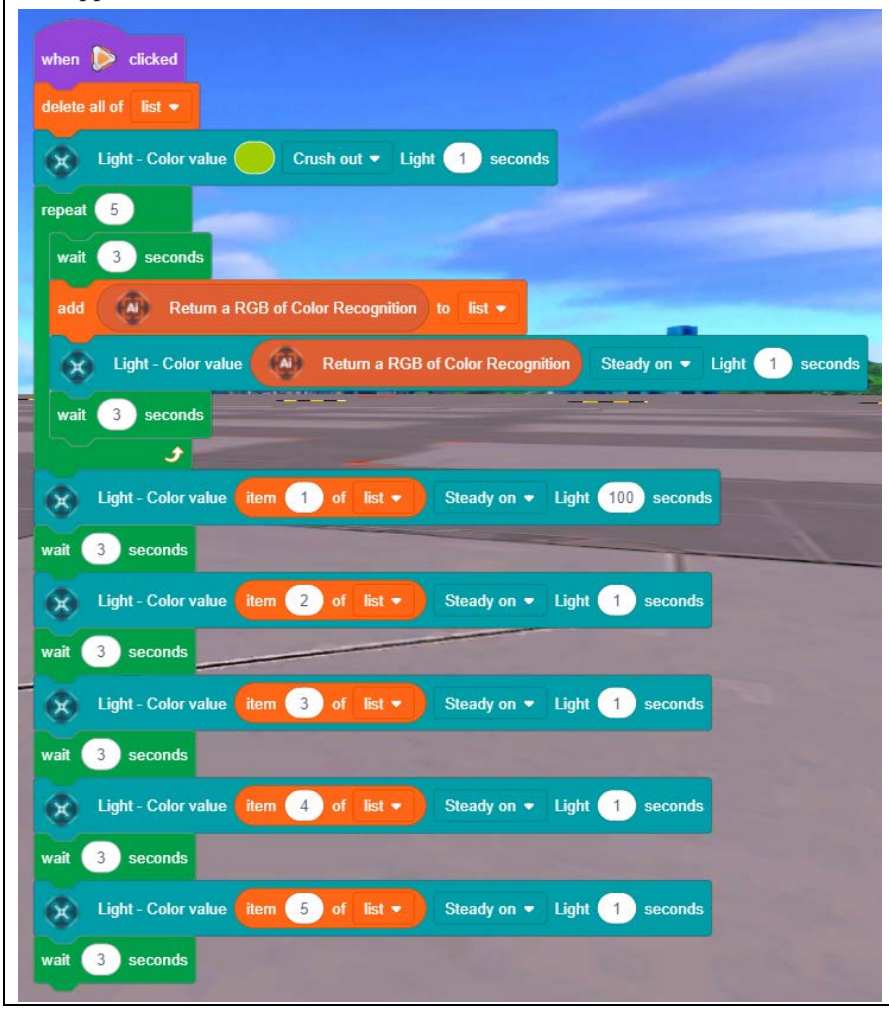
1. Name: replace item [1] of [list] with [thing]
2. Function: It is an instruction for substituting the existing item at the position indicated by the number [1] in the given list with the new specified item.

3. Type: execution block (list)
4. Example: Replace the first item "A" in the list with "D" to obtain lists D, B, and C..

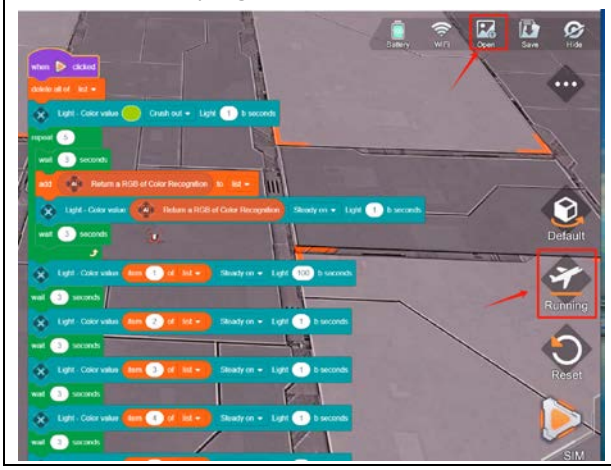


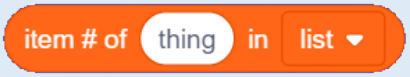


1. Name: item [1] of [list]
2. Function: It is an instruction for retrieving the data stored in the item at the specified position indicated by the number [1] in the given list.
3. Type: parameter block (variable-type data)
4. Example: The drone learns five colors and lights the corresponding colors in the order in which they appear.

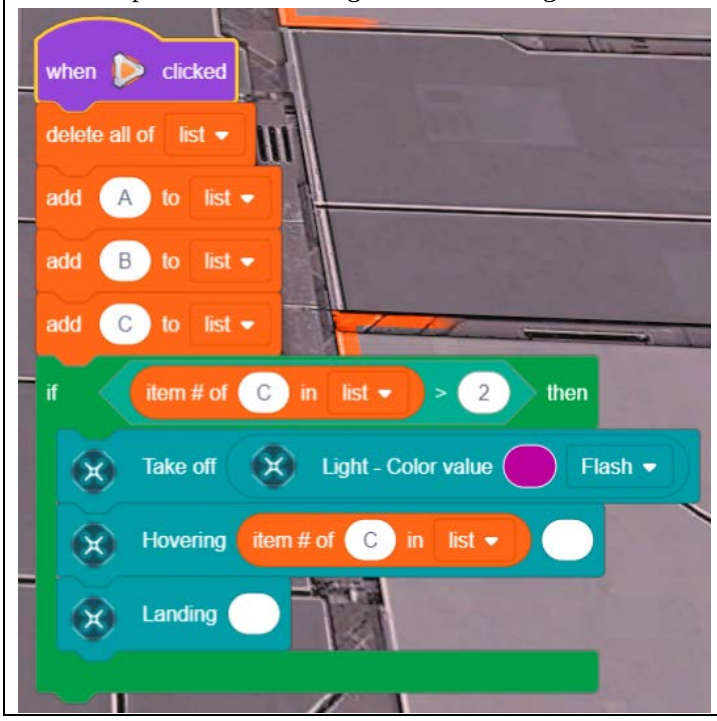


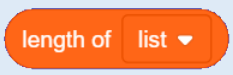
Note: For color recognition, click  to open the image transmission first, and then click  to make the aircraft run the program.





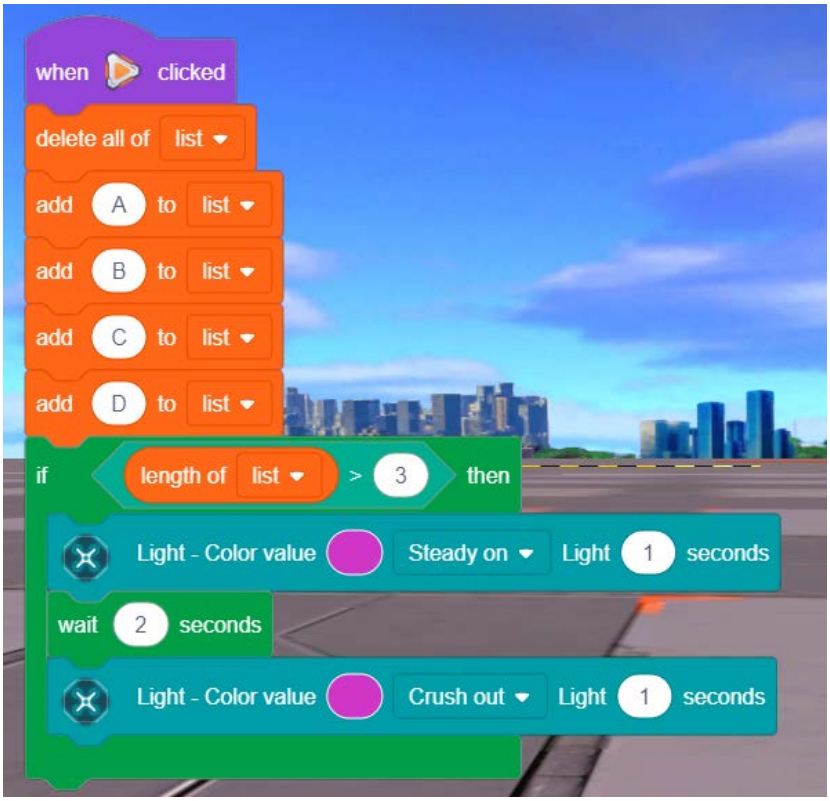
1. Name: item # of [thing] in [list]
2. Function: It is an instruction for getting the position of the specified thing within the given list.
3. Type: parameter block (variable-type data)
4. Example: Aircraft's taking-off and hovering is controlled by C's number in the list.



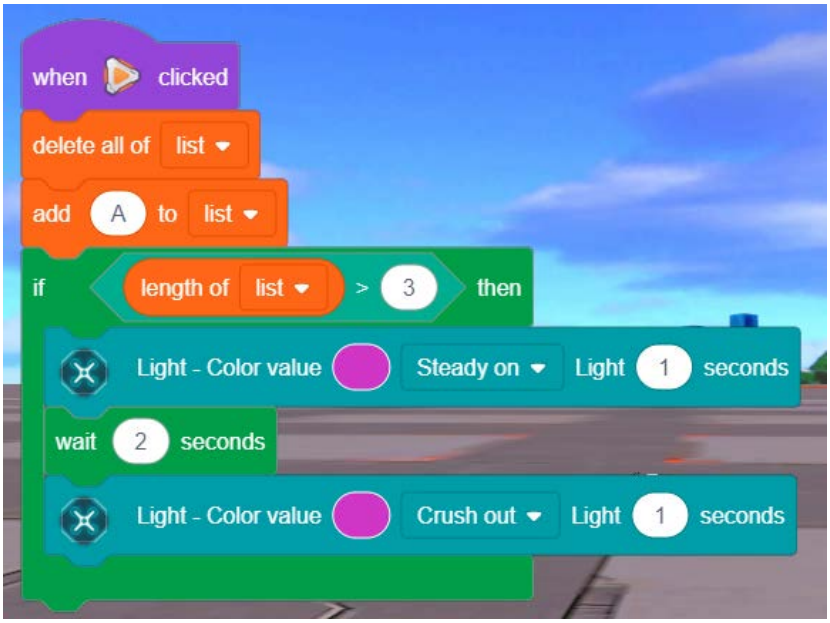


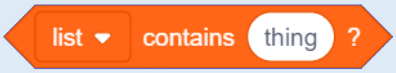
1. Name: length of [list]
2. Function: It is an instruction for getting the count of items contained within the list.
3. Type: parameter block (variable-type data)

4. Example: When you know that there are more than 3 items in the list, the control plane lights up.



If there is only one item in the list, no light is On.



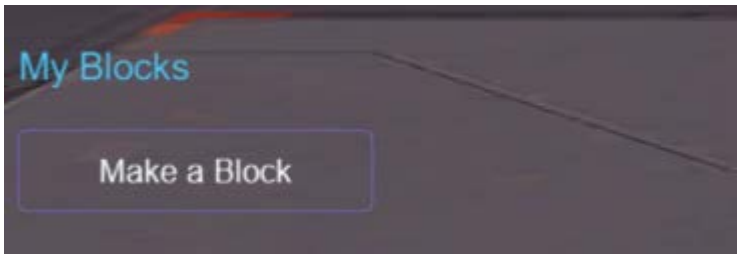


1. Name: [list] contains [thing]?
2. Function: It is an instruction for determining whether some value is contained in the given list. If the list contains the input value, it returns "True". If not, it returns "False".
3. Type: parameter block (variable-type data)

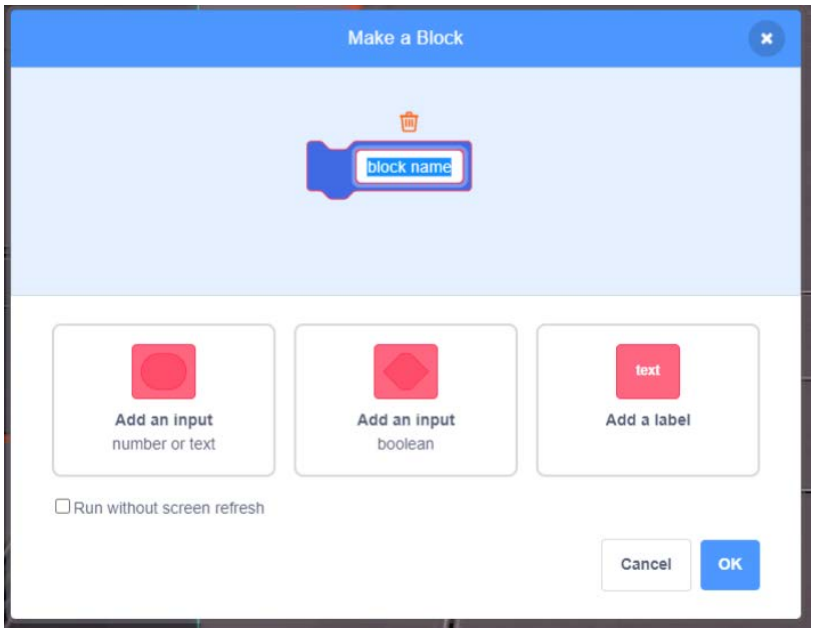
4. Example: If the list contains "D", the light is on.



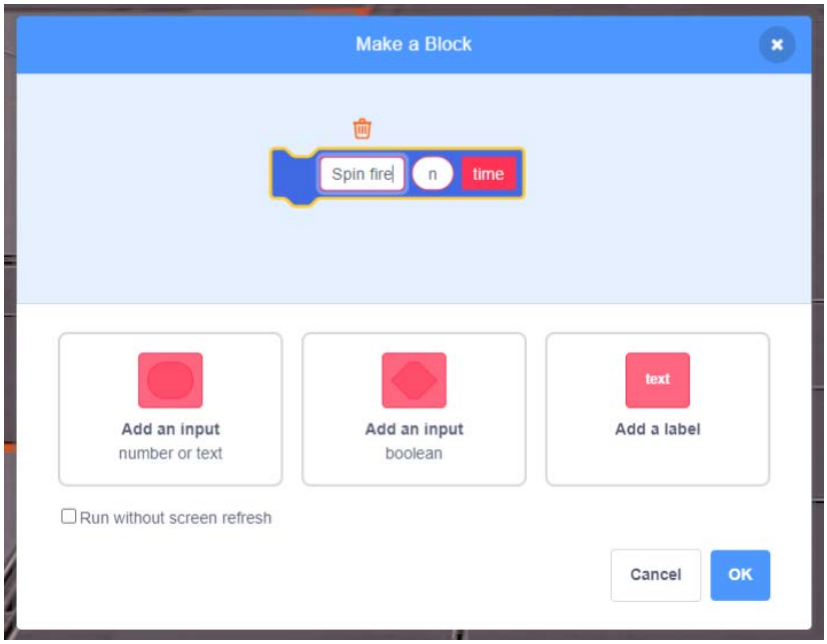
8.2.11 My Blocks



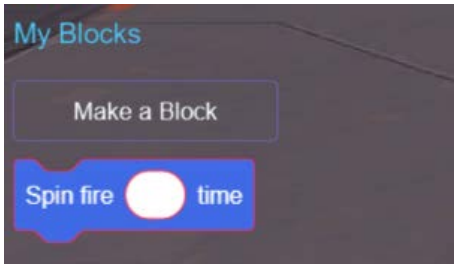
Make a block with user-defined functions.



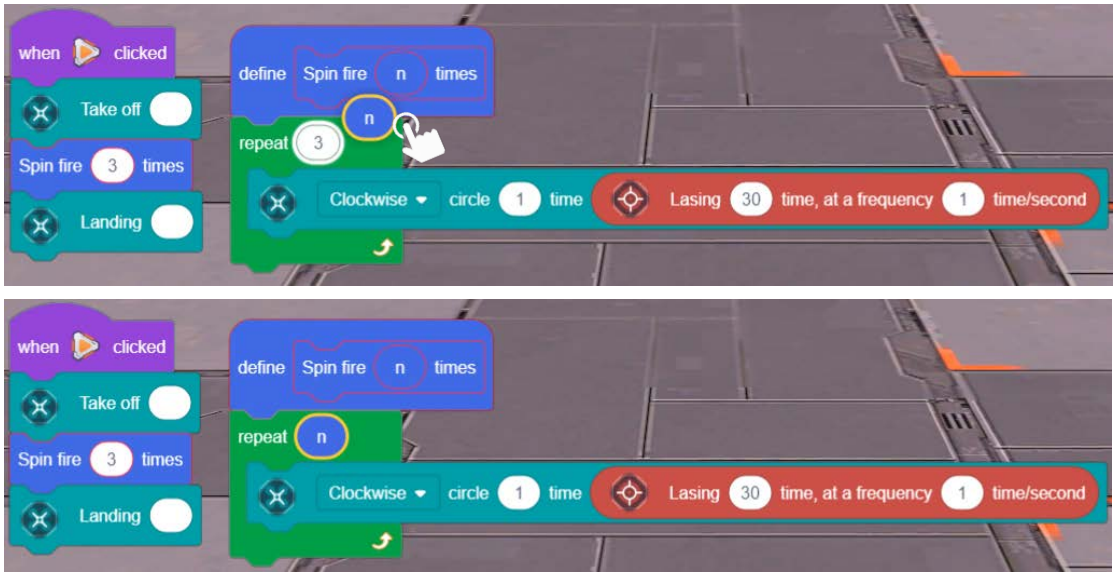
Click on the option [Add an input] and type "n" at the custom block input panel. Once the block is created, "n" will be defined as a variable.



Click to select an input slot, and you can remove the item entirely by clicking the "trash can" icon above the corresponding slot.



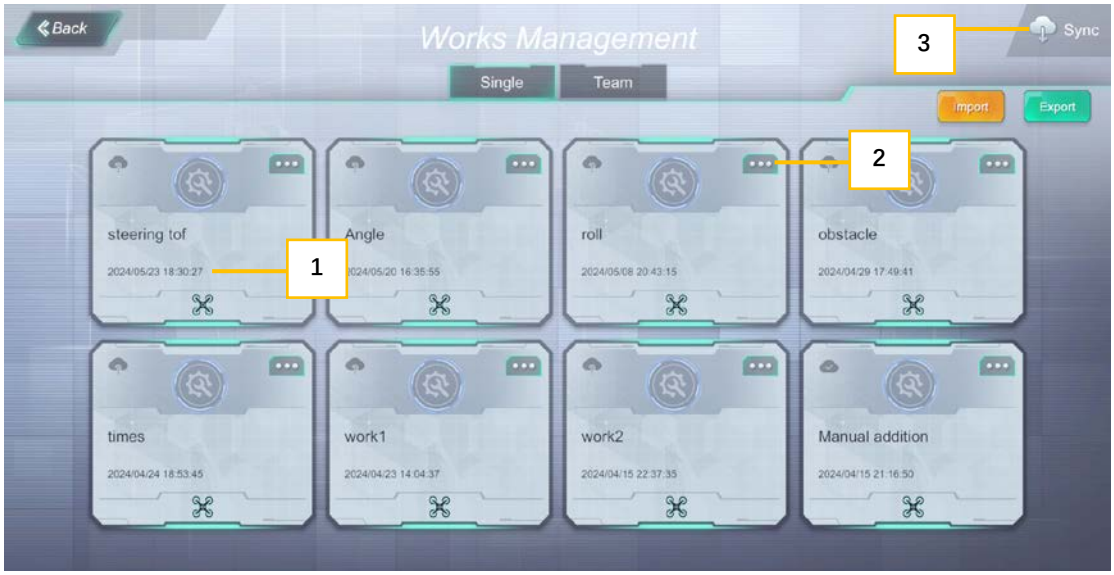
After clicking "OK", you will get a block named "Spin fire [n] times". Define this block as follows:



In the block "define Spin fire [n] times", you can tap and drag the variable block [n] and embed it within a repeat loop.

Spin fire [3] times, where n = 3.

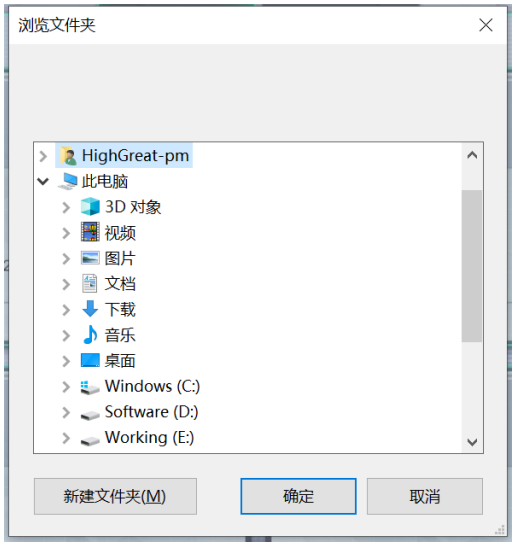
8.3 Work management



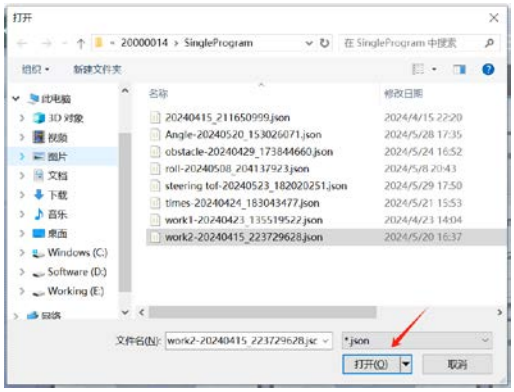
1. Click your work to enter the programming interface. After saving, you can modify the work.
2. Click the horizontal "three dots" icon to open a pop-up window where you can Upload your work to the cloud and rename, copy, and delete it.
3. Synchronize all works in the cloud to the local. Before synchronous download, upload the locally edited works to the cloud first, otherwise there is the risk of being overwritten by the cloud works.
4. PC side supports exporting works, click [Export] multi-selection works, click confirm,



Select export path



5. On the PC, click [Import], select the work file, and click [Open] to import the work.



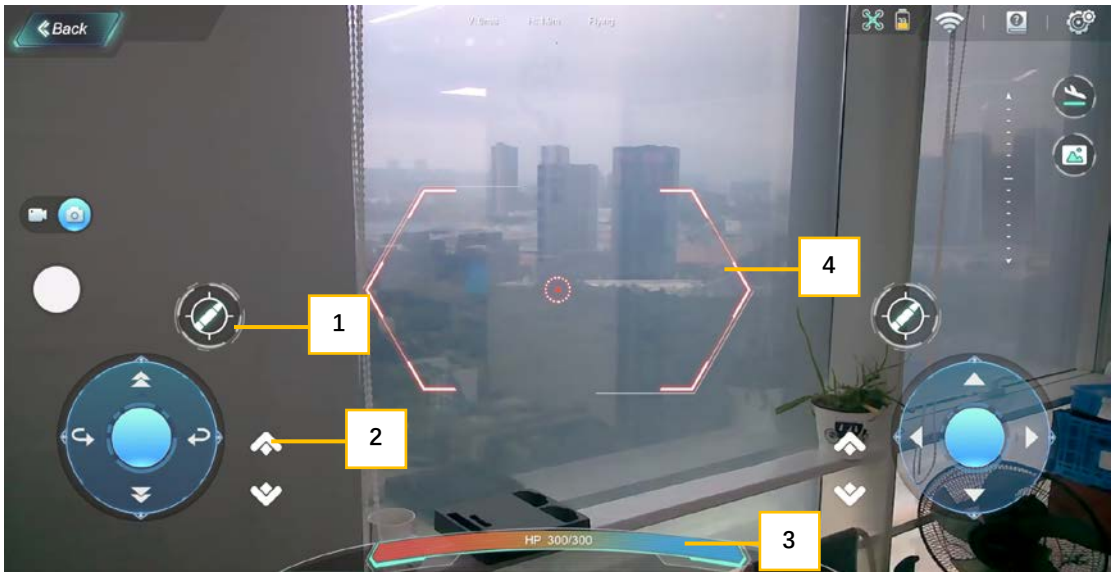
Programming works are saved locally, please cloud backup or export saving.

9. Battle mode

In the battle mode, the aircraft will default to turning off its lights and the obstacle avoidance function.

9.1 Melee mode

Players can directly connect to the aircraft through the APP. They can enter the [Melee mode] and engage in drone battles.



Game rules:

1. Players can directly connect to the aircraft through the APP. In the APP, they can select the [Melee mode] and engage in drone battles. Any other player who enters the melee mode can engage in drone battles.
2. In melee mode, each player fights for themselves, and there is no team formed.
3. After entering the game, the player can control the aircraft to emit laser lights and hit and defeat enemy aircraft.
4. Each player starts with 300 virtual health points (HP). 100 health points will be deducted every time the drone gets hit. When the health points ≤ 0, the drone is considered to be in a defeated state. In this state, the drone cannot emit a laser light and will not take further damage either.
5. A defeated aircraft needs to wait for a certain time before it can respawn and rejoin the battle.
6. **Operation instruction:**

   (1) **Open fire:** Click to fire and emit a laser light once.

   (2) **Gimbal adjustment:** Adjust the gimbal angle up and down for precise aiming assistance.

   (3) **Health points (HP):** Health points are deducted when hit.

   (4) **Crosshair frame:** It is the aiming crosshair for the aircraft's laser, used for aiming assistance.

9.2 Scrimmage

Players can set the aircraft to networking mode, with both the aircraft and the mobile device connecting to the same router's WiFi. They can then enter the [Scrimmage] mode to engage in the team battle.

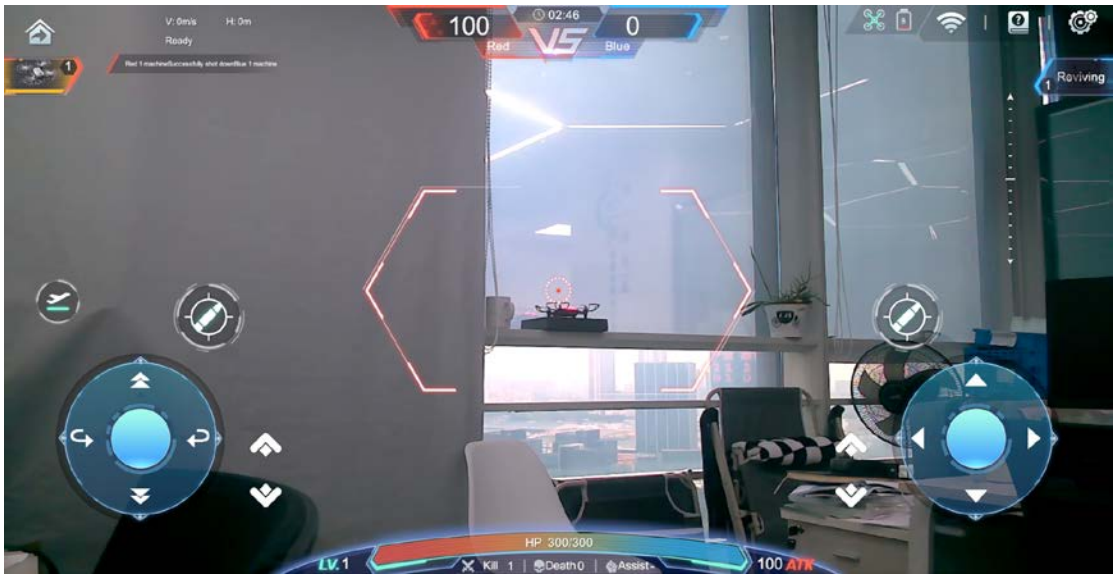
9.2.1 Create a room



Create a room and the rules for entering the room:

7. All players participating in the battle must be connected to the same WLAN (Wireless Local Area Network) to engage in the [Scrimmage] mode.
8. The room is divided into Red and Blue parties. Once the game starts, players from the Red party will compete against players from the Blue party in a drone battle.
9. The first player to enter the [Scrimmage] mode will be considered the room-owner. Other players should wait for 10 seconds after the room-owner creates the room before entering.
10. The room-owner has the authority to adjust the game duration, game points, flight speed, and NO of players.
11. All players can change to an empty slot within the room before the game starts.
12. Once all players are ready, the room-owner can click on the [Start] button to begin the game.

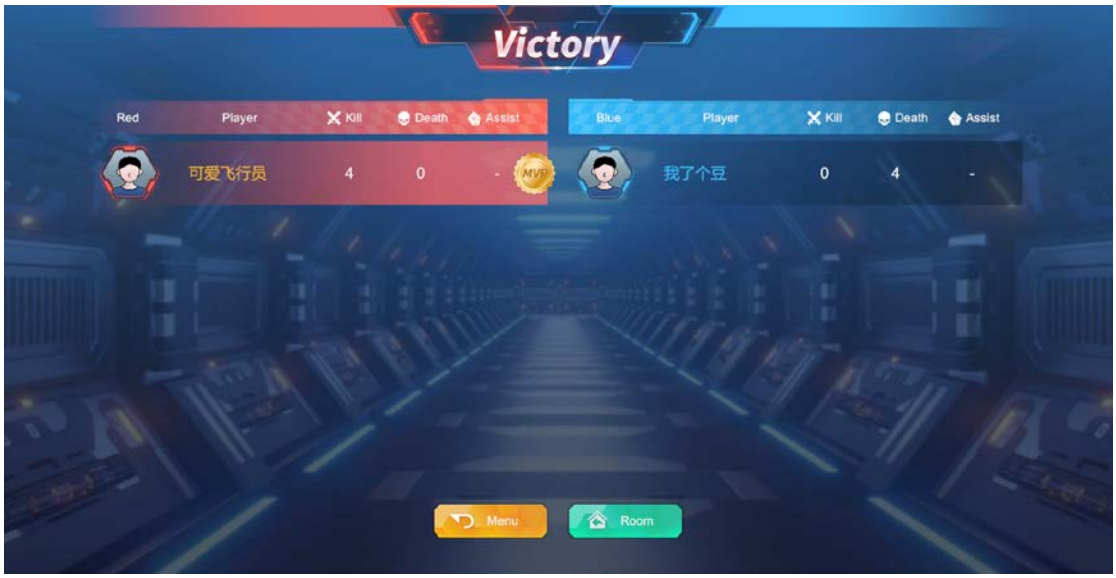
9.2.2 Battle driving interface



Game rules:

1) The game is divided into two parties, Red and Blue. Players decide which party to join in accordance with which one they belong to in the Scrimmage room.
2) Drones from the Red and Blue parties engage in combat by emitting laser lights at each other. Each player starts with 300 virtual health points (HP). 100 health points will be deducted every time the drone gets hit. When the health points ≤ 0, the drone is considered to be in a defeated state. In this state, the drone cannot emit a laser light and will not take further damage either.
3) When you defeat enemy drones, points will be accumulated for your own party, with 10 points added for each defeat.
4) A defeated aircraft needs to wait for a certain time before it can respawn and rejoin the battle.
5) When the game duration is up, the party with higher points wins. If the points are tied, it's considered a draw.
6) If either party achieves the preset highest points for a single round before the game duration is up, the game will end early, and the party with the highest points wins this game.

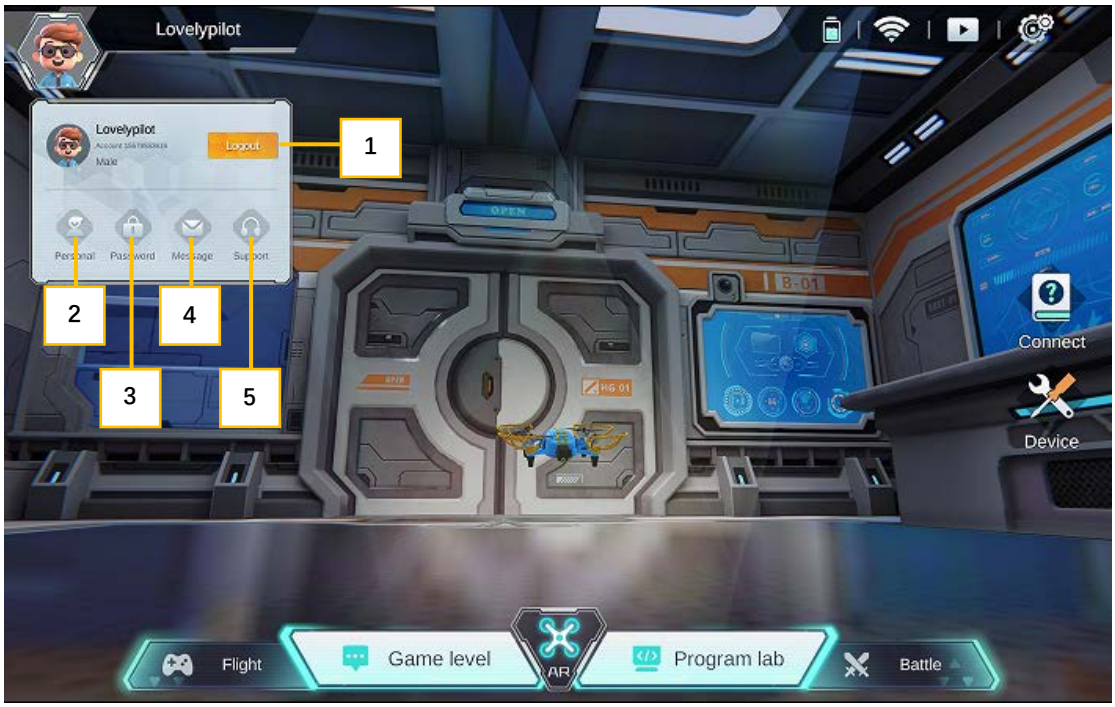
9.2.3 Settlement



Settlement rules:

1) After a match ends, the winner is determined based on the higher battle points of the two parties.
2) After a match ends, the system will compile all players' data on KDA (Kill, Death, and Assist), which will be displayed on the settlement interface.
3) Based on the battle performance in this match, MVP players from both the Red and Blue parties will be selected.

10. Profile

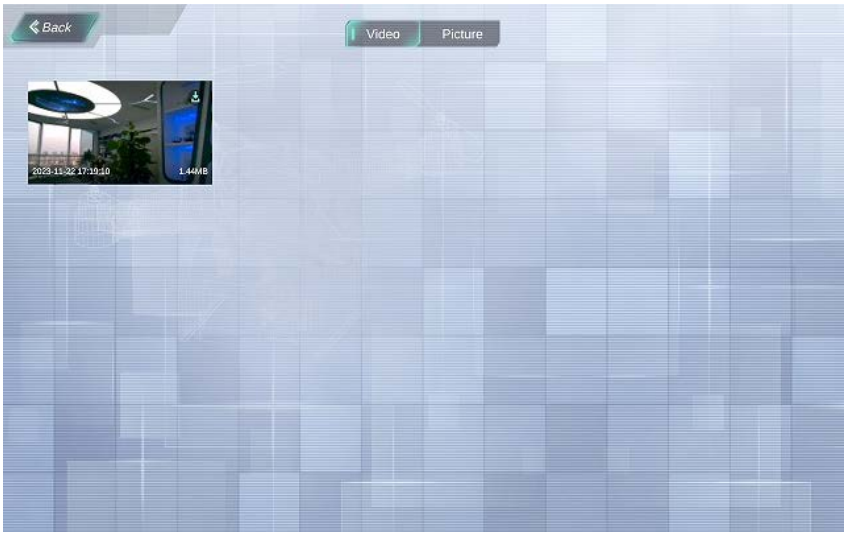


1. Log out: There is a risk of data loss after logging out.
2. Personal info: You can modify your avatar, gender, and username here.
3. Password management: You can change your password via account + verification code here.
4. Message center: You can view system messages here.
5. Technical support: You can view the APP User Manual, provide feedback, and seek after-sales service and technical support here.

Changing accounts might lead to data loss from the previous account. Before logging out, please synchronize data while connected to the Internet.

11. Album

1. This is your video list. Click to download aerial videos from the drone to your local device. Tap on a video file to preview.



2. Tap and hold a video file to evoke the mode of deleting videos. You can select multiple files to delete when the aircraft's storage space is insufficient.
3. This is your photo list. It has a similar management way to that of video.

After downloading videos and photos, please check your phone's "File Manager".

12. Settings

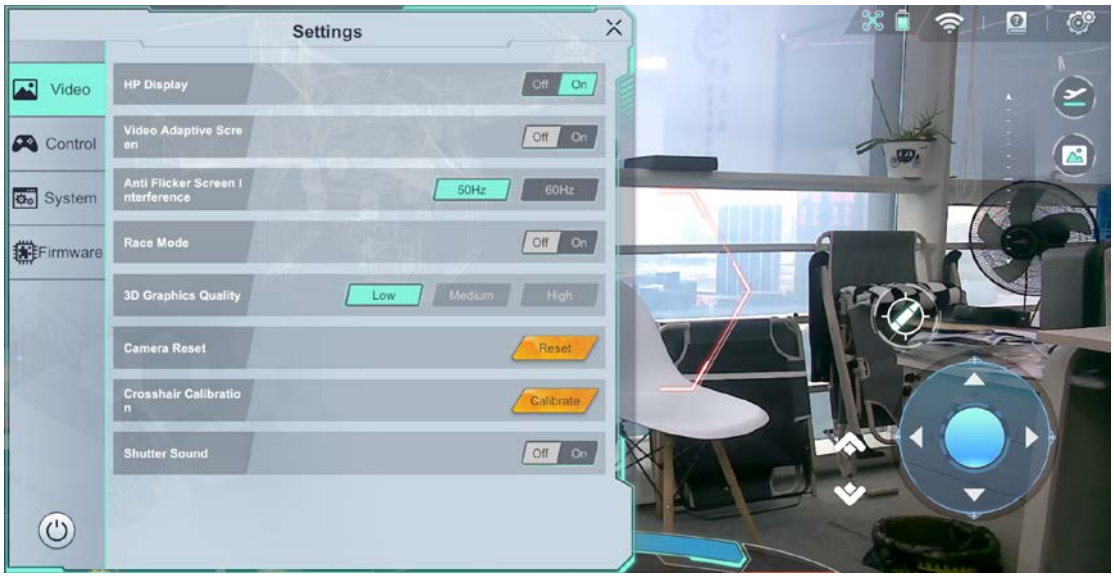
12.1 Image transmission



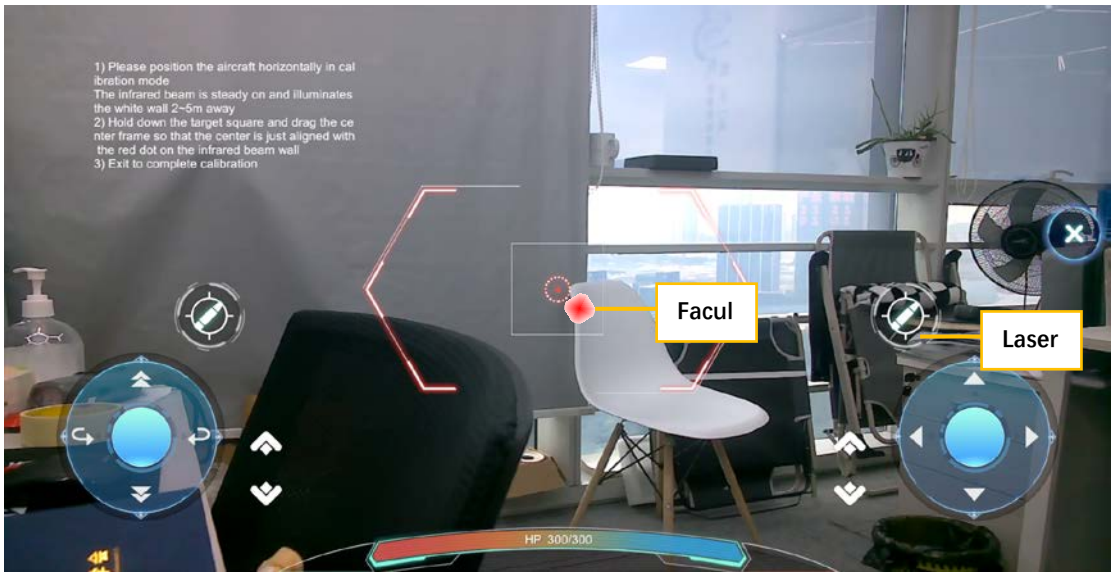
1. **HP Display:** HP Display is set to "On" by default. If it is turned off, the HP bar will not be displayed in battle.
2. **Transferred Image Auto-adapt Screen Switch:** Image transmission is displayed in 16:9

full-screen mode by default. If you want to adapt to other screen ratios, please enable the auto-adapt mode.

3. **Video Resolution:** Available resolution settings include 720/30 fps and 1080 fps.
4. **Anti-flash Screen Interference:** Available frequency settings include 50 Hz and 60 Hz. It is used to prevent flickering interference from fluorescent lights or screens.
5. **3D Image Quality:** You can choose between low, medium, and high-quality settings. Please select the appropriate image quality based on your device's performance.
6. **Camera Reset:** It is used to reset the gimbal to a horizontal position.
7. **Shutter Sound:** It is a switch that controls the camera shutter sound.
8. **Power Off:** You can power off the aircraft when it is in a landed state.
9. **Crosshair Calibration:** Open Settings within Battle Mode - Melee Mode to access crosshair calibration.
10. **Competition mode:** It can be opened after directly connecting the aircraft, and the competition can improve the anti-interference ability of the aircraft during wartime;



Align the crosshair with the facula by aiming at a white wall and emitting the laser light. Click on the facula, and the crosshair frame will move to align with the facula. Click [X] to close and complete calibration.



12.2 Control

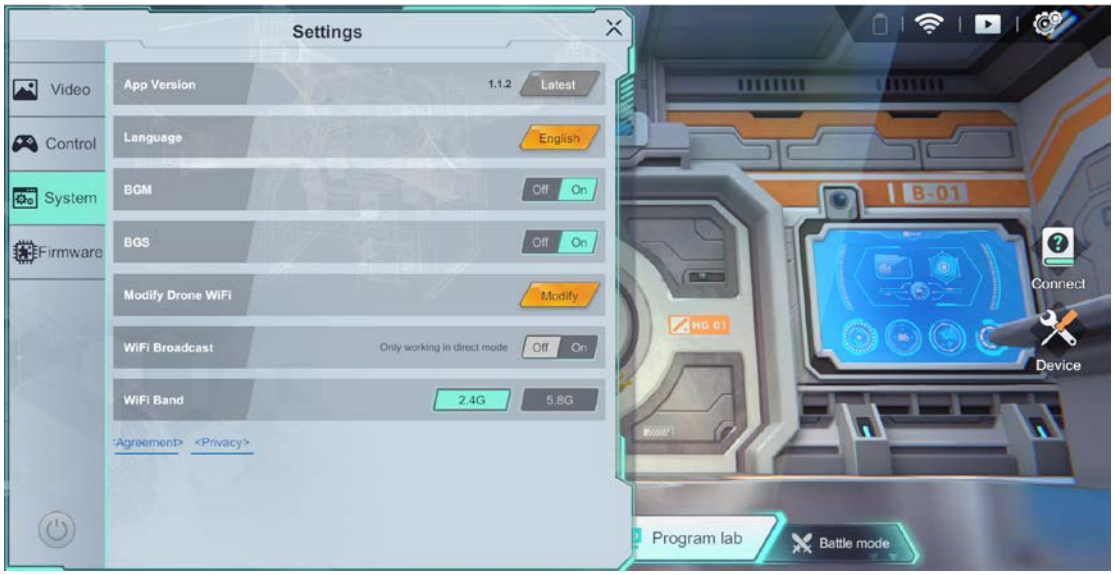


These settings apply only to Single driver and Battle mode and are ineffective in Programming mode. After connecting to the aircraft, it will synchronize the settings parameters of the aircraft.

1. **Movement Speed:** Low 0.5 m/s, Medium 1 m/s, and High 2 m/s.
2. **Obstacle Avoidance Mode Switch:** When enabled, the aircraft will have obstacle avoidance capability and maintain a distance from obstacles.
3. **Infrared Obstacle Avoidance Distance:** Enter the calibration instructions to adjust the detection distance of the infrared obstacle avoidance.
4. **IMU Calibration:** You can calibrate the accelerometer and gyroscope of aircraft here.
5. **Lamp Brightness:** You can adjust the brightness of the lights here.
6. **Infrared Laser Reception:** When it is turned off, the aircraft will not receive laser lights.
7. **Crosshair Type:** You can select the crosshair type here.

8. **Control Way:** You can choose the control way between the Japanese hand and the American hand. After setting, it will become effective when you enter Single driver.
9. **Remote Control Equipment:** On PC, you can choose between the [Mouse + Keyboard] or [Keyboard] button to remote control. Ignore this for Pad and mobile devices.
10. **Vibration Upon Hit:** The Pad/mobile device will vibrate when the aircraft is hit. This applies to devices that support the Vibration function.
11. **Low Battery Auto Landing:** By default, the aircraft automatically lands when the battery is below 10%. Disabling this function may result in a risk of crash due to power loss.
12. **Low Battery Reminder:** The APP will display a reminder when the aircraft's battery is below the specified value.
13. **Track Angle Sensitivity:** The steering angular velocity can be set as Low 45°/s, Medium 90°/s, and High 120°/s.
14. Joystick follow: After opening, the initial position of the virtual joystick will follow the finger on the screen;
15. Button layout: automatic rocker, launch button, fine-tuning PTZ button position;

12.3 System



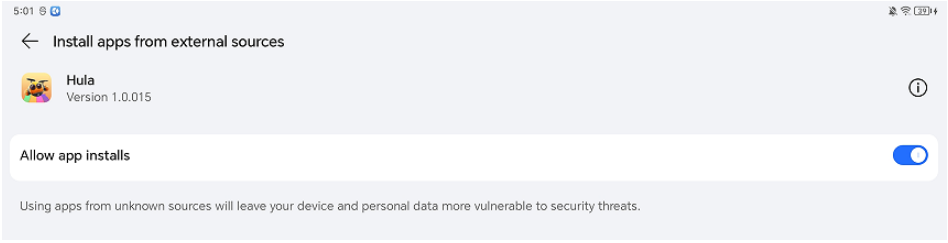
1. **APP Version:** After connecting to the Internet, click on the [Download] button to upgrade the APP.
2. **Language:** Switch between multiple languages.
3. **Modify the aircraft WiFi:** when directly connected to the aircraft, you can modify the aircraft WiFi and password (2.4/5.8G are modified), the aircraft will restart after modification;
4. **WiFi broadcast:** enabled by default. According to national laws and regulations, drones need to broadcast Remote ID;
5. **Switching of WiFi frequency band:** when connecting to the plane directly, you can switch the frequency band of the plane WiFi, and re-enter the password when connecting to the new WiFi of the plane after switching;

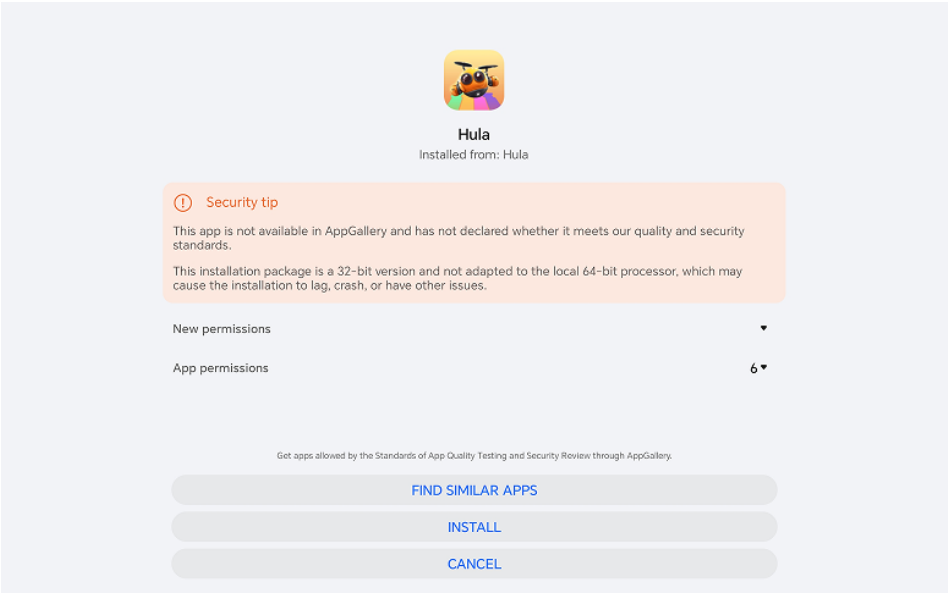
12.3.1 APP upgrade

Every time you open the APP, if a new version is available, a pop-up window will prompt for an upgrade. You can also go to Settings > System > APP Version > Download to open the upgrade pop-up.



1.Click on the [Upgrade Now] button, and the APP will Start downloading the APP update pack,
2.The APP is downloading. Please wait patiently for the download to complete.

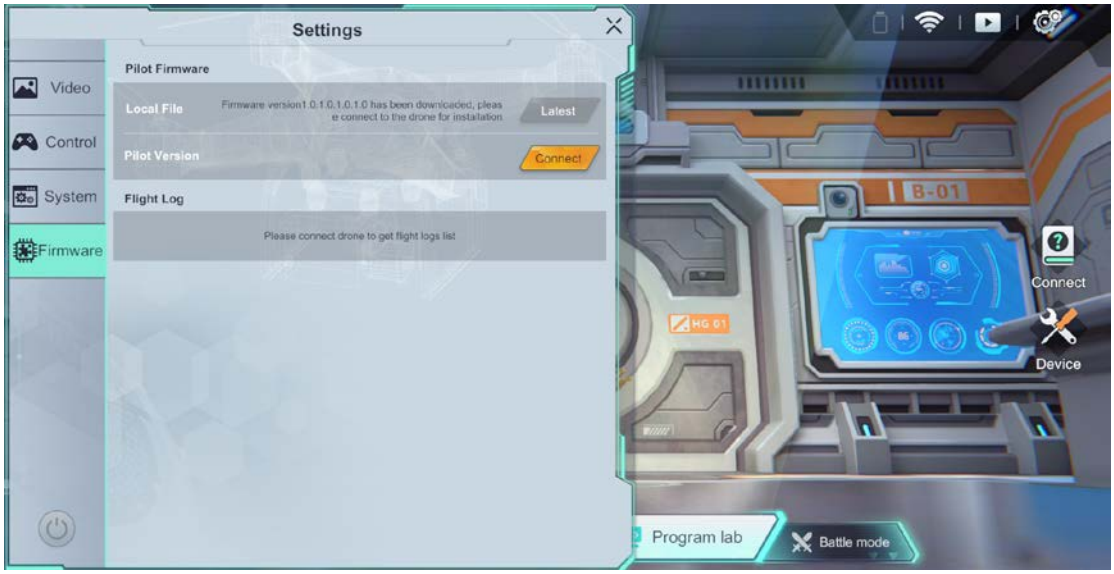




After the APP download is complete, please follow the prompts for the installation process.

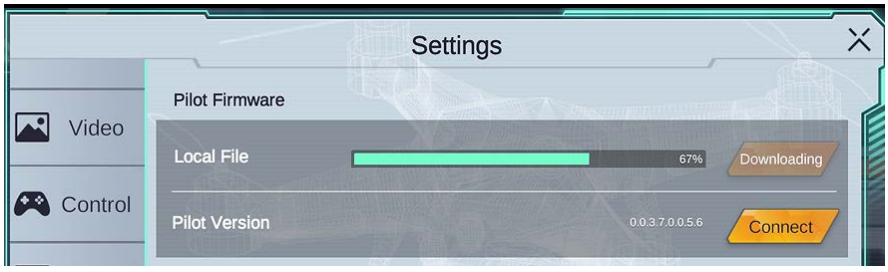
When upgrading the APP, do not disconnect the WiFi, do not close the APP, and do not turn off the control terminal. Otherwise, it may result in upgrade failure.

12.4 Firmware



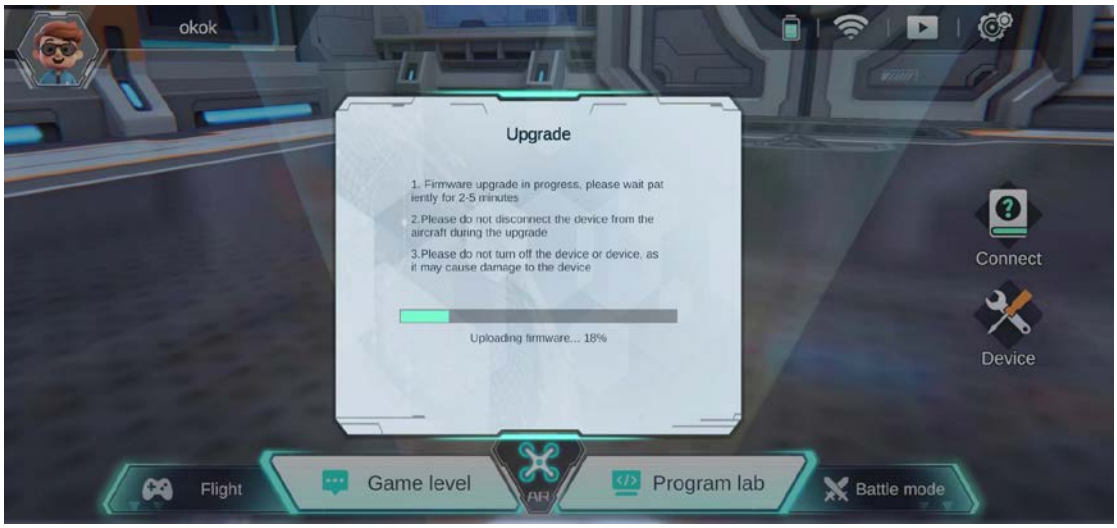
1. **Local Firmware Upgrade:** After connecting to the server, click to download the latest flight control firmware to your local device.
2. **Firmware Version:** If you've already downloaded a new version of the flight control firmware, it will display [Upgrade Flight Control] when connected to the aircraft.
3. **Flight Log:** Please connect to the aircraft to get the flight log list. Flight logs can be downloaded after connecting to the aircraft.

12.4.1 Flight control firmware upgrade

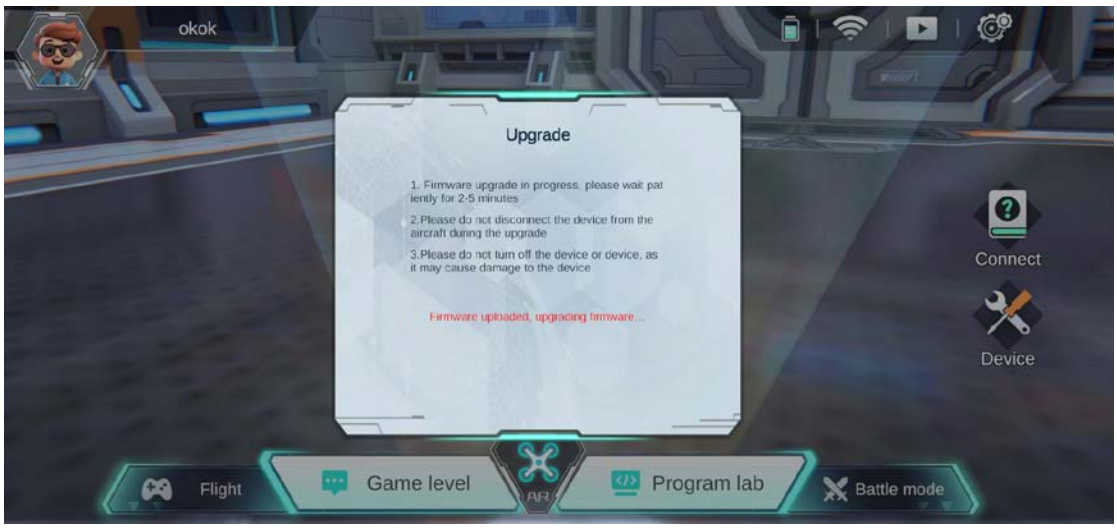




1. Connect to the server and download the latest flight control firmware. Wait for the download to complete.
2. After connecting to the aircraft, click on the [Upgrade Flight Control] button and follow the prompts for the upgrade process.

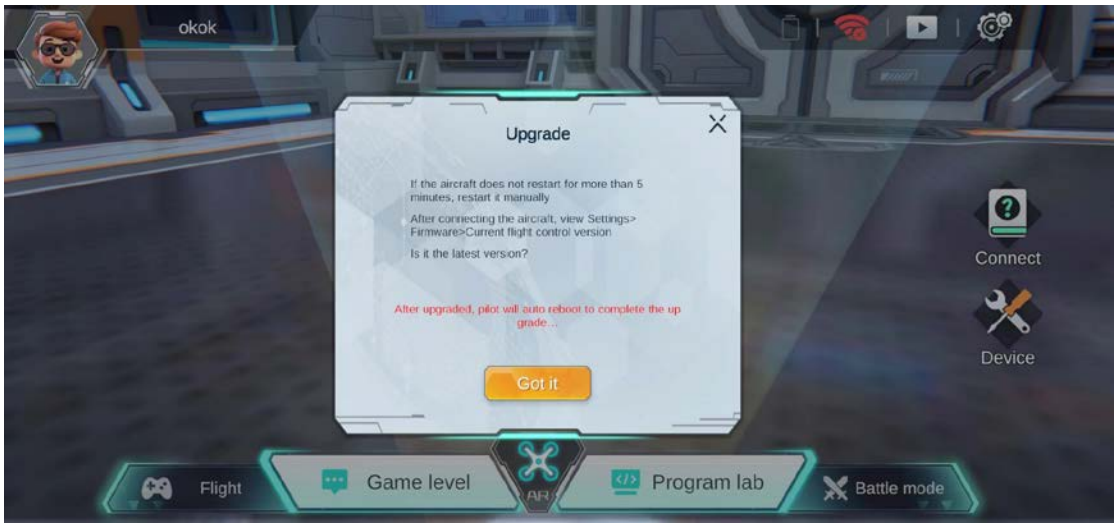


3. The flight control firmware is being uploaded to the aircraft. The aircraft's blue light will flash. Please be patient.



4. After you upload the firmware successfully, please wait a few minutes. The aircraft will automatically complete the upgrade. During the upgrade, the aircraft will turn off its lights

and then light up in red, green, and blue for self-check,The camera has reset action. This indicates a successful upgrade and reboot.



5. Once the upgrade is complete, the pop-up window will close automatically. Please reconnect to the aircraft's WiFi. It will show "Already the latest version".
6. If the upgrade fails, click on the [Re upgrade] button to try the upgrade again.

When upgrading the firmware, do not disconnect the aircraft, do not close the APP, and do not turn off the Control equipment or aircraft. Otherwise, it may result in damage to the aircraft during the upgrade.